PANJAB UNIVERSITY, CHANDIGARH-160014 (INDIA)

(Estd. under the Panjab University Act VII of 1947—enacted by the Govt. of India)

SYLLABI

AND THE

STRUCTURAL OUTLINE

FOR

B.A. & B.Sc. GENERAL SECOND YEAR EXAMINATION, 2015

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STRUCTURAL OUTLINE OF THE B.A./B.SC. (GENERAL AND HONOURS) DEGREE COURSE IN THE 10+2+3 SYSTEM OF EDUCATION FOR FIRST YEAR 2014 EXAMINATION (ALREADY HELD) SECOND YEAR 2015 EXAMINATION

The new B.A./B.Sc. (General & Honours) course (at + 3 stage) in the 10+2+3 system of education has been structured keeping in view certain basic guidelines as under :--

1. (i) The scheme should provide a broad-based, interrelated and comprehensive programme of study, built upon the + 2 stage, which ultimately may act as a sound foundation for postgraduate education, participation in competitive examinations, and joining professional courses like those in Law, Education, etc.

   (ii) It is so structured and organised that it is in conformity with the needs of the country in the present context. Apart from providing good general education, it should have an applied component as part of the course work so that it creates awareness of the practical problems facing the society and generates employability of the youth in diverse spheres of activity.

   (iii) It is so designed and formulated that it provides flexibility and relative freedom of choice of courses to the students according to their special interests and capabilities.

   (iv) It is based on intensive teacher–student interaction.

2. The B.A./B.Sc. courses have been so structured and revised that they are comparable both in content and quality.

3. To cope with the fast increasing knowledge in each discipline, it has been considered desirable to cover each elective subject in two papers.

The two-paper scheme will enable the teachers to teach the subject in detail and examine the students accordingly. Effort has to be made to improve the level and adequacy of content of each course in such a manner that the total scheme provides distinct upgradation of the existing curricula for the first degree. In the absence of such an upgradation, the increase in the total educational period from 14 to 15 years for the B.A./B.Sc. will be unjustifiable.

Keeping in view the above objectives and guidelines, the following structural outline has been introduced for the B.A./B.Sc. (General) degree course :

Admissions to B.A./B.Sc. (General/Honours) course for the second and third year (For the students who have passed First Year under this scheme) for the session 2012-2013 (2013 examination) shall be as hereinafter mentioned.

Note : A person who has passed B.A./B.Sc. 1st year or 2nd year examination from other Universities in India may be allowed to migrate to this University subject to the condition that he shall have to clear the deficient subject but the total number of credits required to be earned shall remain the same.

(ii) STRUCTURAL OUTLINE OF THE B.A./B.SC. (GENERAL/HONOURS) DEGREE COURSE IN THE 10+2+3 SYSTEM OF EDUCATION

B.A. (General)

The B.A. (General) programme of study shall consist of 24 credits, each credit having a value of 100 marks. A subject studied for the whole academic year shall carry 2 credits. All the theory papers and practicals irrespective of their credit value shall be studied throughout the academic year.

The detail of the subjects to be studied shall be as under :

1st Year 2014 Examination (already held)

COMPULSORY

(a) Punjabi – Two Papers — 1 Credit

OR

*History & Culture of Punjab –One Paper — 1 Credit

These papers constitute one subject in each year.

(b) English — 1 Credit

ELECTIVE

+ Any three elective subjects of two credits each including languages and elective vocational subject (To be studied in selected colleges). — 6 Credits

Total : 8 Credits

****Environment and Road Safety Education : 70 marks**

2nd Year 2015 Examination

Same as in the 1st year

Note :

1. In subjects having practicals, the theory papers and practicals together will be of 2 credit value. The candidate will have to pass in theory and practical/s separately.

2. Each paper of one credit shall be allocated 3 hours of teaching per week during each academic year. However, for subjects having practicals, three hours of teaching will be allocated to each theory paper and two hours for a practical per week or as per requirement of the subject concerned.

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* For B.A./B.Sc. History and Culture of Punjab is allowed as an option in lieu of Punjabi (compulsory) to those students who had already offered earlier in the First Year of 2014 examination.

** This is a compulsory qualifying paper, which the students have to study in the B.A./B.Sc. 1st year. The students are required to qualify this paper either in the 1st year/2nd year/3rd year of the course. The examination will be conducted by the University.

3. B.A. (General) degree holders shall be eligible for admission to the Master's course in any of the elective subjects studied by them during all the three years of the programme of study, earning six credits in each, provided they fulfil the eligibility conditions.

4. A student would offer any science subject, including Mathematics, only if he has passed that subject in the qualifying examination or qualifies in the subject as a deficient/additional subject from the concerned Board/University/Council in the Supplementary Examination subsequent to the admission.

   Provided further that a student can offer :

   (a) Statistics only if he takes up Mathematics.

   (b) Applied Statistics only if he takes up other subject(s) excluding Mathematics.

B.Sc. (General)

The B.Sc. (General) programme of study shall consist of 20 credits, each credit having a value of 100 marks. A subject studied for the whole academic year shall carry 2 credits. All the theory papers and practicals irrespective of their credit value shall be studied throughout the academic year.

The detail of the subjects to be studied shall be as under :

1st Year 2014 Examination (already held)

*Punjabi/History & Culture of Punjab	1 Credit
+ Three Elective Subjects of 2 credits each including Elective Vocational Subject (To be studied in selected colleges)	6 Credits
Total :	**7 Credits**

** Environment and Road Safety Education : 70 marks

2nd Year 2015 Examination

(a)	English	1 Credit
(b)	Three elective subjects of 2 credits each including Elective Vocational Subject (To be studied in selected colleges) (the same as in the 1st year)	6 Credits
	Total :	**7 Credits**

* For B.A./B.Sc. History and Culture of Punjab is allowed as an option in lieu of Punjabi (compulsory) to those students who had already offered earlier in the First Year of 2014 examination.

** This is a compulsory qualifying paper, which the students have to study in the B.A./B.Sc. 1st year. The students are required to qualify this paper either in the 1st year/2nd year/3rd year of the course. The examination will be conducted by the University.

(iv) STRUCTURAL OUTLINE OF THE B.A./B.SC. (GENERAL/HONOURS) DEGREE COURSE IN THE 10+2+3 SYSTEM OF EDUCATION

Note :

1. The two credits allocated to an elective subject, to be studied in an academic year as spelled out above, shall be covered in two theory papers and a practical or practicals as per requirements of the subject concerned.

2. In the instructional process, it is desirable to strike a balance in the teaching and learning strategies. A close interaction between the teacher and the student is necessary to achieve the above objectives.

3. Each elective subject of 2 credits for the B.Sc. shall be allocated five hours of teaching and four hours of practical work per week during the course of the academic year or as per requirements of the subject concerned.

4. B.Sc. (General) degree holders shall be eligible for admission to the M.Sc. course in any of the three elective subjects which they have studied during all the three years of the degree course, earning 6 credits in each, provided the candidate fulfils the eligibility conditions.

B.A./B.SC. HONOURS

In addition to the main objectives listed in the preamble to the B.A./B.Sc. (General) degree course, the Honours Course seeks to provide course work of advanced nature in one of the elective subjects already being studied from 1st year onwards by a student. Both in content and quality, it would be of a level as gives adequate specialized knowledge in a subject even to make it a culminating point in the programme of study if a student so chooses. Or else, it should provide strong, additional foundation for pursuing the subject at the M.A./M.Sc. level.

Keeping in view the objectives stated in the preamble to the B.A./B.Sc. (General) degree course and that enunciated above, the structural outline for the Honours degree course shall be as under :

B.A. (Honours)

The B.A. Honours Course shall carry 28 credits (each credit carrying the equivalent of 100 marks). Apart from the structure of the programme of study prescribed for the B.A. (General) degree course, credit value of the subjects/papers and hours of teaching allocated to them as already stated in the scheme, the Honours Course shall include four papers of one credit each of an advanced nature in one of the elective subjects a student shall study in all the three years of the degree course.

A student may offer Honours in Second Year in any one of the elective subjects, to be studied by him in all the three years of the course.

Provided that he has obtained at least 50% marks in the subject of Honours in the First Year of the B.A. (General) Course.

For the B.A. (Honours) Course, the distribution of work during the three years will be as under :-

1st Year : The same as in B.A. (General). (8 credits)

2nd Year : The same as in 2nd Year of B.A. (General). In addition, there shall be two advanced papers of one credit each in the subject in which he seeks to get Honour degree.
(8+2=10 credits)

3rd Year : The same as in 3rd year of B.A. (General). In addition, there shall be two advanced papers of one credit each in the Honours subject.
(8+2=10 credits)

Total Credits : 28

A candidate eligible to appear as a private candidate in B.A. Examination may also offer Honours papers in any of the elective subjects offered for B.A. (General) in all three years provided he fulfils the requirement of the Regulation.

B.Sc. (Honours) :

The B.Sc. (Honours) programme of study shall consist of 24 credits, each credit having a value of 100 marks. A subject studied for the whole academic year shall carry 2 credits. All the theory papers and practicals irrespective of their credit value shall be studied throughout the academic year.

A student may offer Honours in any one of the elective subjects to be studied by him in all the three years of the course provided he has obtained at least 50% marks in the subject concerned in the first year Examination of the B.Sc. (General) course.

Of the 24 credits, each student shall offer courses in each year as under :-

1st Year : The same as for First Year of B.Sc. (General). (7 credits)

2nd Year : The same as for 2nd Year of B.Sc. (General). In addition, there shall be two advanced papers of one credit each in the subject in which he seeks to get Honours degree.
(7+2=9 credits)

3rd Year : The same as for 3rd year of B.Sc. (General). In addition, there shall be two advanced papers of one credit each in the Honours subject.
(6+2=8 credits)

Total credits : 24

Both in B.A. Honours and B.Sc. Honours Courses, each paper of one credit shall be allocated hours of teaching, or as per requirements of the subject concerned.

Note : Instruction through audio and/or video cassettes may form a part of Languages course.

(vi) **GUIDELINES REGARDING CONTINUOUS ASSESSMENT OF REGULAR STUDENTS OF B.A./B.SC./B.COM./B.C.A. COURSES**

IMPORTANT NOTE

(i) In order to incorporate an element of continuous assessment of students, the Colleges will conduct two mandatory House Tests in theory papers – one in the month of September/October and the other in December/January every year.

(ii) *(a)* For September Test, there will be only one paper of one hour's duration in each subject, and for December Test, there will be paper/s on the pattern of annual examination conducted by the University.

There will be a Special Test for those students who could not fulfil the conditions of eligibility. It will not be held to provide an opportunity to all students to improve their earlier score. Those students who are exempted by the Principal of the College from appearing in the House Test/s in September and/or December/January will also be allowed to appear in the Special Test; this Test will determine their eligibility for admission to the examination as well as their score for Internal Assessment.

(b) With a view to meet the grievance of students, if any, on account of scores obtained by them, the answer-books will be shown to them. Difference of opinion on the issue, if any, will be sorted out with the help of respective Heads of departments as well as the Principal of the College.

(iii) Whereas the September House Test will carry weightage of 40 per cent, the December House Test will have weightage of 60 per cent in each subject/paper. The total weightage for both the Tests taken together shall be10 per cent of the total marks in each theory subject/paper. The weightage of 10 per cent marks shall be added to each paper of B.A./B.Sc./B.Com./B.C.A. I, II and III Year which will, henceforth, carry weightage of maximum marks allotted to each paper. A candidate will have to pass in theory and practical/s separately.

(iv) The record of marks secured by the students in the two House Tests will be sent by the respective Colleges so as to reach the office of Controller of Examinations latest by 15th March, failing which the result of the students shall be shown as 'RLA' and the entire responsibility for this would lie with the Principal/s of the College/s.

(v) The Colleges will continue to forward the internal assessment of the students for Practicals. Projects and similar other activities, wherever applicable, to the Controller of Examinations, as usual, so as to reach his office latest by 15th March.

SPECIAL NOTE :

(i) Each theory question paper will be set out of the marks allotted to each theory paper and 10% marks of the maximum marks of each paper will be internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

(iii) It will not be mandatory for the students to separately pass in the internal assessment.

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PANJAB UNIVERSITY, CHANDIGARH

OUTLINES OF TESTS, SYLLABI AND COURSES OF READING IN VARIOUS SUBJECTS FOR B.A. (GENERAL) AND B. SC. (GENERAL) SECOND YEAR EXAMINATION, 2015

ENGLISH (Compulsory)

(FOR B.A. CANDIDATES ONLY)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015
Outlines of tests, syllabi and courses of reading

Max. Marks		: 100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Objective:

The main aim of teaching English (Compulsory) to B.A. II students is to equip them to use language in an efficient manner. The idea is to empower an average student in such a way that learning English becomes a pleasurable endeavour, and ceases to be a nightmare that it is right now. It is with this aim in mind that the new text books have been chosen for all the three years of B.A. English (Compulsory) courses. The focus of the new text books is two-fold: to teach finer nuances of language through an integrated approach; and to teach composition and grammar through activity-based, not rule-based approach. For this reason, we have consciously chosen text books that de-emphasize literature and re-set our focus on the study of language. Given the way English language proficiency levels have touched a new low in our context, it was felt that such a corrective was not only much needed, but was almost inescapable.

Note:

(i) There will be **one paper** in all the three years of B.A. English (Compulsory).
(ii) The paper shall be divided into two sections, i.e. **Section A & B.** The distribution of marks in each section shall be indicated separately against the questions.
(iii) Section A shall deal with the text and Section B with composition and grammar (again based largely upon the text).
(iv) The question should be set strictly in accordance with the pattern of question paper outlined in the syllabus.
(v) For the successful implementation of this syllabus, it is necessary that we reflect objectives of this course, first in our teaching practices, and then in designing question paper/s and evaluating answer scripts of our students. The examination shall be held at the end of the term and/or year, as is recommended by the University, from time to time.

Text Prescribed:

English for Empowerment, Eds., G Damodar, D Venkateshwarlu, M Narendra, M Sarat Babu, G M Sunderavalli, Hyderabad: Orient Blackswan, 2009.

Poetry Section

1. "The Solitary Reaper" by William Wordsworth is replaced by "Ode to Autumn" by John Keats.
2. The Road Not Taken by Robert Frost.
3. "Money Madness" by D.H. Lawrence (from *Strings of Gold,* Editors-Sarjero B.Gare, Nivrutti B. Pawar & Ramesh M.Ingale, Orient Longman 2008) replaces *The Sun Rising* by John Donne.
4. I, Too by Langston Hughes.

5. Goodbye Part for Miss Pushpa T.S. by Nissi Ezekiel.
6. I will embrace only the Sun by Tripuraneni Srinivas.
7. Refuse Mother and Child by Chinua Acheke.
8. This is a Photograph of Me by Margaret Atwood.

Prose Section

No Change in the order in which essay are placed in this section, except essay No.7

7. India's Contribution to World Unity is replaced by Shashi Tharoor's "Freedom of the Press" (from *Strings of Gold*, ed. Sarjerao. B. Sagare, Nivrutti B. Pawar etc. Orient Longman, 2008)

Section-A (Poetry & Prose)

Q.1. The examiner will set fourteen short questions (to be answered each in not more than 60-80 words, in equal number from Poetry & Prose Sections of the prescribed text, out of which a student shall be required to attempt only ten, selecting, at least, five from each section.

(10×2=20 Marks)

Q.2. The examiner shall set eight questions (to be answered each in not more than 160-180 words), again in equal number from the Poetry & Prose Section of the text, out of which a student is required to attempt only five, selecting, at least, two from each section.

(5×4=20 Marks)

Section B (Composition & Grammar)

Q.3. This question shall have an internal choice between poetry and prose and be based on the **post-reading activities** suggested in the prescribed text. In this question, a student shall be asked to write a paragraph on an imaginative, expository or philosophical topic (in not more than 200-250 words). The topics for this purpose may be identified from the prescribed text itself. The examiner shall give three topics, and the student has to attempt one. While setting this question, special care may be taken to borrow ideas from the text or suitably modify them so as to identity viable and feasible topics that the students of B.A II can easily handle. The idea is to test a student's understanding of the text and/or general life-situations, and also devise an effective method of assessing their ability to express themselves in a simple, lucid and correct language. (15 Marks)

Q. 4. (a) A question on note-making based on the text. The examiner may give internal choice in this question. (5 Marks)

(b) A question on report-writing based on the text. The examiner may give internal choice in this question. (5 Marks)

Q.5. This question has to be based on various grammar exercises given in the prescribed text. The question should be set in such a way that most of the activities students have done in the class are adequately covered. (25 Marks)

(a) Appropriate form of words/pronouns/relative clauses etc., as given in the text. (5 Marks)

(b) Using the right form of verbs, phrasal verbs or using nouns as verbs or vice-versa. (5 Marks)

(c) Combining pairs of sentences using words given in the brackets (Simple/ Compound) (5 Marks)

(d) Choosing the correct antonyms/synonyms or picking odd words out. (5 Marks)

(e) Do as directed, which shall cover Active/Passive, Direct/Indirect etc. (5 Marks)

ENGLISH (Compulsory)

(FOR B.Sc. CANDIDATES ONLY)

B.Sc. (GENERAL) SECOND YEAR EXAMINATION, 2015

Outlines of tests, syllabi and courses of reading

Max. Marks	:	100
Theory	:	80 marks
Internal Assessment	:	10 marks
Practical	:	10 marks
Time	:	3 Hours

Objectives:

The objective of teaching English to the science students is to create general awareness among them about literature and its impact on their lives. At the same time, it is expected that the students, on reading this course, shall develop proficiency in reading and writing skills, while acquiring a sensitive and analytical attitude towards literature in particular, and life in general. It is with this aim in mind that the new text has been selected and it is hoped that the objectives of the course will not only be reflected but also realized through necessary shift in the teaching practices, design of the question paper and mode of evaluation.

Note:

(i) There will be one paper of 80 marks, 10marks are reserved for the Internal Assessment and 10 for the Practical Work. Total is 100.

(ii) The paper shall consist of Two Units. Unit I will be text specific and Unit II shall deal with different aspects of communications and language learning skills.

(iii) For Unit I, the prescribed text is **Varieties of Expression,** Ed. A. H. Tak, Foundation Books, which shall replace the existing text **Patterns in Prose** by Jagdish Chander, P.U., Chandigarh. It may be pointed out here that only certain sections of this text i.e **prose and drama** are prescribed. Poetry has been deleted completely. Only five prose and five plays have been recommended for the study. The relevant sections, however, are as follows:

Prose:

I. The Judgement Seat of Vikramaditya, *Sister Nivedita*
II. Engine Trouble, *R. K. Narayan*
III. The Conjurer's Revenge, *Stephen Leacock*
IV. J. C. Bose, *Aldous Huxley*
V. The Position of Women in Ancient India, *Padmini Sen Gupta*

Drama:

VI. *The Rising of the Moon*, Lady Gregory
VII. *Waterloo*, Arthur Conan Doyle
VIII. *The Proposal,* Anton Chekhov
IX. *Riders to the Sea,* J. M. Synge
X. *Lithuania,* Rupert Brooke

(iv) No text book is recommended for Unit II, but a few books that may be used for this Unit are listed towards the end Unit II shall consist of the following:

Communication: It shall focus on different aspects of communication, types of communication, and significance of positive attitude in improving communication.

Writing Skills: This section shall focus on précis-writing, letters of all kinds; curriculum vitae, short, formal reports (not exceeding 200 words); public notices and advertisements relating to product promotion etc.

Modern Forms of Communication: Here special emphasis shall be given to teaching the format of e-mails, fax messages, telegrams, audio-visual aids and power-point presentations. Apart from this, the students shall also be given basic lessons in effective listening, non-verbal communication, how to prepare for an interview and group discussion etc.

Practical work:-

Teacher should assign some project or practical work to the students. This should be in the nature of guided activity, which the students shall have to complete under the direct supervision of the teacher. The students may be given projects on a variety of subjects relating to their discipline i.e. science in general or a specific area of science they are specializing in. Preferably, they should be given minor projects (to be completed within less than two weeks, and length not exceeding 20 pages) in consultation with teachers of science. However, the evaluation of the projects should be done only by the Language Teachers, who must keep all the basic criteria of good writing in mind while doing so.

(Note*:* In case of private candidates and students of University School of Open Learning, the marks obtained by them out of 80 will be proportionately increased out of 100).

Testing Scheme:

The examination paper shall be divided into two sections, corresponding to two units already proposed in the

syllabus. The distribution of questions and marks in Section I shall be as follows:

Section I (It is text-based and corresponds to unit I in the syllabus)

Q.1. It shall consist of *five* short questions (not exceeding 100-120 words) out of which a student will be required to attempt any three. This question shall be based upon the prescribed text **Varieties of Expression** and cover a wide range of issues, topics and problems. It shall consist of **12 marks**.

Q.2. It shall consist of *two* long questions (not exceeding 300-350 words) out of which a student will be required to attempt only one. This question shall have internal choice, be based upon the prescribed text **Varieties of Expression.** This shall carry **10 marks.**

Note: The question 1 & 2 should be so designed as to cover all the chapters prescribed, as well as the major issues and problems listed therein.

Q.3. It shall consist of an **Unseen Passage for Comprehension** (not more than 800 words), with minimum six questions at the end. These questions should be designed in such a way that we are able to test a student's comprehension ability, language/presentation skills and vocabulary etc. This question shall be of **12 marks**.

Q.4. It shall exclusively be a test of vocabulary, but designed strictly on the lines of various exercises given at the end of each chapter in the prescribed text. The candidate shall be given six words in one column and asked to match them with words/meanings in the next column. This shall carry **6 marks.**

Section II (Based upon Unit II)

Q.5 (a) The students shall be asked to write a short survey report on a situation, incident, problem of science or the possibility of starting a new scientific venture (in about 150-200 words). The students shall be given an internal choice in this question. This question shall carry **8 marks.**

Q.5 (b) This question shall be on notices/advertisements of various types (as mentioned in the syllabus). It'll carry **4 marks.**

Q.6. This question shall test a student's ability to write letters of various kinds (in not more than 250 words). Again, there will be internal choice here and the question will be of **8 marks**

Q.7 This will test a student's ability to write a Précis, A passage of about 200 words shall be given and the students shall have to write a précis of about 70 words (including the title). This question shall carry **10 marks**.

Q.8 This question shall test a student's understanding of various aspects of communication and modern forms of communication. It shall be divided into two parts:

(a) Two short questions to be attempted (in not more than 100-120 words each) on different aspects of communication. It'll carry **6 marks.**

(b) Definitions/format of modern forms of communication to be tested. This shall again carry **4 marks.**

Suggested Reading:

1. *Business Communication,* Ed. Om. P. Juneja & Aarti Mujumdar, Hyderabad: Orient Blackswan, 2010.

2. *Textbook of Business Communication,* Anjali Kalkasr, R.B. Suryawanshi, Amlanjyoti Sengupta, Hyderabad: Orient Blackswan, 2010.

ਪੰਜਾਬੀ (ਲਾਜ਼ਮੀ)

(ਸਿਰਫ ਬੀ.ਏ. ਦੇ ਵਿਦਿਆਰਥੀਆਂ ਲਈ)

ਬੀ.ਏ. (ਜਨਰਲ) ਭਾਗ ਦੂਜਾ 2●15 ਦੇ ਇਮਤਿਹਾਨ ਲਈ)

ਪੇਪਰ: ਏ ਕੁੱਲ ਅੰਕ: 5●

ਲਿਖਤੀ: 45

ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ: 5

ਸਮਾਂ: 3 ਘੰਟੇ

ਪਾਠਕ੍ਰਮ

1. ਕਵਿਤਾ ਦਾ ਅਧਿਐਨ — 1● ਅੰਕ
2. ਜੀਵਨੀ ਦਾ ਅਧਿਐਨ — 1● ਅੰਕ
3. ਇਕਾਂਗੀਆਂ ਦਾ ਅਧਿਐਨ — 1● ਅੰਕ
4. ਉਪਰੋਕਤ ਤਿੰਨਾਂ ਪੁਸਤਕਾਂ ਤੇ ਆਧਾਰਤ ਲਘੂ ਉੱਤਰਾਂ ਵਾਲੇ ਪ੍ਰਸ਼ਨ — 15 ਅੰਕ

ਕੋਰਸ

1. **ਮੱਧਕਾਲੀ ਬ੍ਰਿਤਾਂਤ ਕਾਵਿ** (ਸੰਪਾ.), ਪ੍ਰੋ. ਬੀ. ਐਸ. ਘੁੰਮਣ, ਪਬਲੀਕੇਸ਼ਨ ਬਿਊਰੋ, ਪੰਜਾਬ ਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ।

   ਜਰੂਰੀ ਨੋਟ: –

   i. 'ਦਮੋਦਰ' ਦੀ ਰਚਨਾ ਵਿਚੋਂ "**ਸਹਿਤੀ ਦੀ ਗਿਰਾਂਇ ਸੁਹਾਵਾ**"

   ii. ਵਾਰਿਸ ਸ਼ਾਹ ਦੀ ਰਚਨਾ ਵਿਚੋਂ ਲੈ ਕੇ ਸੱਠ ਸਹੇਲੀਆਂ ਹੀਰ ਚੱਲੀ **"ਤੇਰੇ ਬਿਰਹਾ ਫਿਰਾਕ ਨੇ ਕੁੱਠੀਆਂ ਮੈਂ**"

   iii. ਹਾਸ਼ਮ ਸ਼ਾਹ ਦੀ ਰਚਨਾ ਵਿੱਚੋਂ **"ਕੁਟਣੀ ਧ੍ਰੋਹ ਕਮਾਇਆ**"

   iv. ਕਾਦਰਯਾਰ ਦੀ ਰਚਨਾ ਵਿੱਚੋਂ "**ਸੂਰਤ ਦੇਖ ਕੇ ਉਸ ਦੀ ਸਿੱਕ ਧਾਣੀ**"

   ਅਤੇ

   ਭਗਵਾਨ ਸਿੰਘ ਦੀ ਕੋਈ ਵੀ ਕਵਿਤਾ ਸਿਲੇਬਸ ਵਿਚ ਸ਼ਾਮਿਲ ਨਹੀਂ ਹੈ।

2. **ਸ਼ਹੀਦ ਭਗਤ ਸਿੰਘ**, (ਲੇਖਕ) ਸੁਹਿੰਦਰਬੀਰ ਸਿੰਘ, ਕਸਤੂਰੀ ਲਾਲ ਐਂਡ ਸੰਨਜ਼, ਅੰਮ੍ਰਿਤਸਰ
3. **ਛੇ ਛੱਲਾਂ** (ਸੰਪਾ.) ਡਾ. ਆਤਮਜੀਤ ਸਿੰਘ, ਪਬਲੀਕੇਸ਼ਨ ਬਿਊਰੋ, ਪੰਜਾਬ ਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ।

   (1), (2) ਅਤੇ (3) ਉੱਤੇ ਆਧਾਰਤ ਲਘੂ ਪ੍ਰਸ਼ਨ

ਯੂਨਿਟ ਅਤੇ ਥੀਮ

1. ਮੱਧਕਾਲੀ **ਬ੍ਰਿਤਾਂਤ** ਕਾਵਿ ਵਿਚੋਂ ਪ੍ਰਸੰਗ ਸਹਿਤ ਵਿਆਖਿਆ (ਦੋ ਵਿਚੋਂ ਇਕ) — 5 ਅੰਕ
2. ਮੱਧਕਾਲੀ **ਬ੍ਰਿਤਾਂਤ** ਕਾਵਿ ਵਿਚੋਂ ਕਿਸੇ ਇਕ ਕਵਿਤਾ ਦਾ ਸਾਰ (ਦੋ ਵਿਚੋਂ ਇਕ) — 5 ਅੰਕ
3. ਜੀਵਨੀ ਦੇ ਵਿਸ਼ੇ, ਕਿਸੇ ਕਾਂਡ ਦਾ ਸਾਰ ਲਿਖਣਾ (ਤਿੰਨਾਂ ਵਿਚੋਂ ਇਕ) — 1● ਅੰਕ
4. ਕਿਸੇ ਇਕ ਇਕਾਂਗੀ ਦਾ ਸਾਰ / ਵਿਸ਼ਾ ਜਾਂ ਪਾਤਰ (ਦੋ ਵਿਚੋਂ ਇਕ) — 1● ਅੰਕ
5. ਨਿਰਧਾਰਤ ਮੱਧਕਾਲੀ ਬ੍ਰਿਤਾਂਤ ਕਾਵਿ ਵਿਚੋਂ ਲਘੂ ਉੱਤਰਾਂ ਵਾਲੇ ਪ੍ਰਸ਼ਨ (ਅੱਠਾਂ ਵਿਚੋਂ ਪੰਜ) — 5 ਅੰਕ
6. ਪਾਠ-ਕ੍ਰਮ ਵਿਚਲੀ ਜੀਵਨੀ ਸੰਬੰਧੀ ਲਘੂ ਉੱਤਰਾਂ ਵਾਲੇ ਪ੍ਰਸ਼ਨ (ਅੱਠਾਂ ਵਿਚੋਂ ਪੰਜ) — 5 ਅੰਕ
7. ਇਕਾਂਗੀ-ਸੰਗ੍ਰਹਿ ਵਿਚੋਂ ਲਘੂ ਉੱਤਰਾਂ ਵਾਲੇ ਪ੍ਰਸ਼ਨ (ਅੱਠਾਂ ਵਿਚੋਂ ਪੰਜ) — 5 ਅੰਕ

ਨੋਟ: ਪੇਪਰ ਸੈਟਰ ਲਈ : ਰਚਨਾ ਦੇ ਵਿਸ਼ੇ, ਰਚਨਾ ਦੀ ਵਿਚਾਰਧਾਰਾ, ਪਾਤਰ ਅਤੇ ਘਟਨਾਵਾਂ ਜਾਂ ਤੱਥਾਂ ਨਾਲ ਸੰਬੰਧਿਤ ਸਵਾਲ ਪੁੱਛੇ ਜਾਣ। ਉੱਤਰ ਇਕ ਤੋਂ ਤਿੰਨ ਸਤਰਾਂ ਤਕ ਸੀਮਿਤ ਹੋਵੇ।

ਪੇਪਰ: ਬੀ

ਕੁੱਲ ਅੰਕ: 5●
ਲਿਖਤੀ: 45
ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ: 5
ਸਮਾਂ: 3 ਘੰਟੇ

ਪਾਠਕ੍ਰਮ

1.	ਪਾਠਕ੍ਰਮ ਵਿਚਲੇ ਲੇਖਕਾਂ ਦਾ ਜੀਵਨ, ਰਚਨਾ ਤੇ ਯੋਗਦਾਨ	5 ਅੰਕ
2.	ਚਿੱਠੀ-ਪੱਤਰ (ਸਰਕਾਰੀ, ਵਪਾਰਕ ਤੇ ਸੰਪਾਦਕ ਨੂੰ)	1● ਅੰਕ
3.	ਪੈਰ੍ਹਾ ਪੜ੍ਹ ਕੇ ਪ੍ਰਸ਼ਨਾਂ ਦੇ ਉੱਤਰ ਦੇਣੇ	1● ਅੰਕ
4.	ਪੰਜਾਬੀ ਭਾਸ਼ਾ: ਜਨਮ ਤੇ ਵਿਕਾਸ ਅਤੇ ਪੰਜਾਬੀ ਦੀਆਂ ਉਪਭਾਸ਼ਾਵਾਂ	8 ਅੰਕ
5.	ਵਿਆਕਰਣ	12 ਅੰਕ

ਯੂਨਿਟ ਅਤੇ ਥੀਮ

1.	ਕਿਸੇ ਇਕ ਲੇਖਕ ਦਾ ਜੀਵਨ, ਰਚਨਾ ਅਤੇ ਸਾਹਿਤਕ ਯੋਗਦਾਨ: ਨਿਰਧਾਰਤ ਲੇਖਕਾਂ ਵਿਚੋਂ ਦੋ ਕਵੀ ਅਤੇ ਦੋ ਇਕਾਂਗੀਕਾਰ (ਚਾਰ ਵਿਚੋਂ ਇਕ ਕਰਨਾ ਹੈ) **ਲੇਖਕ**: ਆਈ. ਸੀ. ਨੰਦਾ, ਹਰਚਰਨ ਸਿੰਘ, ਬਲਵੰਤ ਗਾਰਗੀ, ਹਰਸਰਨ ਸਿੰਘ, ਅਜਮੇਰ ਸਿੰਘ ਔਲਖ, ਆਤਮਜੀਤ, **ਕਵੀ**: ਗੁਰੂ ਗੋਬਿੰਦ ਸਿੰਘ, ਦਮੋਦਰ, ਵਾਰਿਸ ਸ਼ਾਹ, ਕਾਦਰਯਾਰ, ਨਜਾਬਤ, ਸ਼ਾਹ ਮੁਹੰਮਦ।	5 ਅੰਕ
2.	ਚਿੱਠੀ-ਪੱਤਰ (ਦੋ ਵਿਚੋਂ ਇਕ)	1● ਅੰਕ
3.	ਪੈਰ੍ਹਾ ਪੜ੍ਹ ਕੇ ਪ੍ਰਸ਼ਨਾਂ ਦੇ ਉੱਤਰ ਦੇਣੇ (ੳ) ਪੈਰ੍ਹੇ ਦਾ ਸਿਰਲੇਖ (ਅ) ਤਕਰੀਬਨ ਚਾਰ ਉਲੀਕੇ ਸ਼ਬਦਾਂ ਦੇ ਅਰਥ (ੲ) ਨਾਂਵ, ਕਿਰਿਆ, ਵਿਸ਼ੇਸ਼ਣ ਅਤੇ ਪੈਰ੍ਹੇ ਨਾਲ ਸੰਬੰਧਤ ਦੋ ਹੋਰ ਪ੍ਰਸ਼ਨ ਪੁੱਛੇ ਜਾਣਗੇ।	1● ਅੰਕ
4.	ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਦੇ ਜਨਮ, ਨਾਮਕਰਣ, ਵਿਕਾਸ ਪੜਾਅ, ਪੰਜਾਬੀ ਉਪਭਾਸ਼ਾਵਾਂ ਅਤੇ ਪੰਜਾਬੀ ਭਾਸ਼ਾ 'ਤੇ ਹੋਰਨਾਂ ਭਾਸ਼ਾਵਾਂ ਦੇ ਪ੍ਰਭਾਵ ਬਾਰੇ ਪ੍ਰਸ਼ਨ (ਚਾਰ ਵਿਚੋਂ ਦੋ)	4+ 4= 8 ਅੰਕ
5.	ਸਾਧਾਰਨ, ਸੰਯੁਕਤ, ਮਿਸ਼ਰਤ ਵਾਕਾਂ ਦਾ ਆਪਸੀ ਵਟਾਂਦਰਾ	4 ਅੰਕ
6.	ਬਹੁਤੇ ਸ਼ਬਦਾਂ ਦੀ ਥਾਂ ਇਕ ਸ਼ਬਦ	4 ਅੰਕ
7.	ਅਗੇਤਰ/ਪਿਛੇਤਰ	4 ਅੰਕ

ਸਹਾਇਕ ਪੁਸਤਕਾਂ

1. ਪੰਜਾਬੀ ਸੰਚਾਰ ਯੋਗਤਾ ਅਭਿਆਸ, ਪੰਜਾਬ ਸਟੇਟ ਯੂਨੀਵਰਸਿਟੀ ਟੈਕਸਟ ਬੁੱਕ ਬੋਰਡ, ਚੰਡੀਗੜ੍ਹ।
2. ਪੰਜਾਬੀ ਭਾਸ਼ਾ ਦਾ ਇਤਿਹਾਸ, ਪ੍ਰੇਮ ਪ੍ਰਕਾਸ਼ ਸਿੰਘ, ਪੈਪਸੂ ਬੁੱਕ ਡਿਪੂ, ਪਟਿਆਲਾ।
3. ਕਾਲਜ ਪੰਜਾਬੀ ਵਿਆਕਰਣ, ਡਾ. ਹਰਕੀਰਤ ਸਿੰਘ ਤੇ ਗਿਆਨੀ ਲਾਲ ਸਿੰਘ, ਪੰਜਾਬ ਸਟੇਟ ਯੂਨੀਵਰਸਿਟੀ ਟੈਕਸਟ ਬੁੱਕ ਬੋਰਡ, ਚੰਡੀਗੜ੍ਹ।

ਨੋਟ: ਹਫਤੇ ਦੇ 6+ 3 = 9 ਪੀਰੀਅਡ

HISTORY AND CULTURE OF PUNJAB

(For B.A. candidates only)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

General Instructions :

1. In all, ***nine*** questions will be set. Each question will carry 18 marks.
2. First question shall be Short Answer type containing 15 short questions spread over the whole syllabus. Candidates will attempt any ***nine*** out of the fifteen questions in about 25 to 30 words each. Each short question will carry 2 marks, totalling $9 \times 2 = 18$ marks. The first question is **compulsory.**
3. Rest of the paper shall contain 4 units. Each unit shall have two essay type questions and the candidates will attempt one question from each unit – 4 in all.
4. For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
   The paper setter must put note (4) in the question paper.

HISTORY AND CULTURE OF PUNJAB 1200-1849 A.D.

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Objectives : To introduce the students to the history of Punjab and its Culture in medieval times.

Pedagogy : Lectures, library work and discussions.

Unit-I

1. Society and Culture in Punjab during the Turko-Afghan rule.
2. The Punjab under the Great Mughals.
3. Salient features of the Bhakti movement and Sufism in Punjab.

Unit-II

4. Guru Nanak : His teachings, concept of Langar and Sangat.
5. Development of Sikhism (1539-1581): Contribution of Guru Angad Dev, Guru Amar Das and Guru Ram Das for the development of Sikhism.
6. Transformation of Sikhism : Compilation of Adi-Granth; Martyrdom of Guru Arjan Dev; Guru Hargobind's New Policy.

Unit-III

7. Martyrdom of Guru Tegh Bahadur; the foundation of the Khalsa by Guru Gobind Singh.
8. Banda Bahadur and his achievements; Sikh Struggle for sovereignty from 1716 to 1765 and the role of Dal Khalsa, Rakhi, Gurmata and Misls.
9. Ranjit Singh's rise to power, civil and military administration, relations with the British.

Unit-IV

10. Annexation of the Punjab with special reference to the causes and consequences of the Anglo-Sikh wars.
11. New developments in language, literature, education, art and architecture in the Punjab during the Medieval period.
12. Social life with special reference to the position of women, fairs, festivals, folk music, dance and games in the Punjab.

Suggested Readings :

1. Singh, Kirpal : *History and Culture of the Punjab,* Part II (Medieval Period), Publication Bureau, Punjabi University, Patiala, 1990 (3rd edn.).

N.B. : The required detail and depth would conform to the treatment of the subject in the above survey. (This book will also form the basis of the short answer questions).

2. Singh, Fauja (ed.) : *History of the Punjab*, Vol. III, Punjabi University, Patiala, 1972.
3. Grewal, J.S. : *The Sikhs of the Punjab,* The New Cambridge History of India, Orient Longman, Hyderabad, 1990.
4. Singh, Khushwant : *A History of the Sikhs*, Vol. I : 1469-1839, Oxford University Press, Delhi, 1991.
5. Chopra, P.N., Puri, B.N., and Das, M.N. : *A Social, Cultural & Economic History of India*, Vol. II, Macmillan, Delhi, 1974.
6. Hussain, Yusuf : *Glimpses of Medieval Indian Culture,* Asia Publishing House, Bombay, 1973 (rep.).

ENGLISH (Elective)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Outlines of Tests, Syllabi and Courses of Reading

Objective:

The main aim of teaching English (Elective) to B.A. II students is to enable them to approach a wide variety of literary texts and genres with critically sensitive and analytical understanding. The idea is to introduce the students to the basic concepts of literature and also empower them to read, analyze and write about a poem, prose, essay or drama in an independent manner. It is with this modest aim in mind that the new text books have been selected for all the three years of B.A English (Elective) courses. The focus of the new text books is two-fold: to teach finer nuances of literature and language through an integrated approach; and to help and motivate the students to develop basic tools of analyzing a variety of literary texts/genres.

Note:

(i) There will be two papers in all the three years of B.A. English (Elective) courses. Rather than divide the papers vertically, there will be horizontal division in terms of texts, composition and grammar. As two books are prescribed for each course, these two books shall be distributed across two papers and shall not be included in the same paper. So, **An Anthology of English Verse** shall figure in **Paper A** and **Girish Karnad's *Hayavadana*** shall be included in **Paper B.** Questions on composition and grammar shall also be evenly distributed in both the papers. This is being done to help the students, who shall now find both the papers proportionately divided and so easily manageable.

(ii) Each paper shall be of 90 marks, and 10 marks in each paper shall be reserved for Internal Assessment. Each paper shall be further sub-divided into Sections, I & II. The mode of distribution of marks shall, however, vary from paper to paper. There shall be six questions in each paper. All the questions will be compulsory. Though internal choice may be offered in some of the questions, there will be no external choice.

(iii) Each paper shall include minimum 15 literary terms/concepts appropriate to the particular literary genre included in that paper.

(iv) The questions should be set strictly in accordance with the pattern of question paper outlined in the syllabus.

(v) For the successful implementation of this syllabus, it is necessary that we reflect objectives of this course, first in our teaching practices, and then in designing question paper/s and evaluating answer scripts of our students. The examination shall be held at the end of the term and/or year, as is recommended by the University, from time to time.

Texts Prescribed:

1. **An Anthology of English Verse, Ed.** Dept. of English, Deen Dayal Upadhayaya, Gorakhpur University, New Delhi, OUP, 2004 (Only 15 poems are prescribed)

There is no change in the prescribed text books. Only a few poems (from the prescribed text *An Anthology of English Verse* ed, Dept. of English & Modern European Language, Deen Dayal Upadhayaya University, Gorakhpur, OUP 2004), which have already been included at some earlier stage are being

replaced or changed. 15 prescribed poems shall now be as under:

Texts Prescribed:

1. John Donne: *The Sun Rising*
2. Alexander Pope: *From Essay on Man*
3. Thomas Gray: *Elegy Written in the Country Churchyard*
4. William Blake: *The Tiger*
5. William Wordsworth: *Tintern Abbey*
6. John Keats: *Ode to a Nightingale*
7. Tennyson: Ulysses
8. Browning: *My Last Duchess*
9. Matthew Arnold: *Dover Beach* is replaced by his *To Marguerite* (included in the same anthology)
10. Hookins: *Pied Beauty*
11. Wilfred Owen: *Anthem for the Doomed Youth* is replaced by A.K. Ramanujan's *History* (Included in the same anthology)
12. Yeats: *A Prayer for My Daughter*
13. T.S. Eliot: *Journey of the Magi*
14. W. H. Auden: *The Unknown Citizen* is replaced by Thomas Hardy's *The Darkling Thrush*
15. Philip Larkin: *The Trees*

2. **Girish Karnad: *Hayavadana***, in **Three Plays,** New Delhi: Oxford University Press, 2005.

Paper A:

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Section I

{Literary Terms/Concepts: *Allegory, Allusion, Antithesis, Epic, Epithet, Hyperbole, Internal Rhyme, Rhyme Royal, Terza Rima, Metaphor, Metonymy, Medias Res, Oxymoron, Mood, Tone, Personification, Stanza, Spensarian Stanza, Satire, Verse Paragraph*}

Q.1. It shall be on literary terms/concepts. Eight terms shall be given in all, and the students will be required to do *five.* (15 Marks)

Q.2. The examiner will set seven short text-based questions (each to be answered in not more than 40-60 words) from the prescribed **Anthology,** out of which a student shall be required to attempt only *five*. (15 marks)

Q.3. Long questions *five* out of *seven,* again based on the **Anthology** (and each of these is to be answered in about 100-120 words) (15 marks)

Section II

Q.4. Dialogue Writing: The examiner will give one descriptive passage of about 500 words, and ask the students to re-write the same in dialogue form. However, the examiner must ensure that he/she gives such a passage as can easily be converted into dialogue form. (10 marks)

Q.5. Précis Writing with a suitable title: (A passage of about 240-250 words) (10 marks)

Q.6. Do as directed:

(a) Correction of different types of sentences (5 marks)

(b) Choose the correct meaning of the word (5 marks)

(c) Completing the incomplete sentences (5 marks)

(d) One word substitution (5 marks)

(e) Use of Idioms/Phrases in sentences (5 marks)

Paper B

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Section I

(*Literary terms/concepts: Drama, Myth of Dionysus, Liturgical Drama, Miracle Plays, Mystery Cycles. Interludes, Mimesis, Catharsis, Tragic Hero, Organic Unity, Revenge Tragedy, Poetic Drama, Verse Drama, Myth & Drama, Ritual & Drama, Yaksgana, Rasa Theory, Indian Theory of Drama, Puppetry, Ardhnarishwra)*

Q.1. Literary terms/concepts, any *Five out of Eight* (15 marks)

Q.2. Short text-based questions, *five out of seven*, on **Girish Karnad's *Hayavadana's*** play (each of the questions to be answered in about 50-60 words) (15 marks)

Q.3. Long questions *two out of three*, again based on **Girish Karnad's *Hayavadana*** dealing with the theme, incident, character(s) or use of symbols etc. (each to be answered in about 180-200 words) (15 marks)

Section II

Q.4 Write an essay (of about 500 words) on any topic of national interest. The students shall be given three topics, out of which they have to choose *one.* (20 marks)

Q.5 Letter Writing (Letters to the Editors or Commercial/Business Letters) (15 marks)

Q.6 Translation from Vernacular into English (a paragraph of about 250 words)

Or

A paragraph on any *one* of the *two* given topics (for foreign students only) (10 marks)

Note: Allotment of periods:
Texts 6 per week in a class of 60 students
Composition & Grammar: 3 per week in a class of 30 students

हिन्दी (ऐच्छिक)

बी. ए. (सामान्य) द्वितीय वर्ष परीक्षा, 2015

पत्र–एक **पूर्णांक : 90+10 = 100**

समय : 3 घण्टे

1. **तरंगिणी**–मनोहर लाल आनंद, पंजाब यूनिवर्सिटी, पब्लिकेशन ब्यरो, चण्डीगढ. अंक : 22

निम्नलिखित कवि पाठ्यक्रम में निर्धारित है :–

मैथिलीशरण गुप्त, जयशंकर प्रसाद, सूर्यकान्त त्रिपाठी 'निराला', सुमित्रानन्दन पंत, महादेवी वर्मा, अज्ञेय, धर्मवीर भारती।

(क) 6–6 अंकों की दो संदर्भ–सहित व्याख्याएँ करनी होंगी। कुल चार व्याख्याएँ पूछी जायेंगी।

(ख) कवि–परिचय, सार और उद्देश्य संबंधी कुल दो प्रश्न पूछे जायेंगे। प्रत्येक उत्तर की शब्द सीमा 200 होगी। 10 अंकों का एक ही प्रश्न करना होगा।

2. **एक सत्य हरिश्चन्द्र** – डॉ. लक्ष्मी नारायण लाल, प्रकाशक, राजपाल एण्ड संज, नई दिल्ली। अंक : 16

(क) संदर्भ सहित व्याख्या के लिए दो प्रश्न पूछे जायेंगे, उत्तर एक का ही देना होगा। इसके लिए 6 अंक निर्धारित किये गये हैं।

(ख) पात्रों के चरित्र–चित्रण, तत्वों के आधार पर नाटक की समीक्षा तथा समस्या संबंधी दो प्रश्न पूछे जायेंगे। उत्तर एक का ही देना होगा। इसके लिए 10 अंक निर्धारित हैं।

3. इस खंड में तीन–तीन अंकों के चार लघु–उत्तरापेक्षी प्रश्न करने होंगे। कुल आठ प्रश्न पूछे जायेंगे। प्रत्येक उत्तर की शब्द सीमा 50 होगी। ये प्रश्न इस पत्र के पूर्वोक्त दो खंडों (तरंगिणी व एक सत्य हरिश्चन्द्र') पर आधारित होंगे। अंक : 12

4. **हिन्दी साहित्य का इतिहास** : रीतिकाल और आधुनिक काल (केवल काव्य–खंड) अंक : 20

(क) रीतिकाल की परिस्थितियाँ, नामकरण, सीमा–निर्धारण, प्रवृतियाँ, रीतिबद्ध और रीतिमुक्त काव्य की प्रमुख विशेषताएँ, रीतिकाव्य के प्रमुख कवि – केशव, बिहारी, घनानंद के संबंध में 10 अंकों का एक प्रश्न करना होगा। कुल दो प्रश्न पूछे जायेंगे।

(ख) आधुनिक काल के केवल कविता–इतिहास खंड में से भारतेन्दु युग, द्विवेदी युग, छायावाद, प्रगतिवाद, प्रयोगवाद और नई कविता की केवल प्रमुख प्रवृतियों से संबंधित 10 अंकों का एक प्रश्न करना होगा। कुल दो प्रश्न पूछे जायेंगे।

5. **वस्तुनिष्ठ प्रश्न** : अंक : 20

इस पत्र के पूर्वोक्त चारों खंडों के आधार पर 2–2 अंकों के दस वस्तुनिष्ठ प्रश्नों के उत्तर देने होंगे। कुल पन्द्रह प्रश्न पूछे जायेंगे।

आंतरिक मूल्यांकन : अंक : 10

पत्र–दो **पूर्णांक : 90+10 = 100**

समय : 3 घण्टे

नोट : पत्र दो में भाषा–शिक्षण का व्यावहारिक प्रयोग होगा तथा छात्रों पर व्यक्तिगत रूप से ध्यान देना अनिवार्य होगा। अतः कक्षा–अनुभाग की छात्र संख्या 30–40 के बीच सीमित होगी।

1. **आदर्श एकांकी संग्रह,** सं0 डॉ. संसार चन्द, पंजाब यूनिवर्सिटी पब्लिकेशन ब्यूरो, चण्डीगढ़, द्वारा प्रकाशित। अंक : 10

   (क) एकांकी के तत्त्वों के आधार पर समीक्षात्मक प्रश्न (सार–लेखन, चरित्र–चित्रण, उद्देश्य संबंधी)। 10–10 अंकों के दो प्रश्न पूछे जायेंगे। एक का उत्तर देना होगा।

   (ख) संदर्भ सहित व्याख्या नहीं पूछी जायेगी।

2. **संलग्न 50 टिप्पणियों का अभ्यास** : (पूछी गई दस टिप्पणियों में से पाँच का उत्तर देना होगा)। अंक : 10

3. **समीक्षा सिद्धान्त :** केवल नाटक और एकांकी अंक : 16

   परिभाषा, तत्त्व और वर्गीकरण पर आधारित, नाटक तथा एकांकी दोनों विधाओं से संबंधित दो–दो प्रश्न पूछे जायेंगे, जिनमें से 8–8 अंकों का एक–एक प्रश्न करना होगा।

4. **व्यावहारिक व्याकरण** अंक : 20

   (क) समाकृति, भिन्नार्थक शब्द–युग्म अंक : 4

   (ख) स्वर–संधि एवं व्यंजन–संधि अंक : 4

   (ग) संधि–विच्छेद (केवल व्यावहारिक) अंक : 4

   (घ) वाक्य–शोधन अंक : 4

   (ड) विराम–चिह्न अंक : 4

5. **सार–लेखन** अंक : 8

6. **शासकीय पत्र–लेखन** अंक : 8

7. **विस्तारण** अंक : 8

8. **तकनीकी शब्दावली (केवल प्रशासकीय शब्दावली)** अंक : 10

नोट : 15 प्रश्न पूछे जायेंगे, जिनमें से 10 का उत्तर देना होगा।

आंतरिक मूल्यांकन : अंक : 10

प्रशासनिक शब्दावली

1.	Acceptance	स्वीकृति
2.	Account	लेखा, खाता, हिसाब
3.	Accuse	अभियोग लगाना
4.	Acknowledgement (fact etc.)	अभिस्वीकार करना, मानना, कबूल करना
5.	Acknowledgement due	रसीद, पावती–सहित
6.	Addressee	पानेवाला
7.	Adjournment	स्थगन
8.	Administration	प्रशासन
9.	Admissible	ग्राह्य, स्वीकार्य
10.	Affidavit	शपथ लेना, हलफनामा
11.	Agent	अभिकर्ता, एजेन्ट
12.	Agitation	आन्दोलन
13.	Agreement	करार, अनुबंध सहमति
14.	Allowance	भत्ता
15.	Amenity	सुख–सुविधा
16.	Anti-Corruption Officer	भ्रष्टाचार निरोध अधिकारी
17.	Appeal	अपील, अपील करना
18.	Appointment	नयुक्ति
19.	Appoint	नियुक्ति करना
20.	Approval	अनुमोदन
21.	Article	अनुच्छेद, नियम, वस्तु
22.	Association	संघ, समाज, संगम
23.	At par	सममूल्य पर
24.	Attendant	परिचर
25.	Attestation	साक्ष्यांकन, अनुप्रमाणन
26.	Audit	लेखा–परीक्षा
27.	Ballot Paper	मतपत्र, मतपर्ची
28.	Ban	प्रतिबंध, रोक, पाबंदी
29.	Bonafide	सद्भावी, वास्तविक, असली

30.	Book-fair	पुस्तक मेला
31.	Bureau	कार्यालय, ब्यूरो
32.	Cabinet	मंत्रिमंडल
33.	Candidate	अभ्यर्थी, प्रार्थी, उम्मीदवार
34.	Care-taker	रखवाला, अवधायक
35.	Cashier	रोकड़िया
36.	Censure	निंदा प्रस्ताव, परिनिंदा
37.	Certificate of Medical Fitness	आरोग्य प्रमाण–पत्र
38.	Character Certificate	चरित्र प्रमाण–पत्र
39.	Charge Sheet	आरोप पत्र, फर्द, जुर्म
40.	Circular	परिपत्र, गशती–चिट्ठी
41.	Circulation of Traffic	यातायात परिचालन
42.	Circus	क्रीड़ा रंग, रंगमंडप
43.	Citation	प्रशस्ति, उद्धरण, अनुलेखन
44.	City Booking Office	नगर टिकटघर, नगर बुकिंग–कार्यालय
45.	City Compensatory Allowance	नगर भत्ता
46.	Civil	नागरिक
47.	Civil poll	नगर निर्वाचन
48.	Civil Sense	नागरिक भावना
49.	Civil-air-craft	असैनिक वायुयान
50.	Claimant	दावेदार, दावी
51.	Collector	समाहर्ता, संग्रहक, संकलनकर्ता, कलेक्टर
52.	Colony	बस्ती, उपनिदेश
53.	Colossal	विशाल, बृहत्काय
54.	Colour Blindness	वर्णान्धता
55.	Communique	विज्ञप्ति
56.	Complaint	शिकायत
57.	Complementary	पूरक
58.	Completion Report	समापन रिपोर्ट
59.	Certified Copy	प्रमाणित प्रतिलिपि

60.	Complementary Copy	उपहार प्रति, मानार्थ प्रति
61.	Composite	मिला–जुला, सम्मिश्रण, संश्लिष्ट
62.	Comprehension	व्यापक कार्य बोध
63.	Contingencies	आकस्मिक व्यय
64.	Contractor	ठेकेदार, संविदाकार
65.	Confiscate	जब्त करना, अधिहरण करना
66.	Corporation	निगम
67.	Custody	अभिरक्षा
68.	Decorum	शिष्टता, शालीनता
69.	Defacto	वस्तुतः
70.	Defaulter	त्रुटिकारी, चूक करनेवाला, बाकीदार
71.	Defendant	प्रतिवादी
72.	Depreciation Charge	मूल्याह्रास प्रभार, अवक्षय प्रभार
73.	Design	अभिकल्प, अभिकल्पना, रूपांकन, डिजाइन
74.	Dispatch	प्रेषण, रवानगी, प्रेषित करना, रवाना करना
75.	Dignitory	उच्च पदधारी, महिमान्वित
76.	Director	निदेशक
77.	Disbursement	संवितरण
78.	Discipline	अनुशासन
79.	Discrepancy	विसंगति, गलती
80.	Discretion	विवेक, विवेकाधिकार
81.	Dissent	विसम्मति, असहमति
82.	Disqualified	अयोग्य, अनर्ह
83.	Division	विभाजन, मंडल, श्रेणी, प्रभाग, डिवीजन
84.	Eligible	पात्रता प्राप्त, पात्र
85.	Emergency	आपात्, आपात्–स्थिति, आपातिक
86.	Emigrant	उत्प्रवासी
87.	Employment	रोजगार, नौकरी, नियोजन
88.	Employment Exchange	रोजगार कार्यालय, रोजगार दफतर
89.	Employment Officer	रोजगार अधिकारी

90.	Enquiry	पूछताछ, जाँच
91.	Enrolment Number	नामांकन संख्या
92.	Enrolled	नामांकित
93.	Evacuee	निष्क्रांत
94.	Executive Engineer	कार्यपालक अभियंता, कार्यपाल इंजीनियर
95.	Faculty	संकाय
96.	Finance	वित्त, रूपया लगाना
97.	Gazetted Officer	राजपत्रित अधिकारी
98.	Gazetteer	राजविवरणिका
99.	Gazetted post	राजपत्रित पद
100.	Grant	अनुदान, स्वीकार करना
101.	Incentive	प्रोत्साहन
102.	Index	सूचक अनुक्रमणी
103.	Initials	आद्यक्षर
104.	Insured Letter	बीमा किया हुआ पत्र
105.	Interim	अन्तरिम
106.	Intelligence	गुप्तवार्ता, आसूचना
107.	Intelligentia	प्रबुद्ध वर्ग, बुद्धिजीवी वर्ग
108.	Intensive reading	गहन पठन
109.	Intention	अभिप्राय, आशय
110.	Judgment	निर्णय
111.	Jurisdiction	अधिकार–क्षेत्र
112.	Ledger	खाता
113.	Lessee	पट्टेदार
114.	Liaison Officer	संपर्क अधिकारी
115.	Maintenance	अनुरक्षण, भरण–पोषण, रख–रखाव
116.	Major	व्यस्क, बालिग, मेजर, प्राप्तव्य
117.	Manager	प्रबंधक, व्यवस्थापक, मैनेजर
118.	Minor	अवयस्क, नाबालिग, अप्राप्तव्य
119.	Monopoly	एकाधिकारी, एकाधिपत्य

120.	Motion	प्रस्ताव
121.	Nationalization	राष्ट्रीयकरण
122.	Negotiation	संधिवार्ता, समझौते की बातचीत
123.	Note of Dissent	विसम्मति लेख, असहमति लेख
124.	Notification	अधिसूचना
125.	Oath Commissioner	शपथ आयुक्त
126.	Offender	अपराधी
127.	Permissible	अनुमेय, अनुज्ञेय, क्षम्य
128.	Planning Commission	योजना आयोग
129.	Precedence	पूर्वता, अग्रता
130.	Procedure	कार्यविधि
131.	Public	सार्वजनिक, आम, सरकारी, लोक
132.	Quorum	गणपूर्ति, कोरम
133.	Receipt	पावती, प्राप्ति रसीद
134.	Recruitment	भर्ती
135.	Receiver	पानेवाला
136.	Reminder	स्मरण–पत्र, अनुस्मारक
137.	Representative	प्रतिनिधि
138.	Senior	वरिष्ट, ज्येष्ठ
139.	Sine die	अनिश्चित काल के लिए
140.	Statutory	कानूनी, विधिक, सविधिक
141.	Stenographer	आशुलिपिक
142.	Subordinate	अधीन, अधीनस्थ
143.	Tender	निविदा, टेंडर
144.	Transfer	बदली, स्थानांतरण, अंतरण
145.	Treasurer	कोषपाल
146.	Unofficial	अशासकीय, अशासिक
147.	Vacancy	रिक्ति
148.	Vigilance Officer	सतर्कता अधिकारी, निगरानी अधिकारी
149.	Warrant	अधिपत्र, वारंट

50 टिप्पणियाँ

1.	A brief note is placed below	संक्षिप्त टिप्पणी नीचे प्रस्तुत है।
2.	Acknowledge receipt of this	इसकी पावती भेजिए।
3.	Action as proposed may be taken	यथा प्रस्तावित कार्यवाही की जाए।
4.	Agenda of the meeting is put up	बैठक की कार्यसूची प्रस्तुत है।
5.	Application may be rejected	आवेदन अस्वीकार कर दिया जाए।
6.	Approved as proposed	प्रस्ताव के अनुसार अनुमोदित
7.	Administrative approval may be obtained	प्रशासनिक अनुमोदन प्राप्त किया जाए।
8.	Await reply	उत्तर की प्रतीक्षा करें।
9.	Await further report	और विवरण की प्रतीक्षा करें।
10.	Ascertain this position please	कृपया स्थिति का पता लगायें।
11.	Amended draft is submitted for approval	संशोधित प्रारूप अवलोकनार्थ प्रस्तुत है।
12.	Brief resume of the case is given below	मामले का संक्षिप्त सार नीचे दिया गया है।
13.	Call for explanation	स्पष्टीकरण माँगे
14.	Call for report	रिपोर्ट मँगवाए।
15.	Comply with the orders	आदेशों का पालन करें।
16.	Clarify the position please	कृपया स्थिति को स्पष्ट करें।
17.	Copy enclosed for ready reference	सुलभ संदर्भ के लिए प्रतिलिपि संलग्न है।
18.	Copy is enclosed	प्रतिलिपि संलग्न है।
19.	Copy forwarded for information and necessary action	प्रतिलिपि सूचनार्थ एवं आवश्यक कार्यवाही के लिए प्रेषित है।
20.	Case may be kept pending	मामले को अभी अनिर्णीत रखा जाए।
21.	Delay should be avoided	विलम्ब न किया जाए।
22.	Disciplinary proceedings may be initiated	अनुशासनिक कार्यवाही शुरू की जाए।
23.	Draft reply is put up	उत्तर का मसौदा प्रस्तुत है।
24.	Enquiry may be conducted	जाँच की जाए।
25.	Expedite action	शीघ्र कार्यवाही करें।
26.	Explanation may be called for	स्पष्टीकरण माँगा जाए।
27.	Facts for the case may be put up	मामले के तथ्य प्रस्तुत करें।
28.	For perusal and return	देखकर लौटाने के लिए।

29.	For comments please	कृपया टिप्पणी दें।
30.	For sympathetic consideration	सहानुभूतिपूर्ण विचार के लिए।
31.	Forwarded and recommended	प्रेषित और संस्तुत ।
32.	I concur with the proposal	मैं इस प्रस्ताव से अपनी सहमति प्रकट करता हूँ।
33.	I have no remarks to offer	मुझे कोई टिप्पणी नहीं करनी है।
34.	Inform all concerned	सभी संबंधित व्यक्तियों को सूचित करें।
35.	Issue a circular	परिपत्र जारी करें।
36.	Keep pending	निर्णय के लिए रोक रखें।
37.	Kindly accord sanction	कृपया स्वीकृति दीजिए।
38.	Kindly confirm	कृपया पुष्टि करें।
39.	Matter is under consideration	मामला विचाराधीन है।
40.	No action is necessary	कोई कार्यवाही अपेक्षित नहीं है।
41.	Needful has been done	आवश्यक कार्यवाही की गई है।
42.	Order may be issued	आदेश जारी किया जाए।
43.	Please see the proceeding notes	कृपया पिछली टिप्पणियाँ देख लें।
44.	Put up the relevant papers	संबंधित कागज प्रस्तुत करें।
45.	Reminder may be sent	अनुस्मारक भेजा जाए।
46.	Report compliance immediately	अनुपालन करके तुरन्त सूचित करें।
47.	Specific reason may be given	विशिष्ट कारण दिया जाए।
48.	Submitted for information	सूचना के लिए प्रस्तुत है।
49.	This may be treated as confidential	इसे गोपनीय समझा जाए।
50.	Verified and found correct	जाँचा और सही पाया।

ਪੰਜਾਬੀ (ਇਲੈਕਟਿਵ)

ਬੀ. ਏ. (ਜਨਰਲ) ਭਾਗ ਦੂਜਾ 2●15 ਦੇ ਇਮਤਿਹਾਨ ਲਈ

ਪੇਪਰ : ਏ

ਕੁੱਲ ਅੰਕ: 1●●
ਲਿਖਤੀ: 9●
ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ: 1●
ਸਮਾਂ: 3 ਘੰਟੇ

ਪਾਠਕ੍ਰਮ

1.	ਮੱਧਕਾਲ ਦੀ ਚੋਣਵੀਂ ਪੰਜਾਬੀ ਕਵਿਤਾ	4● ਅੰਕ
2.	ਸਫ਼ਰਨਾਮੇ ਦਾ ਅਧਿਐਨ	25 ਅੰਕ
3.	ਕਹਾਣੀ-ਸੰਗ੍ਰਹਿ	25 ਅੰਕ

ਕੋਰਸ

1. **ਮੱਧਕਾਲੀ ਕਾਵਿ ਸੁਗੰਧੀਆਂ,** (ਸੰਪਾ.) ਡਾ. ਧਰਮ ਸਿੰਘ, ਪਬਲੀਕੇਸ਼ਨ ਬਿਊਰੋ, ਪੰਜਾਬ ਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ।
2. **ਪੱਥ ਹੀ ਮੰਜ਼ਿਲ ਹੈ,** (ਲੇਖਕ) ਮਨਮੋਹਨ ਬਾਵਾ, ਚੇਤਨਾ ਪ੍ਰਕਾਸ਼ਨ, ਲੁਧਿਆਣਾ।
3. **ਕਥਾ-ਪ੍ਰਵਾਹ**, (ਸੰਪਾਦਕ) ਸੁਰਿੰਦਰ ਕੁਮਾਰ ਦਵੇਸ਼ਵਰ, ਪਬਲੀਕੇਸ਼ਨ ਬਿਊਰੋ, ਪੰਜਾਬ ਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ।

ਯੂਨਿਟ ਅਤੇ ਥੀਮ

1.	(ੳ)	**ਮੱਧਕਾਲੀ ਕਾਵਿ ਸੁਗੰਧੀਆਂ** ਪੁਸਤਕ ਵਿਚੋਂ ਪ੍ਰਸੰਗ ਸਹਿਤ ਵਿਆਖਿਆ (ਚਾਰ ਵਿਚੋਂ ਦੋ)	1● +1●= 20 ਅੰਕ
	(ਅ)	ਕੇਂਦਰੀ ਭਾਵ (ਤਿੰਨਾਂ ਵਿਚੋਂ ਇਕ)	1● ਅੰਕ
	(ੲ)	ਕਵਿਤਾ ਦਾ ਸਾਰ (ਤਿੰਨਾਂ ਵਿਚੋਂ ਇਕ)	1● ਅੰਕ
2.	**ਸਫ਼ਰਨਾਮਾ** :		
	(ੳ)	ਲਘੂ ਉੱਤਰਾਂ ਵਾਲੇ ਪ੍ਰਸ਼ਨ (ਅੱਠਾਂ ਵਿਚੋਂ ਪੰਜ)	5×3 =15 ਅੰਕ
	(ਅ)	ਸਫ਼ਰਨਾਮੇ ਦਾ ਵਿਸ਼ਾ, ਪ੍ਰਮੁੱਖ ਘਟਨਾਵਾਂ, ਸਾਹਿਤਕ ਪਰਖ (ਦੋ ਪ੍ਰਸ਼ਨਾਂ ਵਿਚੋਂ ਇਕ)	1● ਅੰਕ
3.	**ਕਹਾਣੀ-ਸੰਗ੍ਰਹਿ** :		
	(ੳ)	ਲਘੂ ਉੱਤਰਾਂ ਵਾਲੇ ਪ੍ਰਸ਼ਨ (ਅੱਠਾਂ ਵਿਚੋਂ ਪੰਜ)	5×3 =15 ਅੰਕ
	(ਅ)	ਕਹਾਣੀ ਦਾ ਵਿਸ਼ਾ/ਸਾਹਿਤਕ ਆਲੋਚਨਾ	1● ਅੰਕ

ਨੋਟ: ਹਫਤੇ ਦੇ 6 + 6 = 12 ਪੀਰੀਅਡ

ਪੇਪਰ : ਬੀ

ਕੁੱਲ ਅੰਕ: 1●●
ਲਿਖਤੀ: 9●
ਇੰਟਰਨਲ ਅਸੈਸਮੈਂਟ: 1●
ਸਮਾਂ: 3 ਘੰਟੇ

ਪਾਠਕ੍ਰਮ

1. ਪੰਜਾਬੀ ਸਾਹਿਤ ਦਾ ਇਤਿਹਾਸ (17●1-19●● ਈ. ਤੱਕ) 3● ਅੰਕ
2. ਸਾਹਿਤ ਆਲੋਚਨਾ ਅਤੇ ਭਾਰਤੀ ਕਾਵਿ-ਸ਼ਾਸਤਰ 2● ਅੰਕ
3. ਛੰਦ-ਪ੍ਰਬੰਧ 1● ਅੰਕ
4. ਸਾਹਿਤ ਦੇ ਰੂਪ 2● ਅੰਕ
5. ਸਭਿਆਚਾਰ ਅਤੇ ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ 1● ਅੰਕ

ਕੋਰਸ

1. ਪੰਜਾਬੀ ਸਾਹਿਤ ਦਾ ਇਤਿਹਾਸ (17●1ਈ. ਤੋਂ 19●● ਈ. ਤੱਕ), ਪੰਜਾਬੀ ਯੂਨੀਵਰਸਿਟੀ, ਪਟਿਆਲਾ
2. ਸਾਹਿਤ ਆਲੋਚਨਾ ਅਤੇ ਭਾਰਤੀ ਕਾਵਿ-ਸ਼ਾਸਤਰ
3. ਛੰਦ-ਪ੍ਰਬੰਧ
4. ਸਾਹਿਤ ਦੇ ਰੂਪ
5. ਸਭਿਆਚਾਰ ਅਤੇ ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ

ਯੂਨਿਟ ਅਤੇ ਥੀਮ

1. ਪੰਜਾਬੀ ਸਾਹਿਤ ਦਾ ਇਤਿਹਾਸ (17●1ਈ. ਤੋਂ19●● ਈ. ਤੱਕ) 15 ਅੰਕ

   (ੳ) ਸੂਫ਼ੀ ਕਾਵਿ-ਧਾਰਾ, ਕਿੱਸਾ-ਕਾਵਿ ਧਾਰਾ, ਵਾਰ ਤੇ ਜੰਗਨਾਮਾ ਕਾਵਿ-ਧਾਰਾ, ਵਾਰਤਕ ਸਾਹਿਤ ਨਾਲ ਸੰਬੰਧਤ ਇਤਿਹਾਸਮੂਲਕ ਤੇ ਪ੍ਰਵਿਰਤੀਮੂਲਕ ਪ੍ਰਸ਼ਨ
   (ਦੋ ਵਿਚੋਂ ਇਕ)

   ਨੋਟ:-ਪੇਪਰ ਸੈਟਰ ਵੱਲੋਂ ਇਕ ਪ੍ਰਸ਼ਨ ਇਤਿਹਾਸਮੂਲਕ ਤੇ ਇਕ ਪ੍ਰਵਿਰਤੀਮੂਲਕ ਪੁੱਛਿਆ ਜਾਵੇ।

   (ਅ) 17●1ਈ. ਤੋਂ 19●● ਈ. ਤਕ ਦੇ ਪੰਜਾਬੀ ਸਾਹਿਤ ਦੇ ਇਤਿਹਾਸ ਵਿਚੋਂ ਲਘੂ-ਉੱਤਰਾਂ ਵਾਲੇ ਪ੍ਰਸ਼ਨ (ਵੀਹ ਵਿਚੋਂ ਪੰਦਰਾਂ) 15×1=15 ਅੰਕ

2. ਸਾਹਿਤ ਆਲੋਚਨਾ ਅਤੇ ਭਾਰਤੀ ਕਾਵਿ-ਸ਼ਾਸਤਰ 1●+1●=2● ਅੰਕ

   (ੳ) ਸਾਹਿਤ ਆਲੋਚਨਾ ਦੀ ਪਰਿਭਾਸ਼ਾ, ਸਰੂਪ, ਮਹੱਤਵ

   (ਅ) ਭਾਰਤੀ ਆਲੋਚਨਾ : ਰਸ ਸੰਪਰਦਾਇ -ਰਸ ਦੀ ਪਰਿਭਾਸ਼ਾ, ਭਾਵ, ਵਿਭਾਵ, ਅਨੁਭਾਵ, ਸੰਚਾਰੀ ਭਾਵ, ਸਥਾਈ ਭਾਵ, ਰਸ ਦੇ ਪ੍ਰਕਾਰ

   (ਏ) ਧੁਨੀ ਸੰਪਰਦਾਇ: ਧੁਨੀ ਦੀ ਪਰਿਭਾਸ਼ਾ, ਸ਼ਬਦ ਸ਼ਕਤੀਆਂ, ਅਭਿਧਾ, ਲਕਸ਼ਣਾ ਅਤੇ ਵਿਅੰਜਨਾ

   [(ੳ), (ਅ) ਅਤੇ (ਏ) ਭਾਗਾਂ ਦੇ ਚਾਰ ਸਵਾਲਾਂ ਵਿਚੋਂ ਕੋਈ ਦੋ ਕਰਨੇ ਅਤੇ ਹਰੇਕ ਭਾਗ ਦੇ ਅੰਕ ਬਰਾਬਰ ਹੋਣਗੇ]

3. ਛੰਦ-ਪ੍ਰਬੰਧ: ਦੋਹਿਰਾ, ਸੋਰਠਾ, ਕੋਰੜਾ ਅਤੇ ਦਵੱਈਆ, ਸਵੱਈਆ, ਕਬਿੱਤ, ਸਿਰਖੰਡੀ, ਨਿਸ਼ਾਨੀ, ਬੈਂਤ, ਚੌਪਈ (ਚਾਰ ਵਿਚੋਂ ਦੋ) 5+5 =10 ਅੰਕ

4. ਸਾਹਿਤ ਦੇ ਰੂਪ 1●+1● =20 ਅੰਕ

   ਕਿੱਸਾ, ਵਾਰ, ਕਾਫੀ, ਜੰਗਨਾਮਾ, ਸ਼ੀਹਰਫ਼ੀ, ਸਿੱਠਣੀਆਂ, ਸੁਹਾਗ, ਘੋੜੀਆਂ
   (ਚਾਰ ਪ੍ਰਸ਼ਨਾਂ ਵਿਚੋਂ ਕੋਈ ਦੋ)

5. ਸਭਿਆਚਾਰ ਅਤੇ ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ : 1● ਅੰਕ

ਸਭਿਆਚਾਰ ਦੀ ਪਰਿਭਾਸ਼ਾ, ਸਰੂਪ, ਲੱਛਣ ਅਤੇ ਪੰਜਾਬੀ ਸਭਿਆਚਾਰ ਦੇ ਵਿਭਿੰਨ ਪੱਖ
(ਦੋ ਵਿਚੋਂ ਕੋਈ ਇਕ ਸਵਾਲ)

ਸਹਾਇਕ ਪੁਸਤਕਾਂ :

1. ਪੰਜਾਬੀ ਸਾਹਿਤ ਦਾ ਇਤਿਹਾਸ (17●1-19●●), ਪੰਜਾਬ ਯੂਨੀਵਰਸਿਟੀ, ਚੰਡੀਗੜ੍ਹ।

2. ਪੰਜਾਬੀ ਸਾਹਿਤ ਦਾ ਇਤਿਹਾਸ (17●7-19●●), ਪੰਜਾਬੀ ਯੂਨੀਵਰਸਿਟੀ, ਪਟਿਆਲਾ।

3. ਪੰਜਾਬੀ ਸਾਹਿਤ ਦੀ ਉਤਪਤੀ ਤੇ ਵਿਕਾਸ, ਪਰਮਿੰਦਰ ਸਿੰਘ ਤੇ ਕਿਰਪਾਲ ਸਿੰਘ ਕਸੇਲ, ਲਾਹੌਰ ਬੁੱਕ ਸ਼ਾਪ, ਲੁਧਿਆਣਾ।

ਵਿਸ਼ੇਸ਼ ਨੋਟ : ਸਮੁੱਚੇ ਪਾਠਕ੍ਰਮ ਲਈ ਹਫਤੇ ਵਿੱਚ 6+6=12 ਪੀਰੀਅਡ।

संस्कृत (इलैक्टिव)

बी.ए. (जनरल) द्वितीय वर्ष, 2015

PAPER-A : गीता एवं नाटक **पूर्णांक: 90 + 10= 100**

(आन्तरिक परीक्षा- 10, लिखित परीक्षा- 90) **समय: 3 घण्टे**

टिप्पणी - प्रश्न पत्र का माध्यम हिन्दी होगा । उत्तरों का माध्यम संस्कृत, हिन्दी, पंजाबी या अंग्रेजी में से कोई एक भाषा होगी ।

उद्देश्य –

- विद्यार्थियों को भारतीय दर्शन के अद्वितीय ग्रन्थरत्न श्रीमद्भगवद्गीता (चतुर्थ अध्याय) में प्रतिपादित निष्काम कर्मयोग, समत्व योग में स्थित योगी पुरुषों का आचरण एवं उनकी महिमा, विभिन्न प्रकार के ज्ञानयज्ञ तथा ज्ञान का महत्त्व इत्यादि विषयों का अध्ययन करवाना ।
- संस्कृत नाट्यविद्या के स्वरूप और प्रवृत्तियों से परिचित कराना।
- इसके अतिरिक्त व्यावहारिक शब्दावली द्वारा विद्यार्थियों की संस्कृत के प्रति रुचि जागृत करना।

(क) गीता (चतुर्थ – अध्याय) **40 अंक**

(i) सूक्ति/श्लोक : सप्रसंग अनुवाद एवम् व्याख्या $3 \times 10 = 30$ अंक

(ii) गीता (चतुर्थ अध्याय) पर आधारित समीक्षात्मक प्रश्न $1 \times 10 = 10$ अंक

(ख) दूतवाक्यम् – भास **30 अंक**

(i) श्लोक/सूक्ति : सप्रसंग अनुवाद व व्याख्या $2 \times 10 = 20$ अंक

(ii) समीक्षात्मक प्रश्न (लेखक परिचय, रचनाएँ, विषयवस्तु, पात्रचरित्रचित्रण) $1 \times 10 = 10$ अंक

(ग) व्यावहारिक संस्कृत शब्दावली **20 अंक**

(i) वस्त्र, आभूषण, श्रृंगारपरक $5 \times 2 = 10$ अंक

1. कंबल	- कम्बलः	2. कुरता	- कञ्चुकः
3. कोट	- प्रावारः	4. चादर	- प्रच्छदः, शय्याच्छादनम्
5. जूता	- उपानत्	6. तकिया	- उपधानम्
7. दरी	- आस्तरणम्	8. दुपट्टा	- उत्तरीयम्
9. धोती	- अधोवस्त्रम्	10. पगड़ी	- शिरस्त्रम् उष्णीकम्
11. परदा	- यवनिका	12. पायजामा	- पादयामः
13. पैंट	- आप्रपदीनम्	14. रुमाल	- करवस्त्रम्
15. शेरवानी	- प्रावारकम्	16. साडी	- साटिका
17. स्वेटर	- उर्णावरकम्	18. कपडा	- वस्त्रम्, वसनम्, चीरम्
19. अंगूठी	- अङ्गुलीयकम्, मुद्रिका	20. कंगना	- कंकणः
21. कान की बाली	- कुण्डलम्	22. गहना	- अलङ्कारः, आभरणम्
23. घुंघरु	- किंकिणी	24. चूड़ी	- काचवलयः, काचवलयम्
25. नथ	- छोलिका	26. पाजेब	- नूपुरः, नूपुरम्, शिञ्जा

27. बाजूबंद(ब्रेस लेट)	- केयूरम्	28. हार	- हार:
29. सोने का कड़ा	- कटक:	30. आइना(शीशा)	- दर्पण:, आदर्श:
31. इत्र (सैंट)	- गन्धतैलम्	32. उबटन	- उद्वर्तनम्
33. कंघी	- प्रसाधनी	34. काजल	- अञ्जनम्, कज्जलम्
35. क्रीम	- शर:	36. ड्रेसिंग टेबल	- श्रृंगारफलकम्
37. तिलक	- तिलकम्	38. दाँत का ब्रुश	- दन्तधावनम्, दन्तकूर्च:
39. नेल पोलिश	- नखरंजनम्	40. बिन्दी	- बिन्दुक:
41. पाउडर	- चूर्णकम्	42. मंजन	- दन्तचूर्णम्
43. मेंहदी	- मञ्जिष्ठा	44. लिपस्टिक	- ओष्ठरंजनम्
45. साबुन	- फेनिलम्, फेनकम्	46. सिंदूर	- सिन्दूरम्
47. दांत कूरेदने की सूई	- दन्तशोधनी	48. महावर	- अलक्तका
49. रूज	- कपोलरंजनम्	50. सिंगारदान	- श्रृंगारधानम्, श्रृंगारपिटकम्

(ii) अन्न, भोजन एवं तत्सम्बद्ध पात्र $5 \times 2 = 10$ **अंक**

1. अरहर	- आढकी	2. उड़द	- माष:
3. गेहूँ	- गोधूम:	4. गेहूँ का आटा	- गोधूमचूर्ण:
5. चना	- चणक:	6. जौ	- यव:
7. चावल	- तण्डुल:, व्रीहि:	8. ज्वार	- यवनाल:
9. तिल	- तिल:	10. दाल	- द्विदलम्
11. धान	- धान्यम्, शालि:	12. बाजरा	- प्रियङ्गु:
13. बेसन	- चणकचूर्णम्	14. मसूर	- मसूर:
15. मूंग	- मुद्ग:	16. सत्तु	- सक्तु:
17. सरसों	- सर्षप:	18. हल्दी	- हरिद्रा
19. अचार	- सन्धितम्	20. गरम भोजन	- उष्णभोजनम्
21. खिचड़ी	- कृशर:	22. चटनी	- अवलेह:
23. चिकना	- चिक्कणम्	24. ठंडाभोजन	- शीतभोजनम्
25. भरता	- भर्ता	26. भात	- ओदन:, ओदनम्
27. मट्ठा	- तक्रम्	28. रसोई	- रसवती, पाकशाला, महानस:
29. रायता	- राज्यक्तम्	30. रोटी	- रोटिका
31. शक्कर	- शर्करा	32. समोसा	- समोष:
33. सौंफ	- मधुर:	34. हींग	- हिंगु:
35. अंगीठी	- हसन्ती	36. कटोरा	- कटोरम्
37. कड़ाही	- कटाह:, स्वेदनी	38. कांच का गिलास	- काचकंसम्, काचचषक:
39. गिलास	- कंसम्, चषक:	40. घड़ा	- घट:, कुम्भ:
41. चम्मच	- चमस:	42. चीमठा	- सन्दंश:
43. टव	- द्रोणी	44. थाली	- स्थालिका, थालिका
45. प्लेट	- शराव:	46. बाल्टी	- उदञ्चनम्
47. लोटा	- करक:	48. स्टोव	- उद्ध्मानम्
49. बेलन	- वेलनम्	50. चाकू	- छुरिका

नोट - 1. पत्र का अध्ययन समय छह पीरियड प्रतिसप्ताह होगा ।
2. सभी प्रश्नों में शतप्रतिशत विकल्प आवश्यक हैं ।

PAPER-B **व्याकरण तथा छन्द** **पूर्णांकः 90 + 10= 100 अंक**

(आन्तरिक परीक्षा- 10, लिखित परीक्षा- 90) **समयः 3 घण्टे**

टिप्पणी प्रश्न पत्र का माध्यम हिन्दी होगा । उत्तरों का माध्यम संस्कृत, हिन्दी, पंजाबी या अंग्रेजी में से कोई एक भाषा होगी ।

उद्देश्य –

छात्रों के व्याकरण सम्बन्धी ज्ञान को और भी उन्नत करने तथा भाषा पर वैशिष्ट्य प्राप्त्यर्थ व्यंजन सन्धि, समास, तद्धित प्रत्यय, शब्दरूप, धातुरूप, कृदन्त तथा छन्दों का ज्ञान कराना ।

(क) व्यञ्जन सन्धि $5 \times 2 = 10$ **अंक**

(ख) समास - (द्वन्द्व, तत्पुरुष) $5 \times 2 = 10$ **अंक**

(ग) शब्दरूप - $2 \times 8 = 16$ **अंक**

(i) तत्, यत्, किम् व सर्व (तीनों लिंगों में)

(ii) महत् व बलवत् (पुंल्लिंग में)

(iii) कति, राजन् व चन्द्रमस्

(घ) धातुरूप - अस्, दा, कुप्, शक्, प्रच्छ्, मिल्, कृ, क्रीञ्, ग्रह्, ज्ञा, श्रु, तथा कथ् $2 \times 9 = 18$ **अंक**

(केवल लट्, लोट्, लृट्, लङ् व विधिलिङ् लकार में)

(ङ) कृदन्त प्रत्यय - क्तः, क्तवतु, क्त्वा, तुमुन् प्रत्यय $5 \times 2 = 10$ **अंक**

(गम्, पठ्, पत्, क्रीड्, वद्, पा, लिख्, नी, दृश्, भू, त्यज्, स्मृ, अस्, दा, कुप्, शक्, प्रच्छ्,

मिल्, कृ, क्री, ग्रह्, ज्ञा, श्रु, कथ्- इन धातुओं के योग में)

(च) तद्धित-प्रत्यय अण्, मतुप्, तरप्, तमप् (केवल उदाहरण ही प्रष्टव्य हैं) $4 \times 1 = 4$ **अंक**

(छ) छन्द - अनुष्टुप्, वंशस्थ, इन्द्रवज्रा, उपेन्द्रवज्रा, उपजाति, शिखरिणी, मन्दाक्रान्ता, मालिनी, वसन्ततिलका तथा भुजंगप्रयात (दो के ही सोदाहरण लक्षण प्रष्टव्य हैं) $2 \times 6 = 12$ **अंक**

(ज) कारक (केवल सामान्य नियम- अशुद्ध वाक्यों को शुद्ध करना) $5 \times 2 = 10$ **अंक**

सहायक पुस्तक :- शीघ्रबोध, चौखम्बा प्रकाशन, वाराणसी

नोट- 1. पत्र का अध्ययन समय छह पीरियड प्रतिसप्ताह होगा ।

2. सभी प्रश्नों में शतप्रतिशत विकल्प आवश्यक हैं ।

URDU (Elective)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Outlines of tests, syllabi and courses of reading

Paper-A 100 Marks
Time : 3 Hrs.

1. Explanation of Verses : 30 marks
   - (i) Ghazal 20 marks
   - (ii) Nazm 10 marks
2. Explanation of Prose 30 marks
3. Central Idea of a lesson or a poem 20 marks
4. Introduction and literary contribution of the following poets & prose writers : 20 marks

Poets :

1. Mir Taqi Mir
2. Ghalib
3. Nazir Akbarabadi
4. Chakbast
5. Jigar Muradabadi

Prose Writers :

1. Sir Sayyed
2. Mohd. Hussain Azad
3. Hali
4. Prem Chand
5. Rashid Ahmed Siddiqui

Book Prescribed

Khyaban-e-adab compiled by Azim-ul-Haq Junedi, published by Educational Book House, A.M.U., Market, Aligarh.

The following selections from the book Khyaban-e-Adab are prescribed in the course :

1. Ghazliyat
   Mir, Dard, Zauq, Ghalib, Momin, Hasrat Mohani, Fani, Jigar, Shad, Azeemabadi and Firaq.

2. Qasida by Ghalib
3. Masnavi by Naseem
4. 'Marsiya-e-Gokhle' by Chakbast
5. Rubaiyat

   Akbar Allahabadi and Anis Lucknavi
6. **Nazmen :**

   Iqbal :

   1. Chand Aur Tare
   2. Naseehat
   3. Mulla Aur Behisht

   Nazir Akbarabadi :

   1. Mele Ki Sair
   2. Holi Ki Bahar

   Chakbast :

   Ramayan Ka Ek Scene :

Paper-B 100 Marks

Time : 3 hrs.

1. Novel - one question each on its plot, characterization — 40 marks
2. Explanation of a paragraph from the Novel — 40 marks
3. Rajinder Singh Bedi Ki Novel Nigari Ki Khususiat — 20 marks

   Or

   Novel Ke Ajza-e Tarkibi

Book Prescribed

"Ek Chadar Maili Si" by Rajinder Singh Bedi.

PERSIAN (Elective)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Note : There will be two papers of 100 marks each of three hours duration.

Paper-I : PROSE AND POETRY Max. Marks : 100

Time : 3 hrs.

1.	Translation of prose passages into English, Hindi, Urdu, Punjabi or Persian (with choice).	30 marks
2.	Explanation of poems into English, Hindi, Urdu, Punjabi or Persian.	30 marks
3.	Summary of any One lesson or poem into English, Hindi, Urdu, Punjabi or Persian.	10 marks
4.	Basic knowledge of Elements and kinds of Poetry. (with definitions only)	15 marks
5.	Knowledge of the elements and kinds of Prose. (with definitions only)	15 marks

Shairi Asnaaf :

1. Ghazal
2. Qasida
3. Masnawi
4. Marsiya
5. Rubai
6. Qata
7. Nazm
8. Khumariyat
9. Hamad
10. Ikhaqi Shayeri

Nasari Asnaaf

1. Dastan
2. Novel
3. Afsana
4. Tazkerah Nawisi
5. Taarikh Nawisi
6. Tanaz-o-Mizah
7. Drama
8. Hikayat Nawisi

Book Prescribed :

Farsi Bray-e-ghair Farsi Zebanan az "Samineh Baghci wan" available at Maktab-e-Jamia, Ansari Road, Near Jama Masjid, Delhi.

Paper-II : TRANSLATION AND LITERARY HISTORY

Max. Marks : 100

Time : 3 hrs.

Note : All the questions carry equal marks.

1. Translation of simplest passage/sentences from Persian into English, Hindi, Punjabi or Urdu.

2. Translation of simplest passage/sentences from English, Hindi, Punjabi or Urdu.

3. Basic knowledge of the different periods of the Literary History of Persian Literature and Medieval Indian History only the following periods :

   Samanid, Ghaznavid, Seljuqiyeh, Temurid, Safavid, Moghaliya (Baber to Aurangzeb)

4. The direct and simple questions on the most important, famous and eminent poets and writers.

5. Short note on any One most important and famous books from the above given Historical Periods. Basic and elementary knowledge of the kinds of prose and poetry (Asnaaf-e-Nasr-o-Nazm)

Book prescribed for this paper :

1. Zaban-e- Adabeyat-e-Farsi (Danishgahi)

   Movallephaan Dr. Malik Mohd. Farrukh Zaad, Dr. Hussan Ali Mohmdi and Manizheh Falahi, Published from Tehran, Iran.

2. Tarikh-e-Adabeyat-e-Farsi by Raza Zadeh Shafaq.

3. Amuzish-e-sada-ye-zabaan-e-Farsi (easy way of learning Persian by Majid Ahmedi).

ARABIC (Elective)

(Kept in Abeyance)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

There will be two papers of 100 marks each

Paper-A		100 Marks
		Time : 3 hours
1.	Grammar Mirraat-ul-Arabiya by Dr. M.S. Rehman published by Kutab Khanna Rashidyya, Delhi-6.	50 marks
2.	Translation of Arabic into English/Hindi/Urdu.	30 marks
3.	Use of words into Arabic sentences.	20 marks

Paper-B		100 Marks
		Time : 3 hours
1.	Prose The following topics of Al' Abarat by Al Manfluti; Al-Yatim and Al-Hijab	60 marks
2.	Poetry	40 marks

Nazam-wa-Nasr

The following poets from the book 'Majmu' at Minan :

1. Hafiz Ibrahim
2. 'Abul' Ala'al Mu'arri
3. Al-Farazadaq
4. Hassan B. Thabit

(All books can be Had from Kutab Khana, Rashidyya, Urdu Bazar, Jama Masjid, Delhi-6)

FRENCH (Elective)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A : Introduction to French Civilization, Prose, Poetry

Max. Marks		: 100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

I *Selected Readings in Poetry :*
Questions, explanation of stanzas or of poems, central ideas and summaries of poems to be asked and answered in French. 20 marks

II (a) Questions (pertaining to the prescribed text)of 4 marks each. 20 marks

(b) Questions on French Civilization in the form of fill in the blanks and multiple choice (wherever possible) simple answers of 1-2 sentences of 2 marks each. 20 marks

(c) Dialogue based on the subject or theme from the text. 10 marks

III *Prose :* 20 marks
Direct simple questions based on the prescribed text to be asked and answered in French, Questions should be of 5 marks each.

CHOICE TO BE GIVEN IN ALL QUESTIONS

Paper-B : APPLIED GRAMMAR, CREATIVE WRITING AND VIVA–VOCE

Max. Marks	:	100
Theory	:	60 marks
Internal Assessment	:	10 marks
Time	:	3 Hours
Viva-Voce	:	30 marks

1. Translation from French into English of an unseen passage or short sentences. 10 marks
2. Translation from English into French of an unseen passage or short sentences. 10 marks
3. A composition or letter of about 150-200 words. 10 marks
4. Questions on grammar including conjugation (in applied form) pertaining to the prescribed text. 30 marks
5. Viva-Voce : 30 marks
   (a) Dictation (an unseen passage of about 100 words), easier than the prescribed Textbook. 10 marks
   (b) Conversation. 10 marks
   (c) Reading (unseen passage). 10 marks

Courses of Reading :

Textbook-Campus-II, Jacky Girardet, et Jacques Pécheur, CLE International, Paris, 2002.

(a) Livre de l' élève.

(b) Cahier d exercices.

(c) Cassettes.

Poetry :

Comprehension, explanation, literary appreciation and criticism of the poems to be studied.

The following 10 poems are to be studied:

1. Pour Hélène-quand vous serez bien vieillie-Pierre Ronsard – French Poetry for students-A.W. Bains.
2. Le Savetier et le financier-La Fontaine, French Poetry for students – A.W. Bains.
3. Le Rat qui s' est retiré du monde – La Fontaine - French Poetry for students – A.W. Bains.
4. L'isolement – Lamartine-French Poetry for students – A.W. Bains.
5. Puisque j'ai mis ma lèvre – V Hugo French Poetry for students – A.W. Bains.
6. La mort du Loup – A De vigny-French Poetry for students – A.W. Bains.
7. Recueillement-Baudelaire-French Poetry for students – A.W. Bains.
8. Page d'écriture- J. Prévert-Paroles.
9. Si tu t'imagines-Raymond Queneau-L'Instant Fatal.
10. Mon rêve familier – P. Verlaine-collection littéraire. Lagarde et Michard. XIX siècle.

Note : 1. The latest syllabus should be strictly followed.
2. Choice should be given in questions.

Text Books for Reference :

1. French Poetry for students by A. W. Bains.
2. Paroles édition Folio by Jacques Prévert.
3. Le français et la Vie-Tome III G. Mauger/M.Bruezièr e.
4. Collections Littéraires XIX & XX Siècle-Nouveaux classique Larousse.
5. Poèmes choisis-A de. Musset-édition-Nouveaux classique Larousse.
6. A Little Book of French Poetry-Blackie's Little French Classics.

Prose : Tintin Au Tibet Casterman

N. B. : 1. The compositions and the unseen passages should be based on the vocabulary and grammar covered till B.A. Second Year.

2. The paper shall be set and answered in French (except translation).

3. Eight periods of one hour weekly—Six hours for Text and Two hours for Composition.

GERMAN (Elective)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A : WRITTEN

Max. Marks : 100

Time : 3 Hours

10 multiple choice questions of 3 marks each to be asked. This question is compulsory. Question in applied grammar (fill in the blanks) comforming to the prescribed book "Deutsche Sprachlehre für Ausländer" in einem Band.	:	30 marks
4 out of 6 questions to be attempted.	:	20 marks
Texte zum lesen and Nachèrzählen i.e. from page 30 onwards.	:	20 marks
Question on Culture and Civilisation from Blick auf Deutschland (Page No. 30 to 68) excluding lesson 10 a & 10 b.	:	20 marks
Internal Assessment (for regular students)	:	10 marks
		100 marks

Paper-B : (I) WRITTEN

Max. Marks : 100

Time : 3 Hours

Paraphrasing of a poem or stanzas out of the following poems from "German Verse" by B.B. Kulkarni & R.N. Chapekar : : 20 marks

1. An den Mond (Johann Wolfgang von Goethe).
2. Erlkönig (Goethe).
3. Abendphantasie (Friedrich Holderlin).
4. Du bist wie eine Blume (Heinrich Heine).
5. Abschied (Joseph Freiher von Eichendorff).

6. Glück (Hermann Hesse).

7. Der Mai ist gekommen (Emanuel Geibel).

5 simple unseen sentences from German into English (5 out of 7) (unknown words to be given).	:	20 marks
5 sentences from English into German (Meaning of unknown words to be given)	:	20 marks
		60 marks

II : ORAL/PRACTICAL

Reading of text(s) and General Conversation in German.	:	30 marks
Internal Assessment (For regular students). The mode of evaluation for Internal Assessment is to be followed as per University Guidelines.	:	10 marks

Prescribed Text Books :

1. Deutsche Sprachlehre fur Ausländer in einem Band by Dora Schulz und Heinz Griesbach (Chapter 15 to 26).

2. Texte zum lessen und Nachèrzählen by Hans. Joachin Arndt (Max Huber Verlag) (From page 30 to 65).

3. Blick auf Deutschland (page 30 to 68) excluding lesson 10a and 10b.

4. German Verse by B.B. Kulkarni & R.N. Chapekar.

RUSSIAN (Elective)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A :

Max. Marks : 100
Theory : 90 marks
Internal Assessment : 10 marks
Time : 3 Hours

WRITTEN

I Translation from simple Russian into English/Hindi/Punjabi (about 120 words). : 45 marks

II Translation from English/Hindi/Punjabi into Russian (about 80 words). : 45 marks

NOTE : USE OF DICTIONARIES IS ALLOWED

Paper-B : Max. Marks : 100

(I) WRITTEN

Theory : 60 marks
Internal Assessment : 10 marks
Time : 3 Hours

1. *Composition :* (1 out of 3) on the following topics : 16 marks
   - (i) Journey/Excursion.
   - (ii) Indian/Russian Writer.
   - (iii) My Country.
   - (iv) Capital City : (India/Russia).
   - (v) Our Library.
   - (vi) Our University/College.
   - (vii) My Friend.
   - (viii) A Cultural Evening.

2. *Applied Grammar :* 4 questions (out of 7) of 4 marks each. 16 marks

3. *Comprehension :* 6 questions of 3 marks each (out of 8) on a passage from unseen text(s) to be produced in the question paper. 18 marks

4. Question on the texts studied—5 questions (out of 7) of 2 marks each. 10 marks

(II) ORAL 30 marks

Reading of text(s). Conversation/Narration in Russian (day to day incidents/topics).

Books Prescribed :

Wagner, V.N. : RUSSIAN – P.P.H., New Delhi (Lessons 32-45).

OR

Maurya, A.K. : Essential of Russian, University of Delhi, 2004 (Lessons 15-23).

Books Recommended for Additional Readings :

1. Timberlake Reference : Grammer of Russian, Cambridge.
2. Pulkina, I. : RUSSIAN—A Practical Grammar with Exercises, Izdatelstovo, 'Russky Yazik', Moscow, 1980.
3. Anpilogova, B.G. : "ULYBKA" (SMILE), P.P.H., New Delhi.
4. Pulkina, I. : "Uchebnik Ruskova Yazika" Izdatelstovo,Vicshaya Shkola, Moscow, 1964.
5. Dictionaries

BENGALI (Elective)
(Kept in Abeyance)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A 100 Marks

Poetry	60 marks
Drama	40 marks

Poetry : (a) P.N. Bisi & Tarapada Mukhopadhyaya (Ed.) Kavya Vitan.

The following pieces only :-

1. Vidyapati : 6, 9
2. Chandidas : 8,10
3. Govindadas : 10
4. Ramprasad : 1
5. Michael Madhusudan : Atma Bilap
6. Satyendra Nath : Champa
7. Jibanananda Das : Mrityur Age
8. Premendra Nath Mitra : Path
9. Sukanta Bhattacharya : Ranar

Drama : (b) Rabindranath : Kahini

The following pieces :

(i) Karna Kunti Sambad

(ii) Gandharir Abedan

(iii) Bhasa O Chhanda

Paper-B 100 Marks

Poetry	40 marks
Drama	40 marks
Composition	20 marks

Poetry : Rabindranath : Sanchayita

The following pieces only :

1. Nirjharer Swapnabhanga
2. Sonar Tari
3. Swarga Hoite Biday
4. Mukti (Naibedya)
5. Balaka
6. Karna Kunti Sambad
7. Phanki
8. Africa
9. Kalika
10. Bansi

Drama : Sri Madhusudan by Banaphul

Composition : A poem or a part of a poem will be given to the students. They are expected to

(i) Write the central idea of the same.

(ii) Suggest a title.

(iii) Annotate a few words; and

(iv) Comment on the style and diction.

TAMIL (Elective)
(Kept in Abeyance)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A

Max. Marks : 100

Time : 3 hours

Poetry	50 marks
Prose	50 marks
Total :	100 marks

Paper-B

Max. Marks : 100

Time : 3 hours

1.	Collection of Short Stories	50 marks
2.	History of Tamil Literature from 10th Century to 17th Century	50 marks
	Total :	100 marks

Book Prescribed

Poetry :

Selections in Tamil Poetry for B.A. Part II, 1994-95, Published by the University of Madras, Madras-5.

Prose :

Ilakkiya Nenjam by Dr. M.P. Balasubramanian, Published by Paari Nilaiyam, 184, Broadway, Madras-600108.

Collection of Short Stories :

"Guru Peedam" by Jayakantan, published by Meenakshi Puthanga Nilaiyam, Madurai.

KANNADA (Elective)
(Kept in Abeyance)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A 100 marks

1. *Poetry* 50 marks

   Samakalina Kannada Kavite Bhaga : 3, pages 53 to 90 published by Bangalore University, Bangalore.

2. *Prose* 50 marks

   Amerikadalli nanu by Dr. B.G.L. Swami, published by Kavyalaya, Mysore.

Paper-B 100 marks

1. *Grammar* 50 marks

   Sandhi, Samasas, Figure of Speech, Metre are to be studied.

2. *History of Kannada Literature* (Medieval period only) 50 marks

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MALAYALAM (Elective)
(Kept in Abeyance)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A — 100 : Marks

Time : 3 hrs.

1. *Poetry :* — 50 marks

   Malayala Kavy Samgraham (poems 11 to 18, 20 and 27), published by National Book Trust, New Delhi.

2. *Prose :* — 50 marks

   Natappathagal by Punathil (autobiographical essays) published by D.C. Books, Kottayam, Kerala.

Paper-B — 100 : Marks

Time : 3 hrs.

1. Grammar : — 50 marks

   Sandhis, Samasas, Figure of speech and Metre are to be studied.

2. *History of Malayalam Literature* — 50 marks

   (Medieval period only)

TELUGU (Elective)
(Kept in Abeyance)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A 100 : marks

Time : 3 hrs.

Poetry	50 marks
Prose	50 marks
	100 marks

1. *Poetry* : Draupadi Prabhavamu from Andhra Mahabharatamu, Sabha Parvamu, Aswasam II - Verses from 206 to 264 of any addition.

2. *Prose* : Sahityadhyanam, published by Yuva Barati 5, Kingsway, Secunderabad-3.

Paper-B 100 : marks

Time : 3 hrs.

1. *Grammar :* 50 marks
   Rules of Sandhis, Samasas, Figure of Speech and Metre are to be studied.

2. *History of Telugu Literature* (History of Medieval Literature) 50 marks

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PHYSICAL EDUCATION

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

PAPER-A :

Max. Marks	:	65
Theory	:	60 marks
Internal Assessment	:	05 marks
Time	:	3 Hrs.

INSTRUCTIONS FOR THE PAPER-SETTER AND STUDENTS :

(i) There shall be nine questions in all, spread over five units.

(ii) First question/unit is compulsory. It will contain six short answer type questions, spread over the whole syllabus to be answered in brief. It will carry 12 marks.

(iii) Rest of the paper shall contain four units for descriptive questions. Each unit shall have two questions and the students shall be given internal choice i.e. the students shall attempt one question from each unit.

(iv) All questions/units will carry equal marks.

(v) Private candidates and the students of the University School of Open Learning (USOL) will not be allowed to take this subject.

UNIT-I 12 Marks

Entire syllabus given in the Units II to V will be covered to set six short answer type questions in first question/unit of the question paper which is compulsory.

UNIT-II 12 Marks

Psychological Basis of Physical Education :

- Meaning of Psychology and Sports Psychology.
- Psychological factors effecting physical performance.
- Meaning of Learning.
- Laws of Learning.
- Learning curve, its types, characteristics and implications in Physical Education and Sports.
- Psychological characteristics and problems of an adolescent.
- The role of Physical Education and Sports in solving the problems of an adolescent.

UNIT-III 12 Marks

Motivation:

- Meaning, definitions, types and methods of motivation.
- Importance of motivation in Physical Education and Sports.

Transfer of Training:

- Meaning, definitions, types and factors affecting transfer of training.

UNIT-IV 12 Marks

Personality:

- Meaning, definitions, characteristics, dimensions and traits of personality.
- Factors affecting development of personality (Heredity and Environment).
- The role of physical activities in Personality Development.

Sports and Socialization:

- Meaning and definitions of socialization, socialization through sports.

UNIT-V 12 Marks

Politics, Economy and Media and Sports Performance :

- Role of politics and economy in the promotion of games and sports.
- Role of media in promotion of sports.
- Causes of deterioration and suggestions for the improvement of Sports Performance.

Basics of Swimming:

- History.
- Different Strokes and its Events.
- Dimensions of pool.
- Rules and regulations (number of official required and general rules).
- Major tournaments and Arjuna awardees of the game.

References :

1. Blair Jones & Simpson :*"Educational Psychology ",* The MacMillan Co., New York, 1962.
2. Brown, R. : *"Social Psychology",* Free Press, New York, 1965.
3. Bucher, Charles A. : *"Foundations of Physical Education",* St. Louis, 1979, The C.V. Mosby Company.
4. Conger, J.J.: "*Adolescence and Youth : Psychological Development in a Changing World,"* 22nd Edition, 1977, Harper and Row, New York.

5. Clarke, S. K.A. and Fein, G.G. : *"Children: Development through Adolescence"*, John Wiley & Sons, New York, 1983.

6. Cratty, B.J.: *" Psychology and Physical Activity"*, Englewood Cliffs, N.J., Prentice Hall, 1968.

7. Hollander, E.P.: *" Principles and Methods of Social Psychology"*, New York, 1971, Oxford University Press.

8. Linsa, M.I., *"Child-Adults, Peers : Patterns Communication",* Progress Publishers, Moscow, 1985.

9. Linggren, H.E.: *" Educational Psychology in the Class Room"*, John Wiley & Sons, New York, 1962.

10. Loy, John W. and Kenyon Gerald, S. *: "Sport, Cultural and Society : A Reader on the Sociology of Sport",* London, 1969, The MacMillan Co.

11. Hurlock, E.B. : *"Adolescent Development",* Appleton Century Crofts, New York,1956.

12. Rushall, Brent S. and Siedentop : *"The Development and Control of Behaviour in Sport and Physical Daryl Education"*, Philadelphia, 1972, Lea and Febiger.

13. Shaphard, R.J. : *"The Fit Athlete",* Oxford University Press, 1978.

14. Singh, Ajmer et. al. : *Modern Text Books of Physical Education, Health and Sports,* Kalyani Publishers, Ludhiana, 2000.

15. Talamini, John T. and Page & Charles, H. : *"Sport & Society – An Anthology",* Boston, 1973, Little Brown Co.

16. Walia, J.S. : *"Foundations of Educational Psychology",* Paul Publishers, Jalandhar City, 1992.

17. Wilmore, Jack H. : *"Athletic Training and Physical Fitness",* 1977, Allyn and Bacon, Inc., 470, Atlantic Avenue, Boston, Massachusetts.

PAPER-B :

Max. Marks		: 65
Theory	:	60 marks
Internal Assessment	:	05 marks
Time	:	3 Hrs.

INSTRUCTIONS FOR THE PAPER-SETTER AND STUDENTS :

(i) There shall be nine questions in all, spread over five units.

(ii) First question/unit is compulsory. It will contain six short answer type questions, spread over the whole syllabus to be answered in brief. It will carry 12 marks.

(iii) Rest of the paper shall contain four units for descriptive questions. Each unit shall have two questions and the students shall be given internal choice i.e. the students shall attempt one question from each unit.

(iv) All questions/units will carry equal marks.

(v) Private candidates and the students of the University School of Open Learning (USOL) will not be allowed to take this subject.

UNIT-I 12 Marks

Entire syllabus given in the Units II to V will be covered to set six short answer type questions in first question/unit of the question paper which is compulsory.

UNIT-II 12 Marks

Respiratory System :

- Meaning of Respiration, types of Respiration, Organs of the Respiratory System.
- Functions of the Respiratory System. Vital capacity and its measurement.
- Mechanism and Neural Control of Respiration.

Digestive System :

- Meaning, Importance and Organs of Digestive System.
- Functions, processes, mechanism and nerve regulators of Digestive System.

UNIT-III 12 Marks

Circulatory System :

- Meaning of the circulatory system.
- Heart, its structure, functions and control of the heart rate.
- Various types of blood vessels and their functions.
- Cardiac Cycle.

Blood :

- Meaning, functions and composition of blood.
- Maintenance of blood supply.
- Blood groups and their importance.
- Blood Clotting.

UNIT-IV 12 Marks

Communicable Diseases :

Meaning of a communicable disease. Communicable diseases such as HIV/AIDS, Viral Hepatitis—A, B & C and Tetanus, their modes of transmission and methods of prevention.

Yoga:

- Meaning and aim of Yoga.
- Meaning, Principles and Importance of Asanas.

- Meditative poses (Padamasna, Vajrasana, and Sukhasana), their technique, precautions and effects/advantages.

- Cultural poses (Savasna, Halasana, Bhujangasana Sarvangasana, and Dhanurasana), their technique, precautions and effects/advantages.

Basics of Tennis :

- History of the game.
- Basic fundamentals.
- Equipment and specifications.
- Marking / layout of court.
- Rules and regulations (number of players, duration of game, number of officials required and general rules of play).
- Major tournaments and Arjuna awardees of the game.

UNIT-V 12 Marks

Sports Injuries :

- Basis of Sports Injuries.
- Common Sports Injuries, such as sprains, strains, fracture, dislocation, abrasions, contusion, bruise, tennis elbow. Their causes, preventive and remedial measures.
- Treatment and care (RICE) of sports injuries.

Disability and Rehabilitation :

- Meaning, types, causes and preventive measures of disability.
- Problems of the disabled. Physical Activity and health for disabled.
- Meaning and scope of Rehabilitation.

References :

1. Chaurasia, B.D. : *"Handbook of General Anatomy",* CBS Publishers and Distributors, First Edition, 1981.
2. Dandiya, P.C., Jafer, Z.Y. K. and Jafer Afifa: *"Health Education and Community Pharmacy",* Second Edition, 1996, Reprinted in 1998, Vallabh Prakashan, Pitampura, New Delhi.
3. Govt. of India : *"Occupational Diseases : A Guide to Recognition and Notification",* Chief Adviser Factories, Ministry of Labour and Employment, New Delhi, 1955.
4. Park, J.E. & Park, K. : *"Text Book of Preventive and Social Medicine",* 10th Edition, 1985, Banarsi Dass Bhanot, Jabalpur, India.
5. Parrot, J.W. : *"Anatomy and Physiology for Physical Education Teachers",* Lend : Edward Arnold Ltd., 1983.
6. Rao, M.N. and Lundgren, N.P.V. : *"A Review of Occupational Health Research in India",* ICMR, New Delhi, 1955.
7. Rose & Wilson : *"Foundations of Anatomy and Physiology",* 5th Edition, 1981, Reprinted in 1985.

8. Winter Griffith, H. : *"Complete Guide to Sports Injuries",* First Indian Edition, 1989, Metropolitan Book Co. Pvt. Ltd., New Delhi.

9. Sampath, K. & Uma Maheshwar, B. : *"Human Anatomy & Physiology",* First Edition, 1999- 2000, Birla Publications, Delhi.

10. Singh, Ajmer et. al. : *"Modern Text Book of Physical Education, Health and Sports",* Kalyani Publishers, Ludhiana, 2000.

11. Smith, Olive and F. Guthrie : *"Rehabilitation and Remedial Exercise",* Bailliere, Tindall and Cox Limited, London, 1949.

12. Sri Anand : *"The Complete Book of Yoga : Harmony of Body and Minds",* Orient Paper Back, New Delhi, 1980.

13. Staford, T. & Ellen Davis Kelly : *"Preventive and Corrective Physical Education",* Third Edition, 1958, Ronald Press Co., New York.

14. Swami Kuvalyananda : *"Kaivalyadhama",* Lonavala, Bombay, 1933.

15. Vijayendra Pratap : *"A Teacher's Guide for Beginning Yoga",* First Edition, 1987, Sky Foundations, Philadelphia, Pennsylavania, U.S.A.

16. World Health Organisation : *"Health Hazards of the Human Environment",* Geneva,1962.

PRACTICAL

Max. Marks	:	70
Practical	:	60
Internal Assessment	:	10

ATHLETICS

Jumps (Long jump or triple jump or high jump) :

(a) Approach run (b) Take off (c) Flight (d) Landing (e) Brief information of various styles, and practice of any one style depending upon the facilities available (f) Rules and regulations of the jump taught.

GAMES

(Basketball or football and any other one game of the choice of the student)

Basketball :

(a) Measurements (Basketball ground).

(b) Number of players and officials.

(c) Rules and Regulations of the game.

(d) Fundamental and basic skills.

OR

Football:

(a) Measurements (football ground, goal post, ball and flag post).

(b) Number of players and officials.

(c) Fundamental Skills : Types of kicks (low drive, lofted kick, half and full volley and put kick). Dribbling and running with the ball, trapping, receiving and ball control techniques, pass and passing techniques (heading techniques, kicking and shooting practice, goal keeping technique).

Physical Fitness Tests :

More emphasis shall be given on physical fitness with regard to specific physical fitness components (Endurance and Strength).

Test 1 : ENDURANCE : 9/12 min. run and walk test.

Test 2 : STRENGTH : Standing broad jump test.

Division of Practical Marks : Marks for each activity shall be divided as under :

Athletics 15 marks, games 15 marks, participation and achievement in sports/games 10 marks, physical fitness 10 marks, viva voce 10 marks and internal assessment 10 marks based on overall performance of a student during the current academic session which will be assessed by the teacher concerned.

Note :

1. Pole-vault, Hammer Throw, Hurdles, Relay races and steeple chase race are not included in the practical syllabus/course due to the fact that these events are highly technical. Moreover, in the absence of proper facilities required for the events mentioned above may prove to be injurious/fatal to the students.

2. 12 periods per week (6 periods each for theory and practicals) shall be allotted to a class.

3. The theory and practical class shall consist of 60-80 students and 30-40 students respectively.

4. The theory (Paper A and Paper B) papers shall consist of 65 marks each and practical papers shall consist of 70 marks.

5. As per the Panjab University Calendar, Chapter XX (page 298) Volume III, 1996, the maximum teaching work load for a Lecturer in Physical Education for B.A. Pass Course is 24 periods per week, which includes theory as well as practicals.

6. The choice of games by the students shall be confined to the games approved by the Association of Indian Universities.

7. A student is required to prepare a practical note book of athletic events and games given in the syllabus.

EDUCATION

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A : FOUNDATIONS OF EDUCATION

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

GENERAL INSTRUCTIONS FOR THE PAPER-SETTER :

The question paper will consist of five units : I, II, III, IV and V. Unit I, II, III and IV will have two questions from the respective unit of the syllabus and will carry 18 marks each. Unit V will consist of **eight** short answer type questions which will cover the entire syllabus uniformly and will carry 18 marks. The students are required to attempt 6 short answer type questions out of 8 in unit V which will be compulsory. The question paper should preferably carry internal division of marks for all the sub-questions of one main question. Preferably set the words limit for answer (300-350 words for units I, II, III, IV and 75 words for each short answer type question in Unit V).

INSTRUCTIONS FOR THE CANDIDATE :

The students will be required to attempt one question each from Units I, II, III and IV. Unit V will be compulsory. Students are required to attempt 6 short answer type questions out of 8 in Unit V. The words limit will be 300-350 words for Unit I, II, III and IV; and 75 words for each short answer type in Unit V.

Objectives :

1. To acquaint the students with major foundations underlying Education.
2. To enable the students to understand the concept of Philosophy of Education.
3. To enable the students to understand the concept of Educational Sociology and Educational Technology.
4. To enable the students to understand major Philosophies and their Educational implications.

COURSE CONTENTS :

UNIT-I

Philosophical, Sociological, and Technological Foundations of Education—Concept and their Role in Education.

UNIT-II

Philosophy of Education – Definition, Nature and Scope.

Difference between Educational Philosophy and Philosophy of Education.

UNIT-III

Educational Sociology – Definition, Nature and Scope.

Information and Communication Technology—Role in Education.

UNIT-IV

Major Philosophies of the World—Idealism, Naturalism and Pragmatism—Main Features and their Contribution to Education.

Books Recommended :

1. Aggarwal, J.C. : *Theory and Principles of Education,* Vikas Publishing House Pvt. Ltd., New Delhi, 2003.
2. Bhatia, K.K. and Narang, C.L. : *Philosophical and Sociological Bases of Education,* Tondon Publications, Ludhiana, 2002.
3. Sataya, R.N. and Shaida, B.D. : *Development of Educational Thought and Practice,* Dhanpat Rai Publishing Company (P.), New Delhi, 1996.
4. Taneja, V.R. : *Foundations of Education,* Mohindra Capital Publishers, Panchkula, 1998.
5. Singh, K. : *Vidiyak Manovigian,* Gujral Book Depot, Patiala, 1999.

Paper-B : PSYCHOLOGICAL FOUNDATIONS OF EDUCATION

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

GENERAL INSTRUCTIONS FOR THE PAPER-SETTER :

The question paper will consist of five units : I, II, III, IV and V. Unit I, II, III and IV will have two questions from the respective unit of the syllabus and will carry 18 marks each. Unit V will consist of **eight** short answer type questions which will cover the entire syllabus uniformly and will carry 18 marks. The students are required to attempt 6 short answer type questions out of 8 in Unit V which will be compulsory. The question paper should preferably carry internal division of marks for all the sub-questions of one main question. Preferably set the words limit for answer (300-350 words for Units I, II, III, IV and 75 words for each short answer question in Unit V).

INSTRUCTIONS FOR THE CANDIDATE :

The students will be required to attempt one question each from units I, II, III and IV. Unit V will be compulsory. Students are required to attempt 6 short answer type questions out of 8 in Unit V. The words limit will be 300-350 words for Unit I, II, III and IV and 75 words for short answer type question in Unit V.

Objectives :

1. To enable the students to understand the perspective roles of heredity and environment in educational development of a child.

2. To enable the students to understand the concept of personality and its place in Education.

3. To make the students familiar with the concept of Special Education; the special needs of the Gifted, Backward and Delinquent Children and their Educational implications.

4. To make the students familiar with the concepts of Emotions, Stress and Stress management among adolescents.

COURSE CONTENTS :

UNIT-I

Difference between Psychology and Educational Psychology, Significance of Educational Psychology.

Heredity and Environment – Concepts and their role in Educational development.

UNIT-II

Personality : Concept, Assessment and its Educational Implications.

UNIT-III

Special Education : Meaning and Concept.

Types of Exceptional Children with special reference to the Gifted, Backward and Delinquents (Their Characteristics and Educational Implications).

UNIT-IV

Emotions and Stress – Concept, Types of Stress (Physical, Psychological and Social), Stress Management with Special Reference to Adolescents.

Books Recommended :

1. Batra, Promod : *Simple Ways to Manage Stress,* Think Inc., Delhi, 1997.

2. Carewell, M., Clark, L. & Meldrum, C. : *Psychology,* Harper Collins Publishers Ltd., London, 2000.

3. Clare, Sylvia : *Emotions –Releasing Your Child's Potential,* UBS Publishers and Distributors, New Delhi, 2003.

4. Chauhan, S.S. : *Advanced Educational Psychology,* Vikas Publishing House Pvt. Ltd., New Delhi, 2003.

5. Dhaliwal, A.S. : *Vidyak Mnovigian,* Punjabi University, Patiala, 2001.

6. Dutt, Sunil & Sodhi, G.S. : *Teaching Learning – A Process Approach,* Samir Publications, Chandigarh, 1995.

7. Goleman Daniel : *Emotional Intelligence,* Baanam Books, New York, 1995.

8. Kirk, Semual : *Educating the Exceptional Child,* I.B.H. Publishing Co., Calcutta, 1992.

9. Mangal, S.K. : *Advanced Educational Psychology,* Prentice Hall of India Ltd., New Delhi, 2002.

10. Panda, K.C. : *Education of Exceptional Children,* Vikas Publishing House Pvt. Ltd., New Delhi, 2003.

ADULT EDUCATION

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Max. Marks		: 200
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours
Practical	:	100 Marks

GENERAL INSTRUCTIONS FOR THE PAPER-SETTER :
The question paper will consist of five units : I, II, III, IV and V. Units I, II, III and IV will have two questions from the respective units of the syllabus and will carry 18 marks each. Unit V will consist of **8** short answer type questions which will cover the entire syllabus uniformly and will carry 18 marks in all. Each short question will carry 3 marks.

GENERAL INSTRUCTIONS FOR THE CANDIDATE :
The students will be required to attempt at least one question from each unit I, II, III and IV. The students are required to attempt **6** short questions out of **8** in Unit V which will be **compulsory**. The question paper should preferably carry internal division of marks for all the sub-questions of one main question.

OBJECTIVES OF THE COURSE :

The main objectives of the paper are :

1. To expose students to the psychology of adult learner and the strategies used to motivate them.
2. To provide an overview of concept and strategies of Total Literacy Campaign.
3. To provide the basis of suitability of teaching-learning materials and audio visuals material used for neo-literates.
4. To acquaint students with various agencies of adult education.
5. To expose students with the monitoring methods, procedures and levels of monitoring.
6. To acquaint students with evaluation of Adult Education programme.
7. To acquaint students with adult education and its linkage with development.

THEORY :

Unit-I

(i) Psychology of Adult Learners and Motivational Strategies.

(ii) Total Literacy Campaign : Concept and Strategies.

Unit-II

(i) Suitability of Teaching – Learning Materials and Audio Visuals; Material for Neoliterates.

(ii) Suitability of Training Material for Preraks and Nodal Preraks.

Unit-III

(i) Agencies of Adult Education : Government and Non-Government.

(ii) Monitoring of Adult Education Program : The Paradigm Context : General Considerations, Need for Monitoring, Components in Monitoring Methods, Procedures and Levels of Monitoring.

Unit-IV

(i) Evaluation of Adult Education Program : Meaning and Scope, Its Purpose, Preparation of Programs for Evaluation.

(ii) Adult Education and its linkage with Development.

PRACTICAL/FIELD WORK :

Max. Marks : 100
External : 90 Marks
Internal : 10 Marks

Participation of the students in the following :

(i) Monitoring of 5 Adult Education Centres.

(ii) Evaluation of 5 Adult Learners.

Books Recommended :

1. Singh, Madan & Mahanty : *Companion to Adult Education, National Literacy Mission,* Director of Adult Education, Ministry of Human Resources Development, Dept. of Adult Education, New Delhi, 1999.

2. Alan Rogers : *Teaching Adults,* Sterling Publishers Pvt. Ltd., New Delhi, 1989.

3. Mohanty, J. : *Adult & Non-Formal Education,* Deep Publications, New Delhi, 1993.

4. Chopra, R. & Grover, R. : *Adult Education,* the Indian Publications, Ambala Cantt., 1982.

5. Melcoln Tight : *Education for Adults, Vol. I,* Groom Helm, London, 1983.

6. Ahmad Nasim : *Understanding the Adult Learner : Indian Perspective Literacy Mission,* 1994, Vol. XVII.

7. Annonymous : *How to Arouse Interest of Adult Learners and Sustain it over a Period of Time, Literacy Mission,* 1988, Vol. XI, No. 10.

8. Mali, M.G. : *Adult Education in India,* Deep and Deep Publications, New Delhi, 1984.

9. Nimbalkar, M.R. : *Adult Education and its Evaluation System,* Mittal Publications, Delhi, 1987.

10. Goyal, G., and Meera Kakar : *Motivation : An Integral Part of Adult Education : Indian Journal of Adult Education,* 1987, Vol. 48, No. 2, p. 31.

11. Kanwal, R. : Motivation towards Adult Education, Criterion Publications, New Delhi, 1990.

12. Vashist, S. R. : *Adult Education : Issues, Problems and Prospects,* Pointer Publishers & Distributors, 1997.

13. Reddy, Subba, C.N.V. : *Linkages with Developmental Agencies, in the book 'Research in Adult Education',* Edited by Bhatia, S.C. and Patil, B.R., Indian Adult Education Association, New Delhi, 1983.

MUSIC (Vocal)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

General Instructions

1. In case of the private candidates, there will be no internal assessment and the marks obtained in the external assessment of the practical examination shall be proportionately increased.
2. There should not be more than eight students in a batch for practical examination.
3. Harmonium will not be allowed as accompaniment in vocal music, but harmonium can be used while singing Alankars.
4. The candidate can take vocal music along with instrumental music.
5. The candidate can also take instrumental music with tabla.
6. While sending the syllabus to paper-setter in theory, the syllabus prescribed for the practical paper shall also be sent.
7. In all, nine questions will be set. The question paper will be divided into five units. Four units will contain two questions each. The candidates are required to attempt Five questions selecting at least one question from each unit. The ninth question from unit V based on the Notation shall be **compulsory.**
8. Separate Practical paper shall be set for each session from the syllabus prescribed for Practical Paper-B.
9. There would be upto ten students in one section in practical class.

Paper-A : THEORY (3 Hours duration)			:	90 marks
Paper-B : PRACTICAL (20 minutes duration)			:	90 marks
(i)	Viva	.	:	70 marks
(ii)	Gayan		:	10 marks
(iii)	Tabla		:	10 marks
Internal Assessment (Theory + Practical)		(10 +10)	:	20 marks
			Total :	**200 marks**

Paper-A : THEORY (Duration 45 Minutes, 02 Theory periods per week)

Unit–I

1.1 Historical developments in North Indian Music during medieval period (13th to 17th A.D.).

1.2 Brief knowledge of Gram, Murchhana.

1.3 Elementary knowledge of Raga-Ragini Vergikaran.

Unit–II

2.1 Gharana, its importance and contribution towards Khyal Gayan.

2.2 Importance of Tanpura and Sahayak Nada.

2.3 Knowledge of Alap, Tana and its various forms.

Unit-III

3.1 Brief life sketches of the great masters of Music and their contributions.

(i) Ustad Faiyaz Khan Sahib of Agra.

(ii) Sh. Krishan Rao Shankar Pandit.

(iii) Ustad Amir Khan Sahib.

(iv) Sh. Sohan Singh Ji.

3.2 Explain the following :
Upaj, Mukhda, Bol-Alap, Khatka, Mend, Bol-Baant, Gamak.

3.3 Knowledge of the notation system of Pt. V.N. Bhatkhande & Pt. V.D. Paluskar.

Unit-IV

4.1 Description of the Ragas and Talas prescribed in the course :
Ragas—Malkauns, Kedar, Bhairav, Bhimplasi and Bihag.

4.2 Knowledge of writing Talas in single and double.

4.3 Knowledge of the following Non-detailed Ragas :
Chandrakauns, Kalingda, Kamod and Dhani.

Unit-V

5.1 Ability to write the Vilambit and Drut Khayal of ragas prescribed in the course :
Malkauns, Kedar, Bhairav, Bhimplasi and Bihag.

Paper-B : PRACTICAL (Duration 45 Minutes, 06 practical periods per week)

1. One Drut Khayal in each of the following Ragas with Alaps and Tanas :
Malkauns, Kedar, Bhairav, Bhimplasi and Bihag.

2. One Vilambit Khayal in any of the detailed Ragas prescribed in the course with extempore Alaps and Tanas.

3. One Dhrupad in Thah and Dugan layakaries in any of the detailed Ragas prescribed in the course.
4. One Lakshan-Geet in any of the prescribed ragas.
5. Ability to play Ektal and Chautal on Tabla.
6. Ability to recite the following talas in Thah and Dugun layakaries by hand :
   Tilwada, Dhamar, Teental and Roopak.
7. Ability to recognize the prescribed Ragas sung by the Examiner.
8. Knowledge of the following Non-Detailed Ragas : Ability to sing their Aroh, Avroh and Pakads *with Tanpura;* Chandrakauns, Kamod, Kalingda and Dhani.
9. Ability to sing one Drut Khayal of your course with Harmonium.

Books Recommended :

1.	*Sangeet Visharad*	:	Sangeet Karyalaya Hathras (U.P.).
2.	*Rag Parichaya Part II*	:	H.C. Srivastav, Sangeet Sadan Prakashan, 134, South Malaka, Allahabad.
3.	*Sangeet Kaumudi Part IV (Punjabi)*	:	S.S. Kareer, published by Punjabi University, Patiala.
4.	*Hamare Sangeet Ratna*	:	Sangeet Karyalaya Hathras (U.P.).
5.	*Sangeet Sar Part II*	:	Mrs. Veena Mankaran, Raj Publishers (Regd.) Adda Tanda, Jalandhar City.
6.	*Kramik Pustak Malika Part I to IV*	:	Pt. V.N. Bhatkhande, Sangeet Karyalaya, Hathras.
7.	*Rag–Vigyan Part I to VII*	:	Pt. V. Patwardhan.

MUSIC (Instrumental)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

General Instructions

1. In case of the private candidates, there will be no internal assessment and the marks obtained in the external assessment of the practical examination shall be proportionately increased.
2. There should not be more than eight students in a batch for practical examination.
3. Harmonium can be used while singing Alankaras.
4. In Instrumental Music, the candidates have the option to take any one of the following instruments : Sitar, Sarangi, Veena, Sarod, Dilruba, Violin, Guitar, Bansuri, Shehnai.
5. The candidate can take vocal music along with instrumental music.
6. The candidate can also take instrumental music with tabla.
7. While sending the syllabus to paper-setter in theory, the syllabus prescribed for the practical paper shall also be sent.
8. In all, nine questions will be set. The question paper will be divided into five units. Four units will contain two questions each. The candidates are required to attempt Five questions selecting at least one question from each unit. The ninth question from Unit V based on the Notation shall be **compulsory**.
9. Separate Practical paper shall be set for each session from the syllabus prescribed for Practical Paper-B.
10. There would be upto 10 students in one section in practical class.

Paper A : THEORY (3 Hours duration) **(Duration 45 minutes 06 practical + 02 Theory per week)**		:	90 marks
Paper B : PRACTICAL (20 Minutes duration)		:	90 marks
(i) Viva		:	70 marks
(ii) Gayan		:	10 marks
(iii) Tabla		:	10 marks
Internal Assessment (Theory + Practical)	(10 + 10)	:	20 marks
		Total :	**200 marks**

Paper-A : THEORY (Duration 45 minutes 02 Theory periods per week)

Unit-I

1. Historical developments of Indian music during medieval period i.e. 13th to 17th Century.
2. Knowledge of Pt. V.D. Paluskar notation system.
3. A study of Shudh raga, Chhayalag raga, Sankirna raga.

Unit-II

1. Definitions and explanations of the following musical terms :
   Various Bols of Mizrab, Alap, Jod, Kan, Andolan, Chal-Achal Thata, Kampan, Meend, Ghaseet, Zamzama, Krintan, Gamak.
2. Meaning and definition of Gharana and its importance in Indian Classical Music.
3. Classification of Indian Musical Instruments.

Unit-III

1. The life sketches and contributions of the following musicians :
   (i) Ustad Inayat Khan.
   (ii) Ustad Vilayat Khan.
   (iii) Pt. Lal Mani Mishra.
   (iv) Ustad Abdul Halim Zaafar Khan.
2. Knowledge of Avirbhava-Tirobhava and Alpatva—Bahutava with example.

Unit-IV

1. Description of the prescribed ragas and talas :
   Ragas : Bhairav, Bihag, Durga, Malkauns.
   Talas : Ektal, Deepchandi, Tilwara, Tivra.
2. Description of the following Non-detailed ragas :
   Vrindavani - Sarang, Kalingada, Chandrakauns.

Unit-V

Knowledge of writing notation of Vilambit gat and Drut gat in the prescribed Ragas :
Bhairav, Bihag, Durga, Malkauns.

Paper-B : PRACTICAL (Duration 45 minutes, 06 practical period per week)

1. One Razakhani (Drut) Gat with Toras and Jhalas in each of the following ragas :
   Bhairav, Bihag, Durga, Malkauns.
2. One Maseetkhani (Vilambit) Gat with Alaps & Toras in any of the prescribed detailed ragas.
3. A knowledge of the following non-detailed ragas with Aroh, Avroh & Pakad :
   Vrindavani - Sarang, Kalingada, Chandrakauns.

4. One Gat in Madhaya laya Ektal.
5. Use of one Swara Meend and Kan.
6. Ability to play Ektal and Teental on Tabla.
7. Ability to sing five Alankaras of Shudh Swaras with the help of Harmonium.
8. Ability to demonstrate the following talas by hands in Ekgun and Dugun Layakaries :
   Talas : Ektala, Deepchandi, Tilwara, Tivra.

Books Recommended :

1.	*Rag Prichaya Part III and IV*	:	H.C. Srivastava, Sangeet Sadan Prakashan, 134, South Malaka, Allahabad.
2.	*Sangeet Bodh*	:	S.S. Paranjape, Madhya Pradesh Hindi Granth Academy, Bhopal.
3.	*Hamare Sangeet Ratan*	:	Sangeet Karyalaya, Hathras.
4.	*Sangeet Sar-Part II*	:	Mrs. Veena Mankaran, Raj Publishers (Regd.), Adda Tanda, Jalandhar City.
5.	*Sangeet Shastra Vigyan*	:	Panna Lal Madan, Abhishek Publication, Chandigarh.
6.	*Rag-Vigyan*	:	Pt. V. Patwardhan.

TABLA (Instrumental Music)

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

General Instructions

1. In the case of private candidates, there will be no internal assessment and the marks obtained in the external assessment of the practical examination shall be proportionately increased.

2. (a) In all, nine questions will be set from the whole syllabus.

   (b) The question paper will be divided into five units. First four units will contain two questions each, out of which candidates are required to attempt one question from each unit.

   (c) Unit fifth will be **compulsory** which will be based on Notation.

   (d) The candidates are required to attempt five questions.

Paper-A : THEORY (3 Hours duration) : 90 marks

Paper-B : PRACTICAL (20 minutes duration) : 90 marks

(i)	Viva	:	60 marks
(ii)	Harmonium	:	10 marks
(iii)	Tabla	:	10 marks
(iv)	Padhant on Hand	:	10 marks

Internal Assessment (Theory & Practical) (10 + 10) : 20 marks

Total : 200 marks

Paper-A : THEORY

UNIT-I

(a) Origin and Evolution of Tala.

(b) Brief study of Ten Pranas of Tala.

(c) Definition of Tisra, Chatasra, Khand, Mishra and Sankeern jaties.

UNIT-II

(a) Definition with examples of the followings :
Uthan, Gat, Rau, Chakradar, Nagma.

(b) Ability to write in notation the Dugun, Tigun and Chaugun of the following talas :
Teentaal, Ektaal, Jhaptal, Sultal, Dadra, Kehrwa.

(c) Explanation of playing techniques of the following :
Dhage, Tirkit, Ghe-Ghe, Kradhatita, Gadigan, Kdan, Ghednag-Kednag.

UNIT-III

(a) Knowledge of Uttar Bhartiya Taal system.

(b) Knowledge of Karnatak taal system.

(c) Introduction of V.D. Palaskar Tala Notation system.

UNIT-IV

(a) Importance of Gharanas of Tabla-Vadan.

(b) History of Playing Techniques of Lucknow, Farrukhabad Delhi & Ajrada Gharana.

(c) Life sketches and contribution of following :

- Ustad Abid Hussain.
- Ustad Ahmedjan Thirkuwa.
- Ustad Amir Hussain Khan.

UNIT-V

Ability to write in notation of compositions and layakaries in following Talas :
Teentaal, Jhaptal, Roopak, Deepchandi, Tivra.

Paper-B : PRACTICAL

(a) Tala prescribed Rupak, Tivra, Jhaptal, Sultal.
Deepchandi and Jhumra Talas.

(i) Proper Badhat of the following Talas by hand on Tabla or Pakhawaj :
Rupak, Tivra, Jhaptal, Sultal.

(ii) Two laggies in Dadra and two in Keharwa.

(iii) Theka of Jhumra Tal in Vilambit Laya.

(iv) Sultal-Tukras, Parans, Bedam Tehais, Damdar Tehai, Chakradar Paran and Rela.
Jhaptal : One Qaida, One Peshkar, One Gat and Rela.
Rupak : Peshkar, Gat, Qaida, Paran and Chakardar Paran.

(v) Practical knowledge of playing Deepchandi with Thumri.

(vi) Tivra : Playing Theka of Tivra in Ekgun, Dugun and Chaugun layakaries.

(b) Practice of playing the above mentioned Talas with Vocal and Instrumental Music.

(c) Knowledge of Shudh and Vikrit Swaras.

(d) Practice of playing Nagma on Harmonium in Roopak, Sultal and Jhaptal.

(e) Tuning of Tabla.

Books Recommended :

1.	*Taal Prichaya Part I & II*	:	G.C. Shrivastva.
2.	*Taal Martand*	:	B.S. Sharma.
3.	*Bhartiya Talon Ka Shastriya Vivechan*	:	Arun Kumar Sen.
4.	*Bhartiya Sangeet Vadya*	:	L.M. Mishra.
5.	*Hamare Sangeet Ratna*	:	Sangeet Karyalaya Hathras.
6.	*Tabla Vadan – Part-II Publication (Pbi. University, Patiala)*	:	Manmohan Sharma

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INDIAN CLASSICAL DANCE

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

General Instructions

1. In the case of private candidates, there will be no internal assessment and the marks obtained in the external assessment of the practical examination shall be proportionately increased.
2. There should not be more than eight students in a batch for practical examination.
3. In all, nine questions will be set. The question paper will be divided into five units. Four units will contain two questions each. The candidates are required to attempt five questions selecting at least one question from each unit. The ninth question from Unit V based on notation shall be **compulsory.**
4. Accompaniment of Tabla, Harmonium/Sarangi will be allowed.
5. The practical paper shall be set from the syllabus prescribed for Paper-B (Practical).
6. There would be upto 10 students in one section in practical.

Paper-A : THEORY (3 hours duration) **(Duration 45 minutes, 06 practical + 02 Theory per week)**		:	90 marks
Paper-B : PRACTICAL (20 minutes duration)		:	90 marks
(i)	Viva	:	70 marks
(ii)	Harmonium	:	05 marks
(iii)	Tabla	:	05 marks
(iv)	Padhant on Hand	:	10 marks
Internal Assessment (Theory + Practical) (10 +10)		:	20 marks
		Total :	**200 marks**

Paper-A: THEORY (Duration 45 minutes, 02 Theory periods per week)

Unit-I

1. Definition of the following :
   Bharamari, Chari, Astooti, Gati, Paran.
2. Detailed study of Abhinay Darpan Granth.
3. Survey and essential characteristics of Bharta Natyam.

Unit-II

1. Knowledge of Solo and Group dance.
2. Survey of Samyukta Mudra based on Abhinaya Darpan.
3. General study of Gharanas of Kathak dance and their comparison.

Unit-III

1. Biography and the contribution of the following dancers in their respective field of specialization :
   (i) Maharaj Sunder Prasad.
   (ii) Pt. Shambhu Maharaj.
   (iii) Pt. Birju Maharaj.
2. Knowledge of folk dances of Himachal Pradesh.
3. Knowledge of folk dances of Rajasthan.

Unit-IV

1. Brief study of Thumri in dance.
2. Importance of Gayan and Vadan.
3. Guru Shishya Parampara

Unit-V

Knowledge of writing notation of Talas, Amad, Tihai and Lehra in adachautal and Teentaal.

Paper-B : PRACTICAL (Duration 45 minutes 06 practical period per week)

1. Adachautal (Matra-14)

(i)	Simple Tatkar with Dugun and Chaugun Layakaries.		
(ii)	Thaat	:	1
(iii)	Tora	:	4
(iv)	Amad	:	1
(v)	Salami	:	1
(vi)	Paran	:	2
(vii)	Chakardar Paran	:	1

2. Teentaal (Matra-16)

(i)	Two Simple Tatkar		
(ii)	Thaat	:	1
(iii)	Amad	:	1
(iv)	Tora	:	2
(v)	Paran	:	1
(vi)	Chakadar Tora	:	1
(vii)	Kavit	:	1

3. Four Gat Nikas.

4. Padhant of all Tukra, Tora, Paran, Chakardar Paran by hand.

5. Theka of all the Talas in Single, Dugun and Chaugun Layakaries by hand.

6. Ability to play Nagma in Adachautal.

7. Theka of following Talas on Tabla in Ek-gun and Dugun Layakaries :
   (i) Dadra.
   (ii) Keharva.

8. Practical demonstration of the folk dance of Punjab.

Books Recommended :

1. *Nritya Prashan Panjika* : J.N. Pathak
2. *Kathak Nritya Shiksha* : Puru Dadhich.
3. *Nritya Prabhakar Prashanotri* : J.N. Pathak.
4. *Kathak Darpan* : T.R. Azad.
5. *Kathak Shingar* : T.R. Azad.

FINE ARTS

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A : THEORY (History of Art) Max. Marks : 60

Written Paper	:	54 Marks
Internal Assessment	:	06 Marks
Total Marks	:	**60 Marks**

Paper–B : PRACTICAL

This paper consists of three sections :

		Max. Marks	*Max. Time*
1.	Landscape Painting (on the spot)	30	5 hours
2.	Life Sketching	30	5 hours
3.	Poster & Book Cover Design or Clay Modelling	30	5 hours
SESSIONAL MARKS : (Based on work related equally to 3 sections).		50	
	Total	**140**	

Note : Minimum of nine hours' teaching per week be assigned and out of nine hours, six hours be ear-marked for practical classes and three hours for theory classes (per week).

THEORY

Paper-A : History of Art Max. Marks: 54
Time : 3 Hrs.

NOTE : -

1. The paper carries 54 marks.
2. The paper-setter is required to set **9 questions** in all. **The candidate is to attempt 5 questions as per the instructions given in the question paper.**
3. The first question shall be of short answer type containing 9 questions spread over the whole syllabus. Each question is to be answered in about 25 to 30 words. It shall carry 18 marks and shall be a **Compulsory** question.
4. 8 questions are to be set from the entire syllabus consisting of 4 units. Two questions will be set from each unit and the candidates shall be given internal choice i.e. a candidate shall attempt one question from each unit. So in all, the candidate shall attempt 4 questions in all, out of 8 questions. **Each question** would be of **9** marks.

Objectives :

The aim of the paper is to introduce to the students various schools, styles and phases of the developments in painting and sculpture in India and the West. The emphasis will be to make them aware of the different terms, concepts, forms and subject matter of these works.

Unit-I : History of Indian Painting

- Decorative Designs and Paintings of the 1439 Kalpasutra Manuscript.
- Manuscript Paintings of Chaurapanchishika and Nimat Nama.
- Mughal Paintings of Hamza-Nama, South Kensington Akbar Nama, Jahangiri-Portraits and illustrations of birds and animals.
- Deccani Paintings – Tarif-i-Hussain-Shahi from Ahmednagar, Najam-ul-Ulum from Bijapur.

Unit-II : History of Indian Sculpture

- Gandhara– Fasting Buddha and Mahaparinirvana.
- Mathura–Portraits of the Kushana Kings.
- Sarnath–Seated Buddha.
- Sultanganj - Bronze Buddha.
- Deogarh–Vishnu Anantashayana.
- Ellora –Ravana Shaking Mount Kailasha.
- Mahabalipuram–Descent of Ganges (Kiratarjuniyam panel).

Unit-III : History of Western Art

- Early Renaissance: Masaccio-Expulsion, Holy Trinity.
- High Renaissance: Leonardo Da Vinci – Last Supper; Michelangelo— Creation of Adam from the Sistine ceiling, Pieta.
- Northern Renaissance: Durer – Knight, Death and the Devil.
- Baroque: Rubens – Garden of Love.
- Neo-Classicism and Romanticism: David – The Death of Socrates; Constable – The Haywain.
- Realism: Courbet – Artist's Studio.

Unit-IV : Definition of Key Terms and General Concepts

- Art and Craft, Design, Mosaic, Tribal Art, Folk Art, Composition – Volume, Texture, Space Unity, Harmony, Rhythm, Proportion, Piece-mould, Waste Mould, Six Limbs of Indian Painting.

Pedagogy :

The students are expected to familiarize themselves with the art forms as seen from the books, slides and related films.

Suggested Readings

1.	Kramrisch, Stella	:	*Survey of Painting in the Deccan,* Oriental Books Corporation, New Delhi, 1983.
2.	Majumdar, R.C. (editor)	:	*The History and Culture of Indian People,* Vol. I, II and III (Sculpture and Painting Sections only) Bhartiya Vidya Bhawan, Bombay, 1988, 1990, 1996.
3.	Rowland, Benjamin	:	*The Art and Architecture of India,* Penguin Books, Great Britain, 1959.
4.	Saraswati, S.K.	:	*A Survey of Indian Sculpture,* Munshiram Manoharlal Publishers Pvt. Ltd., New Delhi, 1975.
5.	Ray, N.R.	:	*An Approach to Indian Art,* Publication Bureau, Panjab University, Chandigarh, 1974.
6.	Barrett, D. and Gray, B.	:	*Painting of India,* The World Publishing Co., Ohio, 1963.
7.	Brown, Percy	:	*Indian Painting under the Mughals,* Cosmo Publications, New Delhi, 1981.
8.	Aggarwala,V.S.	:	*Heritage of Indian Art,* Publications Division, Ministry of Information & Broadcasting, Govt. of India, New Delhi, 1976.
9.	Aggarwala,V.S.	:	*Indian Art* (English), Varanasi, 1965.
10.	Aggarwala,V.S.	:	*Bhartiya Kala* (Hindi), Prithvi Prakashan, 1977.
11.	Read, Herbert	:	*Meaning of Art,* Faber & Faber, London, 1972.
12.	Janson, H.W.	:	*History of Art,* Thames & Hudson, London, 2001.
13.	Gardener, Helen	:	*Art Through the Ages,* Harcourt Brace & Co., U.S.A., 1991.
14.	Gombrich, E.H.	:	*The Story of Art,* Phaidon Press Limited, New York, 1995.

Paper–B : PRACTICAL

This paper consists of three sections :

		Max. Marks	*Max. Time*
1.	Landscape Painting (on the spot)	30	5 hours
2.	Life Sketching	30	5 hours
3.	Poster & Book Cover Design or Clay Modelling	30	5 hours
	Total :	**90**	

Note : The same paper setter will be requested to set papers in all the three options in Section III.

SECTION-I

LANDSCAPE PAINTING (on the spot)

LANDSCAPE PAINTING : To study relationship of objects, their arrangements in the foreground, middle distance and distance, texture, relative size of masses, tones and colours, use of linear and aerial perspective.

Medium	:	Oil, Water or Pastel colours.
Size	:	½ Imperial.

SECTION-II

LIFE SKETCHING

LIFE SKETCHING : From live Model or Cast-Monochrome in any medium, ½ Imperial Sheet.
Emphasis should be on structure, proportion, foreshortening, textural values, posture and individuality of the model.

SECTION-III

POSTER & BOOK COVER DESIGN OR CLAY MODELLING

POSTER :

Poster design on different themes.

Understanding of colour harmony and contrast.

Imparting understanding of printing purposes.

BOOK COVER DESIGN :

Concept of Lettering : Block lettering, Roman lettering, Script lettering and free hand brush lettering. To design book cover with illustration and title, author's name etc. in three colours (excluding the background colour)

Medium	:	Pen & Ink, Poster Colours.
Size	:	25 cms. × 20 cms.

OR

CLAY MODELLING :

To render animal or human forms in clay in relief. Minimum number of three forms to be composed. Size of slab minimum 30 cms × 25 cms. Creative Pottery—Creative Pot making in clay—any size.

SESSIONAL MARKS : 50 (based on work related equally to 3 sections).

Sessional marks will be given on the basis of work done during the session in all the 3 sections. At least five works will be submitted in each section. Sessional marks shall be given by external and internal examiners jointly. In case of difference of opinion, marking may be done separately by each examiner giving marks out of 50 % of the aggregate of the Sessional Marks.

HISTORY OF ART

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A : HISTORY OF INDIAN PAINTING (from ca. 9th century to ca. 1800 A.D.) AND SCULPTURE (from ca. 4th century A.D. to ca. 6th century A.D.)

Max. Marks : 100
Time : 3 Hours

Note :

1. The paper-setter is required to set 9 questions in all. The candidate is to attempt 5 questions as per the instructions given in the question paper.

2. The first question shall be of short answer type containing 14 questions spread over the whole syllabus. Each question is to be answered in about 25 to 30 words. It shall carry 28 marks and shall be a Compulsory question.

3. 8 questions are to be set from the entire syllabus consisting of 4 units. Two questions will be set from each unit and the candidates shall be given internal choice i.e. a candidate shall attempt one question from each unit. So in all, the candidate shall attempt 4 questions in all out of 8 questions. Each question would be of 18 marks.

Objectives :

The aim of the paper is to introduce to the students various schools, styles and phases of the developments in painting and sculpture in India. The emphasis will be to make them aware of the different terms, concepts, forms and subject matter of these works.

Unit-I

Development of Miniature Painting :

(a) Eastern Indian.

(b) Western Indian.

Unit-II

Painting under the Mughals :

(a) Akbar.

(b) Jahangir.

(c) Shahjahan.

Unit-III

Indian Miniature Painting :

(a) Rajasthani Schools—Mewar, Bundi, Kishangarh.

(b) Pahari Schools—Basohli, Guler, Kangra.

Unit-IV

Indian Sculpture in the Classical Phase :

Sculpture under the Guptas – Mathura, Sarnath, Deogarh, Ajanta.

Pedagogy :
The students are expected to familiarize themselves with the art forms as seen from the books, slides and related films.

Suggested Readings :

1. Kramrisch, Stella : *Survey of Painting in the Deccan,* Oriental Books Corporation, New Delhi, 1983.
2. Majumdar, R.C. (editor) : *The History and Culture of Indian People, Vols. I, II and III* (Sculpture and Painting Sections only) Bhartiya Vidya Bhawan Bombay, 1996, 1990, 1988.
3. Goswamy, B.N. & E. Fischer : *Pahari Masters : Court Painters of Northern India,* Museum Reitberg, Zurich Switzerland, 1992.
4. Khandalvala, Karl : *Pahari Miniature Painting,* The New Book Co. Pvt. Ltd., Bombay, 1958.
5. Rowland, Benjamin : *The Art and Architecture of India*, Penguin Books, Great Britain, 1959.
6. Saraswati, S.K. : *A Survey of Indian Sculpture,* Munshiram Manoharlal Publishers Pvt. Ltd., New Delhi, 1975.
7. Barrett, D. and Gray, B. : *Painting of India,* The World Publishing Co., Ohio, 1963.
8. Brown, Percy : *Indian Painting under the Mughals,* Cosmo Publications, New Delhi, 1981.
9. Aggarwala,V.S. : *Heritage of Indian Art,* Publications Division, Ministry of Information & Broadcasting, Govt. of India, New Delhi, 1976.
10. Aggarwala,V.S. : *Indian Art* (English), Varanasi, 1965.
11. Aggarwala,V.S. : *Bhartiya Kala* (Hindi), Prithvi Prakashan, 1977.

Paper-B : HISTORY OF EUROPEAN PAINTING AND SCULPTURE (from 1300 A.D. to ca. 1850 A.D.) and THEORY AND PRINCIPLES OF ART APPRECIATION

Max. Marks : 100
Time : 3 Hours

Note :

1. The paper-setter is required to set 9 questions in all. The candidate is to attempt 5 questions as per the instructions given in the question paper.
2. The first question shall be of short answer type containing 14 questions spread over the whole syllabus. Each question is to be answered in about 25 to 30 words. It shall carry 28 marks and shall be a Compulsory question.
3. 8 questions are to be set from the entire syllabus consisting of 4 units. Two questions will be set from each unit and the candidates shall be given internal choice i.e. a candidate shall attempt one question from each unit. So in all, the candidate shall attempt 4 questions in all out of 8 questions. Each question would be of 18 marks.

Objectives :

The aim of the paper is to introduce to the students various schools, styles and phases of the developments in painting and sculpture in the West. The emphasis will be to make them aware of the different terms, concepts, forms and subject matter of these works.

Unit-I : History of European Painting and Sculpture

(a) Early Renaissance–Masaccio, Donatello.

(b) High Renaissance–Lenoardo Da Vinci, Michelangelo, Raphael, Titian.

Unit-II

Baroque Period–Caravaggio, Rubens, Rembrandt, Bernini, Claude Lorrain.

Unit-III

(a) Neo-Classicism and Romanticism–David, Ingres, John Constable, W. Turner.

(b) Realism–Courbet.

Unit-IV : Theory and Principles of Art Appreciation

(a) Definition of the term Miniature.

(b) Six limbs of Indian Painting and their manifestations in actual works.

(c) Indian concept of primary colours and their symbolic meaning.

(d) Indian theory of Rasa, Bhava and Beauty.

Pedagogy :
The students are expected to familiarize themselves with the art forms as seen from the books, slides and related films.

Suggested Readings :

1. Janson, H.W. : *History of Art,* Thames & Hudson, London, 2001.
2. Gardener, Helen : *Art Through the Ages,* Harcourt Brace & Co., U.S.A., 1991.
3. Gombrich, E.H. : *The Story of Art,* Phaidon Press Limited, New York, 1995.
4. Turner, Jane (ed.) : *The Dictionary of Art,* Macmillan Publishers Limited, New York, 1996.
5. Aggarwala,V.S. : *Heritage of Indian Art,* Publications Division, Ministry of Information & Broadcasting, Govt. of India, New Delhi, 1976.
6. Aggarwala,V.S. : *Indian Art* (English), Varanasi, 1965.
7. Aggarwala,V.S. : *Bhartiya Kala* (Hindi), Prithvi Prakashan, 1977.
8. Murray, Peter and Linda Murray : *A Dictionary of Art & Artists,* Penguin Reference Book, Great Britain, 1959.
9. Rowland Benjamin (Jr.) : *Art in East and West,* Beacon Press, Toronto, 1964.
10. Ray, Niharanjan : *An Approach to Indian Art,* Publication Bureau, P.U., Chandigarh, 1974.
11. Majumdar, R.C. (editor) : *The History and Culture of Indian People,* Vol. I, II and III. (Sculpture and Painting sections only) Bhartiya Vidya Bhawan Bombay, 1996, 1990, 1988.

ANCIENT INDIAN HISTORY, CULTURE & ARCHAEOLOGY

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A : HISTORY AND CULTURE OF INDIA FROM 320 A.D. TO 650 A.D.

Objectives :

The main objective of this paper is to make students do a detailed study of the political history and cultural development starting from the Gupta period till the death of Harsha.

INSTRUCTIONS FOR THE PAPER-SETTERS AND CANDIDATES :

1. The theory question paper will be of 90 marks and 10 marks will be for internal assessment.

2. For Private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

   The paper-setter must put note (2) in the question paper.

3. The paper-setter is required to set **9** questions in all. All questions shall carry equal marks. The paper shall be of 3 hrs. duration.

4. The first question shall be of short answer type containing 15 short questions spread over the entire syllabus. The candidate is required to answer any 9 short answer type questions. Each short answer type question shall be of 2 marks **OR** A question on map. The map work shall consist of 12 marks for the map and 06 marks for the explanatory notes.

5. The map question shall have the following topics :
   (a) Extent of Chandragupta II's empire.
   (b) Extent of Harsha's empire.
   (c) Distribution of monuments of the Gupta-Vakataka period.

6. The rest of the paper shall contain 4 Units. The entire syllabus has been divided into 4 Units. The paper-setter shall set 2 questions from each unit and the candidate shall be given internal choice i.e. the candidate shall attempt one question from each unit. Each question shall carry 18 marks.

UNIT-I

1. (a) Political condition of Northern India on the eve of the rise of the Guptas.
   (b) Foundation of the Gupta Empire.

2. The Imperial Gupta rulers.

UNIT-II

1. Decline of the Gupta Empire.
2. The Gupta administration.
3. Cultural and Economic achievements during the Gupta period.

UNIT-III

1. The Vakatakas of the Deccan.
2. The Huna Invasions and their impact.

UNIT-IV

1. The Aulikaras of Daspura, the Maukharis and the Later Guptas, Pallavas & Chalukyas of Badami, Vatapi.
2. The Pushyabhutis of Thanesar : Political and Cultural survey.

Pedagogy of the Course Work :
The students are expected to have an idea of the original sources for the reconstruction of the history of the period. Maps and transparencies should be put to use for better understanding by the students.

Essential Readings :

1. Agrawal, Ashvini : *Rise & Fall of the Imperial Guptas,* Motilal Banarsidass, Delhi, 1989.
2. Altekar, A.S. & Majumdar, R.C. : *The Vakataka-Gupta Age,* rpt., Motilal Banarsidass, Delhi, 1967.
3. Majumdar, R.C. & Dasgupta, K.K. (Eds.) : *The Comprehensive History of India, Vol. III,* Pt. 1, People Publishing House, Delhi 1981, (Chapters Dealing with the Gupta and the Post Gupta History).
4. Majumdar, R.C. : *Ancient India,* rpt., Manohar, Delhi, 2003.
5. Mookerji, R.K. : *The Gupta Empire,* Motilal Banarsidass, Delhi, 1989.
6. Tripathi, R.S. : *History of Kanauj,* Motilal Banarsidass, Delhi, 1964.
7. Tripathi, R.S. : *History of Ancient India,* rpt., Delhi, 1977.

Paper-B : HISTORY AND CULTURE OF INDIA FROM 650 A.D. TO 1200 A.D.

Objectives :

The primary objective of this paper is to acquaint the students about the political development in India after the death of Harsha. A detailed study of the various dynasties ruling in the different parts of our country will be made.

INSTRUCTIONS FOR THE PAPER-SETTERS AND CANDIDATES :

1. The theory question paper will be of 90 marks and 10 marks will be for internal assessment.

2. For Private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

   The paper-setter must put note (2) in the question paper.

3. The paper-setter is required to set **9** questions in all. All questions shall carry equal marks. The paper shall be of 3 hrs. duration.

4. The first question shall be of short answer type containing 15 short questions spread over the entire syllabus. The candidate is required to answer any 9 short answer type questions. Each short answer type question shall be of 2 marks **OR** A question on map. The map work shall consist of 12 marks for the map and 06 marks for the explanatory notes.

5. The map question shall have the following topics :

   (a) Extent of Pratihara Empire.

   (b) Distribution of Temples—Lingaraja, Konark, Jagannath, Kandariya Mahadev, Parsvanath, Lakshman, Osian, Mahabalipuram.

   (c) Important Centres of Education—Nalanda, Ujjain, Pushpagiri, Nagarjunakonda.

6. The rest of the paper shall contain 4 Units. The entire syllabus has been divided into 4 Units. The paper- setter shall set 2 questions from each unit and the candidate shall be given internal choice i.e. the candidate shall attempt one question from each unit. Each question shall carry 18 marks.

UNIT-I :

1. Pratiharas
2. Palas

UNIT-II :

1. Paramaras
2. Chahamanas

UNIT-III :

1. Gahadavalas
2. Chandellas

UNIT-IV :

1. Rashtrakutas
2. Cholas.

Pedagogy of the Course Work :

The students should be given an outline of the original sources that help to reconstruct the political history of the dynasties and the kings. Slides and transparencies should be used in the class room lectures.

Essential Readings :

1. Altekar, A.S. : *Rastrakutas and Their Times,* 2nd ed. Oriental Book Agency, Poona, 1967.
2. Bhatia, Pratipal : *The Paramaras,* Munshiram Manoharlal,New Delhi, 1970.
3. Majumdar, R.C. : *Ancient India,* rpt., Manohar, Delhi, 2003.
4. Majumdar, R.C. : *History & Culture of Indian People,* Vol. IV, *The Age of Imperial Kanauj,* Bhartya Vidya Bhawan, Bombay, 1955 (Relevant Chapters) Vol. V, *The Struggle for Empire,* Bombay 1957 (Relevant Chapters).
5. Puri, B.N. : *The History of the Gurjara-Pratiharas,* Delhi, 1986.
6. Ray, H.C. : *Dynastic History of India,* 2 Vols, Calcutta 1931,1936.
7. Sastri, K.A.N. : *A History of South India, 4th ed.* Oxford University Press, Madras, 1976.
8. Tripathi, R.S. : *History of Kanauj,* Motilal Banarsidass, Delhi, 1964.

DEFENCE & STRATEGIC STUDIES

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Note :

1. There will be two theory papers A and B carrying 70 marks each. The internal assessment will be of 10 marks for each of the two papers. 40 marks are kept for the practical test. Each theory paper will have one compulsory short answer type question containing 15 questions of 2 marks each covering the entire syllabus. The candidates will be required to attempt any 10 short answer type questions. In addition, there will be four sections of the question paper. The candidate will be required to attempt one question from each of these sections carrying 12½ marks. Each theory paper will be of three hours duration. The practical test will be of 1½ hours duration.

2. Practical examination will be compulsory for regular, correspondence and private candidates. The serving armed forces and para-military personnel will be exempted from practical examination and marks secured by such candidates out of 140 will be proportionately raised out of 200.

3. The persons appearing as private candidates (except serving armed forces and para-military personnel) shall have to complete the requirement of attending the practical at the Department of Defence & National Security Studies, Panjab University, Chandigarh.

4. For reappearing candidates, who have not been assessed earlier for internal assessment and practical examination, their marks will be increased proportionately upto 100 marks for each of two papers.

Paper-A: EVOLUTION OF WARFARE IN INDIA

Marks : 70
Time : 3 Hrs.

Objective :

This paper deals with the salient features of Indian warfare with emphasis on Military organization, battle technique and leadership.

SECTION-I

1. Battle of Hydaspes and Alexander's Art of War.
2. Kautilya's Philosophy of War.

SECTION-II

3. Military organizations and battle techniques of Rajputs and Turks with particular reference to Battle of Tarrain, 1192 A.D.
4. Military organizations and battle techniques of Mughals and Afghans with particular reference to First Battle of Panipat, 1526 A.D.

SECTION-III

5. Battle techniques of Southern Muslim Sultans with particular reference to Battle of Talikota, 1565 A.D.

6. Military organization and battle techniques of Marathas under Shivaji.

SECTION-IV

7. Military organization and battle techniques of Sikh Army under Maharaja Ranjit Singh.

8. Warfare under East India Company with particular reference to Battle of Assaye 1803 A.D. and Chilianwala 1849 A.D.

9. Indian Armed Forces 1858 to 1947 –General Evolution and Development (A Broad Perspective).

Books Recommended :

1. Bajwa, F.S., The Military Systems of Sikhs, Motilal Banarsidas Delhi, 1964.
2. Dass, S.T., An Introduction to the Art of War Sagar Publications, Delhi, 1970.
3. Erskine, W., Memories of Zehir-Ed-Din-Muhammed Baber: Emperor of Hindustan, Volume1, Biblio Bazaar, 2010.
4. JFC, Fuller, Generalship of Alexander, The Great, Eyre and Spottiswoode, 1958.
5. Kangle, K.P., Kautilya's Arthshastra, Motilal Banarsidas Publishers Pvt. Ltd., Delhi, 1986.
6. Majumdar, B.K., Military Systems in Ancient India, Firma K.L. Mukopadhayay Publishers, Delhi, 1960.
7. Mallesan, Decisive Battle of India, Read Books Design, 2010.
8. Nath, Rajendra, Military Leadership in India, Vedic Period to Indo-Pak Wars, Lancers Books, California, 1990.
9. Sarkar, J.N., Military History of India, Orient Longmans, Virginia, 1960.
10. Sen., S.N., Military System of the Marathas, Orient Longmans, Michigan, 1958.
11. Sharma, Gautam, Indian Army through the Ages, Allied Publishers, University of Michigan, 1966.
12. Singh, Gurcharan, Battles of Panipat, Army Educational Stores, 1966.
13. William, Irvin, Army of the Indian Mughals, Low Price Publishers, 1994.

Paper B : EVOLUTION OF WARFARE IN OTHER COUNTRIES

Marks : 70
Time : 3 Hours

Objective :

This paper deals with salient features of warfare (from 331 B.C. to 1865 A.D.) and changes in military strategy and tactics and the impact of science and society on warfare.

SECTION-I

1. Military organizations and battle techniques of Macedonians, Persians with particular reference to Battles of Arbella, 331 B.C.
2. Military Organization and battle techniques of Romans and Carthagenians with particular reference to the Battles of Cannae, 216 B.C.
3. Military organizations and battle techniques of Barbarians with particular reference to Battle of Adrianople, 378 A.D.

SECTION-II

4. Military organizations and battle techniques of the English and French with reference to the Battle of Hastings, 1066 A.D.
5. The Mongol Art of War under Changez Khan and his successors.

SECTION-III

6. Industrial revolution and its Impact on weapons, communications and tactics.
7. Elements and Principles of Napoleonic Warfare.

SECTION-IV

8. Naval Warfare with particular reference to the Battle of Trafalgar, 1805.
9. American Civil War (1861-65) –causes and events.

Books Recommended :

1. Barrie and Rackett, The Ancient Art of War Vols. I & II., Barrie and Rockliff.
2. Fuller, J.F.C., A Military History of the Western World : From the Earliest Times to the Battle of Lepanto, Da Capo Press, New York 1987.
3. Fuller, JFC, Armament and History, Da Capo Press, New York, 1988.
4. Fuller, JFC, Conduct of War, Eyre & Spottiswoode (Publisher) Ltd., Great Britain, 1962.
5. Fuller, JFC, Machine Warfare, the Infantry Journal, University of Michigan, 1943.
6. Fuller, JFC, Military History of the Western World, Da Capo Press, New York, 1987.
7. Montgomery, F.M., A History of Warfare, World Publication Co. the University of Michigan, 1968.
8. Montress, L., War through the Ages, Harper & Brothers, Indiana University, 1946 New York, 1987.

Paper-C : PRACTICAL Max. Marks : 40

Time : 1 ½ hours

Note :

1. Practical examination will be compulsory for regular, correspondence and private students.
2. There will be five questions in all and candidates will be required to attempt any three questions of ten marks each.
3. Examiners are required to set the question paper at least half an hour before the examination.
4. There will be 3 hours of teaching per week for practical.
5. For practical classes, the number of students in one group shall not exceed fifteen.
6. Practical exercises should be carried out on drawing sheets with explanatory notes or on computer.

SECTION –A : Practical Test **Marks : 30**

1. *Bearing :* Definition, types and inter conversion of bearing in detail.
2. *Liquid Prismatic Compass :* Features and functions of its various parts.
3. Determination of Individual Compass Error.
4. *Setting of Map :* With Compass and without compass.
5. *Marching by Compass :* To set the compass in a particular direction for night march.
6. *Finding Position on the Map :* Determine one's and enemy's position on the map by resection and intersection method.

SECTION-B : Practical Record **:** Marks : 5

SECTION-C : Viva-Voce **:** Marks : 5

HISTORY

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

General Instructions : (For Papers A & B)

INSTRUCTIONS FOR THE PAPER-SETTERS AND CANDIDATES :

(i) *The syllabus has been divided into four units.*
There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 9 short answer type questions carrying 18 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** essay type questions, and the candidate shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 18 marks.

(ii) One question from Unit IV shall be set on the map.

(iii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
The paper-setter must put note (iii) in the question paper.

Explanation :

1. Each essay type question would cover about one-third to one half of a topic detailed in the syllabus.
2. The distribution of marks for the map question would be as under :

Map	:	10 marks
Explanatory Note	:	08 marks

Note : In case, a paper setter chooses to set a question on important historical places, the paper-setter will be required to ask the students to mark 5 places on map of 2 mark each and write explanatory note on any four of 2 marks each.

3. The paper-setter would avoid repetition between different types of questions within one question paper.

Paper A : HISTORY OF INDIA, 1750-1964 A.D.

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objectives : To introduce the students to the History of India in Modern times.

Pedagogy : Lectures, library work and discussions.

Unit-I

1. *Foundation of British Rule :* The battles of Plassey and Buxar and their significance; Reforms of Warren Hastings.
2. *Administrative Reforms :* Reforms of Cornwallis, William Bentinck and Dalhousie.
3. *The Uprising of 1857 :* Political, socio-religious, economic and immediate causes; failure; results.

Unit-II

4. *Economic Changes :* British Agrarian policies; commercialization of agriculture; rural indebtedness; deindustrialization and growth of modern industry; theory of economic drain.
5. *Socio-Religious Reform Movements :* Brahmo Samaj; Aligarh Movement; Arya Samaj; Ramakrishna Mission; Swami Vivekananda's contribution.
6. *Depressed Classes Movement :* Contribution of Jotiba Phule, B.R. Ambedkar and Mahatama Gandhi.

Unit-III

7. *Growth of Political Consciousness :* Causes for the growth of political consiciousness/nationalism; foundation of the Indian National Congress; national movement upto 1919.
8. *Indian National Movement :* Gandhi's emergence; circumstances leading to the non-cooperation movement and its significance; demand for complete independence; the civil disobedience movement.
9. *Rise of Communal Politics :* Factors responsible for the growth of communal politics; Separate electorate; Muslim League and Pakistan Resolution.

Unit-IV

10. *Towards Partition and Independence :* Quit India Movement; British proposals for Independence; Indian Independence Act of 1947.
11. *Significant Developments after Independence,1947-66 :* Making of the Constitution; integration of Princely states; the reorganisation of states.
12. *Map :*
    (a) *Important Historical Places* – Delhi, Calcutta, Madras, Bombay, Goa, Pondicherry, Surat, Plassey, Buxar, Gwalior, Jhansi, Hyderabad, Sabarmati, Amritsar, Lucknow, Lahore and Aligarh.
    (b) Extent of the British Empire in 1856.
    (c) Republic of India in 1950.

Books Recommended :

1. Abdul Kalam Azad, *Azadi Ki Kahani* (Hindi Medium), Delhi : Orient Black Swan, 2007.

2. Bandyopadhyay, Sekhar, *From Plassey to Partition : A History of Modern India,* Delhi : Orient Black Swan, 2007 (Also available in Hindi Medium).

3. Chandra, Bipan, *Communalism in Modern India,* Delhi : Vikas Publishers, 1987.

4. Chandra, Bipan *et al., India's Struggle for Independence, 1875-1947,* Delhi : Viking/Penguin Books, 1989.

5. Chandra, Bipan, *Aitihasik Sangharsh – EPIC Struggle* (Hindi Medium), Delhi : Orient Black Swan, 2007.

6. Nanda, B.R., *Making of a Nation : India's Road to Independence,* New Delhi : Harper-Collins, 1998.

Paper B : HISTORY OF THE PUNJAB, 1469-1966 A.D.

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objectives : To introduce the students to the history of Punjab since medieval times.

Pedagogy : Lectures, library work and discussions.

Unit-I

1. *Guru Nanak Dev :* Life and Teachings of Guru Nanak; concept of langar, sangat, guru gaddi; his views on position of women.

2. *Expansion and Consolidation of Sikhism :* Contribution of Guru Angad, Guru Amar Das and Guru Ram Das.

3. *Guru Arjan Dev :* His contribution to the Panth; compilation of the Adi Granth; causes and significance of the martyrdom of Guru Arjan Dev.

Unit-II

4. *Development of Sikhism :* Guru Hargobind's new policy and its justification; martyrdom of Guru Tegh Bahadur and its significance.

5. *Creation of the Khalsa :* Circumstances leading to the creation of Khalsa and its significance; the new injunctions and symbols of the Khalsa; post Khalsa activities of Guru Gobind Singh.

6. *Banda Singh Bahadur :* Establishment of independent rule; his achievements; socio-economic transformation.

Unit-III

7. *Punjab under Ranjit Singh :* Civil & military administration; land revenue system; Anglo-Sikh relations upto 1839.

8. *British Rule :* Circumstances leading to the annexation of the Punjab; British administrative structure.

9. *Socio-Religious Reform Movements :* Nirankaris, Namdharis; Singh Sabha.

Unit-IV

10. *The Phases of Freedom Struggle :* Agitation of 1907; the Ghadar Movement; the Jallianwala Bagh; the Gurudwara Reform Movement; Circumstances leading to the partition of the Punjab.

11. *Developments After Independence :* Rehabilitation and resettlement after partition; Punjabi Suba movement; the Reorganisation Act of 1966; agrarian changes.

12 *Map :*

(a) *Important Historical Places* – Amritsar, Goindwal, Anandpur Sahib, Chamkaur Sahib, Kiratpur, Kartarpur, Paonta Sahib, Sirhind, Muktsar, Tarn Taran, Lahore, Machhiwara, Ropar and Kapurthala.

(b) Battles of Banda Bahadur.

(c) Ranjit Singh's Kingdom.

Note : In case, a paper setter chooses to set a question of map on important historical places, the paper setter will be required to ask the students to mark 10 places on map of 1 mark each and write explanatory note on any four of 2 marks each.

Books Recommended :

1. Grewal, J.S., *The Sikhs of the Punjab : The New Cambridge History of India,* New Delhi : Cambridge University Press, 1999 (2nd edn.).

2. McLeod, W.H., *Sikhs and Sikhism,* New Delhi : Oxford University Press, 2000.

3. Banerjee, I.B., *Evolution of the Khalsa,* 2 Vols., Calcutta : A Mukherjee & Co., 1979 (2nd edn.).

4. Banga, Indu (ed.), *Five Punjabi Centuries : Polity, Economy, Society and Culture c. 1500-1900,* New Delhi : Manohar, 1997.

5. Sinha, N.K., *Rise of the Sikh Power,* Calcutta : A Mukherjee & Co., 1973 (3rd edn. Rep.).

6. Teja Singh and Ganda Singh, *A Short History of the Sikhs, Vol. I : 1469-1765,* Patiala : Punjabi University, 1983.

POLITICAL SCIENCE

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

GENERAL INSTRUCTIONS FOR THE PAPER-SETTERS AND CANDIDATES :

(i) *The syllabus has been divided into four units :*

There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 9 short answer type questions carrying 18 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** essay type questions, and the candidate shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 18 marks.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
The paper setter must put note (ii) in the question paper.

Paper-A : INDIAN GOVERNMENT AND POLITICS

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours.

Objectives :

This paper provides students with a basic knowledge of the fundamental elements and institutions of government, politics and processes in India at both the centre and state levels.

Unit-I

1. Constituent Assembly and making of India's Constitution.
2. Basic Features of Indian Constitution.
3. Preamble and its importance.
4. Nature of Indian Federalism and Centre State relations.

Unit-II

1. Human Rights : An overview, National Human Rights Commission and Punjab State Human Rights Commission.
2. Fundamental Rights and Fundamental Duties.
3. Directive Principles of State Policy.

Unit-III

1. President : Election, powers, position and changing role.
2. Parliament : Composition, powers and role.
3. Indian Cabinet and Prime Minister : Election, powers, position and changing role.
4. Supreme Court : Judicial Review, Public Interest Litigation, Judicial Activism.

Unit-IV

1. Governor : Appointment, powers and role.
2. State Legislature : Composition, powers and role.
3. Council of Ministers and Chief Minister : Election, powers, position and role.
4. High Court : Composition, powers and role in the Indian Constitutional process.

Paper-B : INDIAN POLITICS

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours.

Objectives :

The aim of this paper is to enrich the student's understanding of the working of the Indian political system with reference to political parties, the party system, elections and voting behaviour. Units III and IV of the paper examine in detail certain key issues and debates in contemporary India.

Unit-I

1. Nature of party system in India : A critical evaluation.
2. National Political Parties [Indian National Congress, BJP, CPI, CPI(M), BSP] : Organization, Ideology and Electoral performance.
3. Regional Political Parties (SAD, NC, DMK, Telegu Desam) : Organization, Ideology and Electoral performance.

Unit-II

1. The Election Commission : A critical evaluation and electoral reforms in India.
2. Pressure Groups in Indian Politics.
3. Voting Behaviour in India.

Unit-III

1. Emerging Trends in Indian Politics.
2. Caste, Religion and Language in Indian Politics.
3. Regionalism in Indian Politics.
4. Liberalisation and Indian Politics.

Unit-IV

1. Basic principles and determinants of Indian Foreign Policy.
2. Non-alignment : Meaning and Relevance.
3. Role of India in Non-alignment Movement.

Books Recommended :

1. Rajiv Sikri, *Challenge and Strategy : Rethinking India's Foreign Policy,* Sage Publications, New Delhi, 2009.
2. S.D. Muni, *India's Foreign Policy : The Democracy Dimension,* Foundation Books, Cambridge University Press India, New Delhi, 2009.
3. Sumit Ganguly (ed.), *India's Foreign Policy : Retrospect and Prospect,* OUP, 2009.
4. C. Raja Mohan, *Crossing the Rubicon : The Shaping of India's New Foreign Policy,* Palgrave Macmillan, 2004.
5. G. Austin, *The Indian Constitution : Corner Stone of a Nation*, Oxford, Oxford University Press, 1966.
6. G. Austin, *Working a Democratic Constitution : A History of the Indian Experience,* New Delhi, Oxford University Press, 1999.
7. D.D. Basu, *An Introduction to the Constitution of India*, New Delhi, Prentice Hall, 1994.
8. C.P. Bhambri, *The Indian State Fifty Years,* New Delhi, Shipra, 1997.
9. P. Brass, *Politics of India Since Independence,* Hyderabad, Orient Longman, 1990.
10. P. Brass, *Caste, Faction and Party in Indian Politics, Vol. II :* Election Studies, Delhi, Chanakya Publications, 1985.
11. P. Brass, *Ethnic Groups and the State*, London, Croom Helm, 1995.

12. P. Brass, *Language, Religion and Politics in North India*, London, Cambridge University Press, 1974.

13. B.L. Fadia, *State Politics in India, Vol. II,* New Delhi, Radiant Publishers, 1984.

14. F.R. Frankel, *India's Political Economy 1947-2004 : The Gradual Revolution,* Oxford, Oxford University Press, 2005.

15. R. Kothari, *State Against Democracy : In Search of Human Governance*, Delhi, Ajanta, 1988.

16. R. Kothari, *Politics in India*, New Delhi, Orient Longman, 1970.

17. R. Kothari, *Party System and Election Studies,* Bombay, Asia Publishing House, 1967.

18. Iqbal Narain (ed.), *State Politics in India*, Meerut, Meenakshi Parkashan, 1967.

19. M.V. Pylee, *Constitutional Government in India,* Bombay, Asia Publishing House, 1977.

20. M.V. Pylee, *An Introduction to the Constitution of India*, New Delhi, Vikas, 1998.

21. S.P. Verma, and C.P. Bhambri (ed.), *Election and Politics Consciousness in India,* Meerut, Meenakshi Parkashan, 1967.

22. Partha Chatterjee (ed.), *State and Politics in India*, Oxford University Press, 1999.

23. Zoya Hasan, E. Sridharan and R. Sudarshan (ed.), *India's Living Constitution : Ideas*, *Practices*, *Controversies,* Permanent Black, New Delhi, 2004.

24. Niraja Gopal Jayal (ed.), *Democracy in India*, Oxford University Press, Delhi, 2001.

25. A.G. Noorani, *Constitutional Questions in India*, Oxford University Press, 2000.

26. Zoya Hasan, *Parties and Party Politics in India*, Oxford University Press, New Delhi, 2002.

ECONOMICS

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper–A : MACRO-ECONOMICS

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Course Objective : This paper aims to familiarize the student with the generally accepted principles of macroeconomics. It deals with aggregates i.e. consumers as a whole, producers as a whole, exporters and importers as a whole, the effects of government spending and taxation, and the monetary policy of the central bank.

The course includes the basic theories of determination of income, consumption, investment, employment, money and interest, inflation, Monetary and Fiscal policies, and business cycles.

INSTRUCTIONS FOR THE PAPER-SETTERS AND CANDIDATES :

(i) *The syllabus has been divided into four units.*

There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 12 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 9 short answer type questions carrying 18 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** essay type questions, and the candidates shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 18 marks.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

The paper setter must put note (ii) in the question paper.

Unit-I

Meaning and Scope of Macro-Economics.

Determination of Income and Employment : Classical and Keynesian Theory of Income, Output, Employment. Say's Law of Markets, Aggregate Demand and Supply Functions.

Unit-II

Consumption and Investment Functions : Average and Marginal Propensity to Consume; Static and Dynamic Multiplier; Marginal Efficiency of Capital. Investment : Types, Investment Demand Schedule and Factors Affecting Investment Decisions.

Unit-III

Theories of Money and Interest :

Money : Its Functions and Role, Money Market and Capital Markets.

Quantity Theory of Money : Fisher's Equation, Cambridge Equation and Keynesian Theory.

Liquidity Preference Theory of Rate of Interest.

Banking : Major Functions of Commercial Banks. Credit Creation and Credit Control.

Unit-IV

Inflation and Macro–Economic Policies; Theories of Inflation; Cost-push and Demand-pull, Measures to Control Inflation. Monetary and Fiscal Policies for Stabilization.

Trade Cycle : Meaning and Phases.

Recommended Readings :

1. T.F. Dernburg and D.M. McDougal : *Macro Economics,* McGraw Hill, New York, 1983.
2. O.P. Sharma (Punjabi Medium) : *Macro Economics,* Punjabi University, Patiala.
3. Ranjit Singh Ghuman : *Antar-Rashtriya Arth Vigyan,* Punjabi University, Patiala.
4. M.C. Vaish (Latest ed.) : *Macro Economic Theory,* Oxford University Press, New Delhi.
5. J. Harvey and M. Johnson : *Introduction to Macro Economics* (1971), McMillan, London.
6. M.L. Jhingan : *Macro Economic Theory,* Varinda Publishers, Delhi, 2003.

Supplementary Readings :

1. Gardner Ackley : *Macro Economic Theory,* 1985, McMillan, New York.
2. C.W. Baird : *Elements of Macro Economics,* 1977, West Publishing Company.

Note : Four to five lectures of 45 minutes each per week are required to complete the syllabi.

Paper– B : PUBLIC FINANCE AND INTERNATIONAL ECONOMICS

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Course Objective : The primary course objective is to introduce the students to the basics of public finance and international trade. The first two units aim to introduce students to the primary functions of government to generate resources from its people to spend money improving their lives. The last two units are concerned with basic theories of international trade and commercial policies, balance of payments, determination of exchange rates and role of international financial institutions.

INSTRUCTIONS FOR THE PAPER-SETTERS AND CANDIDATES :

(i) *The syllabus has been divided into four units.*

There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 12 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 9 short answer type questions carrying 18 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** essay type questions, and the candidates shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 18 marks.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

The paper setter must put note (ii) in the question paper.

Unit-I

Introduction to Public Finance : Nature, Scope and its Importance.

Public Expenditure : Principles of Public Finance (Maximum Social Advantage), Effects of Public Expenditure on Production and Distribution. Causes for the Recent Growth of Public Expenditure in India.

Taxation : Classification and Canons, Characteristics of a Good Tax System.

Unit-II

Public Debt : Its types and Role, Burden and Methods of Redemption of Public Debt.

Deficit Financing : Objectives and Limitations.

Incidence and Impact of Taxation : Demand and Supply Theory.

Taxable Capacity : Absolute and Relative Capacity, Determinants of Taxable Capacity.

Unit-III

Pure Theory of International Trade : Classical, Heckscher and Ohlin Theories; The Principles of Reciprocal Demand and Terms of Trade.

Commercial Policy : Free Trade Vs. Protection, Rationale of Protectionist Policy in Less Developed Countries Impact of W.T.O. on Less Developed Countries.

Unit-IV

Theory of Balance of Payments and Exchange Rate : Meaning, Concepts and Components of Balance of Payments. Disequilibrium in the Balance of Payments : Causes and Measures to correct the disequilibrium.

Rate of Exchange : Meaning and its Determination; Fixed Vs. Flexible Rate of Exchange and Purchasing Power Parity Theory. IMF and IBRD : Objectives, Working and Achievements.

Recommended Readings :

1. A.R. Musgrave and P.B. Musgrave : *Public Finance in Theory and Practice,* McGraw Hills, International Students' Edition, 1976.
2. C.P. Kindleberger : *International Economics,* Richard Irwin Homewood, Illinois, Indian Edition (1987).
3. B.O. Soderston : *International Economics,* McMillan Press Ltd., 1990.
4. O.P. Sharma : *Sarvjnik Vit,* Punjabi University, Patiala.
5. Ranjit Singh Ghuman : *Antar-Rashtriya Arth Vigyan,* Punjabi University, Patiala.

Supplementary Reading :

H. Dalton : *Principles of Public Finance,* Routledge and Kegan Paul, London (1954).

Note : Four to Five lectures of 45 minutes each per week are required to complete the syllabi.

SOCIOLOGY

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A : SOCIAL STRUCTURE AND SOCIAL CHANGE

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

(i) For written paper, the students will be required to attempt five questions in all. Question No. I will be **compulsory** comprising of 12 short answer type questions of 2 marks each and will cover the entire syllabus. The students are required to attempt 9 short answer type questions out of 12 i.e. $9 \times 2 = 18$ marks.

In addition to it, Question Nos. II to IX will consist of long answer (essay type) questions, two questions from each unit with internal choice carrying 18 marks each i.e. $4 \times 18 = 72$ marks. The students are required to attempt one question from each of the four units.

(ii) On an average, 15 hours are to be devoted for each unit.

(iii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

The paper-setter must put note (iii) in the question paper.

Objective :

This paper basically introduces students to both conceptual and theoretical understanding of society. Students are introduced to the structural-functional approach and the contributions of various thinkers are discussed to understand the meaning, process and factors of social change.

Course Content

Unit-I

Social Structure : Meaning, Characteristics, Elements – Status, Role, Norms and Values.
Structural–Functional Approach : Contributions of Radcliffe Brown.

Unit-II

Social Change : Meaning and Features.
Types of Social Change : Evolution (Comte), Revolution (Marx), Development – Changing Connotations.

Unit-III

Factors of Social Change : Economic, Demographic, Scientific and Technological, Educational, Legislative.

Unit-IV

Processes of Social Change : Sanskritization, Westernization, Modernization, Secularization and Globalization.

Essential Readings :

1. Bierstedt, Robert (1970) : *Social Order,* New York : McGraw Hill.
2. Haferkand, H. and Smelser, N.J. (1992) : *Social Change and Modernity,* Berkeley : University of California Press.
3. Macionis, John, J. (2005) : *Society : The Basics,* New York : Prentice Hall.
4. Madhurima (2010) : *Readings in Sociology – Part-2,* Jalandhar : New Academic Publishing House (All Mediums).
5. Moore, Wilbert (1965) : *Social Change*, New Delhi, Prentice Hall.
6. Singh, Yogendra (1993) : *Social Change in India,* New Delhi : Haranand Publications.
7. Srinivas, M.N. (1963) : *Social Change in Modern India,* Berkeley : University of California Press.

Further Readings :

1. Redcliffe-Brown, A.R. (1979) : *Structure and Function in Primitive Society*, London : Routledge and Kegan Paul.
2. Haralambos, M. (1998) : *Sociology : Themes and Perspectives,* New Delhi : Oxford University Press.
3. Talcott, Parsons (1959) : *Social System*, London : Tavistock Publications.
4. Singh, Yogendra (1996) : "Sociology and the Emerging Challenge of Change", *Sociological Bulletin,* Volume 45, No. 1, March, Page 1-14.
5. Singer, M. (1972) : *When a Great Tradition Modernizes : An Anthropological Approach to Indian Civilization,* New York : Praeger Publishers.

Paper-B : SOCIAL STRATIFICATION

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

(i) For written paper, the students will be required to attempt five questions in all. Question No. 1 will be **compulsory** comprising of 12 short answer type questions of 2 marks each and will cover the entire syllabus. The students are required to attempt 9 short answer type questions out of 12 i.e. $9 \times 2 = 18$ marks.

In addition to it, Question Nos. II to IX will consist of long answer (essay type) questions, two questions from each unit with internal choice carrying 18 marks each i.e. 4 × 18 = 72 marks. The students are required to attempt one question from each of the four units.

(ii) On an average, 15 hours are to be devoted for each unit.

(iii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

The paper-setter must put note (iii) in the question paper.

Objective :

All over the world, social groups are differentiated from one another and often ranked in terms of certain criteria. In this paper, students are exposed to the theoretical understanding of social stratification. In the Indian context, it is pertinent to apprise the students to the concept of social mobility and various factors that contribute to it. The major purpose of this course is to prepare the students to understand the hierarchical structure of groups in various societies and help them understand the social mobility.

Course Content

Unit-I

Social Stratification – Meaning, Features and Functions; Inequalities – Social and Natural.

Elements : Differentiation, Hierarchy, Ranking, Reward, Evaluation.

Unit-II

Theories of Social Stratification :

Functional – Davis and Moore.

Conflict – Karl Marx.

Multi-dimensional – Max Weber.

Unit-III

Forms : Caste, Class and Gender.

Features and Interface.

Unit-IV

Social Mobility – Meaning, Features, Types, Factors.

Indicators – Education, Occupation, Income.

Essential Readings :

1. Bendix, R. & Lipset, S.M. (1974) : *Class, Status & Power : Social Stratification in Comparative Perspective,* London : Routledge & Kegan Paul.
2. Ghurye, G.S. (1983) : *Caste, Class and Occupation,* Bombay : Popular Prakashan.
3. Giddens, Anthony (2001) : *Sociology : A Text Book for the Nineties,* London : Polity.
4. Haralambos, M. (1998) : *Sociology : Themes and Perspectives,* New Delhi : Oxford University Press.
5. Johnson, Harry, M. (1998) : *Sociology : A Systematic Introduction*, Delhi : Allied Publishers.
6. Macionis, John J. (2005) : *Society : The Basics,* New York : Prentice Hall.
7. Sharma, K.L. (1980) : *Essays in Social Stratification*, Jaipur : Rawat.
8. Tumin, M. (1987) : *Social Stratification,* New Delhi : Prentice Hall.

Further Readings :

1. Beteille, Andre (ed.) (1978) : *Social Inequality,* Auckland : Penguin Books.
2. Beteille, Andre (1996) : "Varna & Jati", *Sociological Bulletin,* Vol. 45, No. 1 (March) Pages 15-27.
3. Culvert (1982) : *The Concept of Class*, London : Hutchinson.
4. Gupta, Dipankar (1991) : *Social Stratification*, New Delhi : Oxford University Press.
5. Hughes, John *et al* (1995) : *Understanding Classical Sociology,* Marx, Weber & Durkheim, London : Sage.
6. Richardson, C.J. (1977) : *Contemporary Social Mobility*, N.Y., Nicholas Pub. Co.
7. Inglis, David & John Bone (ed.) (2006) : *Dimensions of Social Stratificationn, London : Routledge & Kegan Paul*
8. Levine, Rhonda (ed.) (2006) : *Social Class & Stratification,* U.K. Rowman & Littlefield Publishers.
9. Saunders Peter (1990) : *Social Class & Stratification,* London, Routledge and Kegan Paul.
10. Giddens, Anthony & Sutton W. Philip (2010) : *Sociology : Introductory Readings,* U.K., Polity Press.
11. Schacfer, Richard, T. (2010) : *Sociology : A Brief Introductions,* N.Y., Tata McGraw Hill.

PUBLIC ADMINISTRATION

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Outlines of tests, syllabi and courses of reading

PAPER-A : PERSONNEL ADMINISTRATION (WITH SPECIAL REFERENCE TO INDIA)

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objective of the Paper :

The objective of this paper is to give the student an in-depth understanding of various aspects of public personnel administration particularly recruitment, training and promotion. Contemporary issues like grievance redressal machinery, corruption, morale etc. concerning administration of human resources in government will also be examined.

INSTRUCTIONS FOR PAPER-SETTERS AND CANDIDATES :

(i) For Private/University School of Open Learning (USOL) students who have not been assessed earlier for the internal assessment, the marks secured by them in the paper will proportionately be increased in lieu of the internal assessment.
The paper-setter must put note (i) in question paper in this regard.

(ii) The candidate shall attempt 5 questions in all (one compulsory and one each from four units). The first compulsory question shall comprise of 12 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 9. Each short answer type question will carry 2 marks. Rest of the paper shall contain 4 units, each unit having two questions, out of which the candidate would be required to attempt one. Each question will carry 18 marks.

Unit-I

Personnel Administration : Meaning and Significance.

Civil Service : Meaning, Features and Functions.

Bureaucracy : Meaning, Characteristics, Demerits and Remedies.

Classification : Meaning, types, advantages and disadvantages. Classification of Civil Services in India.

Unit-II

Recruitment : Meaning, Methods, Problems. Recruitment to Higher Civil Services in India.

UPSC : Composition, Functions and Role.

SPSC : Composition, Functions and Role.

Unit-III

Training : Meaning, Types, Methods, Training of Higher Civil Services in India.

Joint Consultative Machinery, Staff Associations.

Conduct, Discipline and Ethics in Public Services.

Promotion : Meaning, Significance and Principles.

Unit-IV

Morale, Causes of Low Morale and Measures for Improvement.

Corruption in Administration : Causes and Remedies.

Central Vigilance Commission : Structure, Functions and Role.

Administrative Tribunals : Meaning, Significance, Types, Advantages and Disadvantages.

Essential Readings :

1. Rustom S. Davar : *Personnel Management and Industrial Relations* (New Delhi : Vikas Publishers, 1976).
2. O. Glenn Stahl : *Public Personnel Administration* (New Delhi : Oxford and IBH Publications, 1975).
3. S.L. Goel : *Public Personnel Administration* (New Delhi : Sterling Publishers, 2008)
4. A. R. Tyagi : *Public Administration: Principles and Practice* (Delhi: Atma Ram and Sons Publishers, 2001)

Further Readings :

1. Paul Pigors and C.A. Myers : *Personnel Administration : A Point of View and Method* (New York, McGraw Hill, 1973).
2. Government of India, Administrative Reforms Commission : *Report on Personnel Administration* (New Delhi : Manager of Publication, 1967).
3. Indian Institute of Public Administration : *Personnel Administration* (New Delhi : IIPA, 1970).
4. V.M. Sinha : *Personnel Administration* (Jaipur : RBSA, 1986).

PAPER-B : FINANCIAL ADMINISTRATION (WITH SPECIAL REFERENCE TO INDIA)

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objective of the Paper :

The objective of this paper is to give the student an in-depth understanding of various aspects of financial administration like budgeting, auditing, centre-state financial relations etc. It would also make them aware of : institutions like ministry of finance, union finance commission, parliamentary committees; and concepts like direct taxes, indirect taxes, deficit financing, public debt.

INSTRUCTIONS FOR PAPER-SETTERS AND CANDIDATES :

(i) For Private/University School of Open Learning (USOL) students who have not been assessed earlier for the internal assessment, the marks secured by them in the paper will proportionately be increased in lieu of the internal assessment.
The paper-setter must put note (i) in question paper in this regard.

(ii) The candidate shall attempt 5 questions in all (one compulsory and one each from four units). The first compulsory question shall comprise of 12 short-answer type questions, covering the whole syllabus, to be answered in 25-30 words each, out of which the candidate would be required to attempt any 9. Each short qnswer type question will carry 2 marks. Rest of the paper shall contain 4 units, each unit having two questions, out of which the candidate would be required to attempt one. Each question will carry 18 marks.

Unit-I

Financial Administration : Meaning and Significance.

Budget : Meaning, Types and Principles.

Budgetary Process : Preparation and Enactment.

Unit-II

Union Ministry of Finance : Organisation, Functions and Role.

Centre-State Financial Relations.

Union Finance Commission *:* Composition and Functions.

Unit-III

Legislative Control over Finance.

Public Accounts Committee.

Estimates Committee.

Committee on Public Undertakings.

Unit-IV

Audit : Concept, Objectives and Types.

CAGI : Appointment, Functions and Role.

Concepts : Direct Taxes, Indirect Taxes including Value Added Tax (VAT); Deficit Financing, Fiscal Deficit, Public Debt and Public Expenditure.

Essential Readings :

1. S.L. Goel : *Financial Management and Administration* (New Delhi : Sterling Publication, 1993).
2. Sahib Singh and Swinder Singh : *Public Personnel and Financial Administration* (Jalandhar : New Academic Publishers, 2007).
3. M.J.K.Thavaraj : *Financial Administration of India* (New Delhi : Sultan Chand Publications, 1978).
4. G. S. Lall : *Financial Administration in India* (Delhi : H.P.J. Kapoor, 4th ed.).
5. S.S. Mookerjee : *Indian Public Finance and Financial Administration* (New Delhi : Surjit Pubications, 1979).

Further Readings :

1. Government of India, Administrative Reforms Commission : *Report on Financial Administration* (New Delhi : Manager of Publications, 1967).
2. Government of India, Administrative Reforms Commission : *Report on Financial Accounts and Audit* (New Delhi : Manager of Publications, 1967).
3. Government of India, Administrative Reforms Commission : *Report on Centre-State Relations* (New Delhi : Manager of Publications, 1967).
4. R.K. Sinha : *Centre-State Financial Relations in India : Various Aspects, Irritants, Framework for Action* (New Delhi : Deep & Deep Publications, 1986).

PHILOSOPHY

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Outlines of tests, syllabi and courses of reading

Paper-I : INDIAN ETHICS (Theory)

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours
Lectures	:	75

Aims and Objectives :
This paper highlights the ethical philosophies propounded in the different Indian Philosophical Systems. It exposes the students to the main tenets of Buddhism, Sikhism, Gandhism, Gita and Vedic Culture.
This paper gives an insight into the nature of Ethics, moral notions and basic moral theories as propounded by Western Ethical Philosophers. This paper also deals with problems of applied ethics.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

(i) *The syllabus has been divided into four units.*
There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 9 short answer type questions carrying 18 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** essay type questions, and the candidate shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 18 marks.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
The paper-setter must put note (ii) in the question paper.

Unit-I

1. *Salient Features of Indian Ethics with reference to Purusartha :* Yuga Dharma and Sanatan Dharma.
2. *Varna Vyastha and Ashrama Dharma :* A Contemporary Analysis.
3. Concept of Sreyas and Preyas.

Unit-II

4. *Buddhist Ethics :* Four Noble Truths, Eight Noble Paths, Ahimsa and Karma Bhava Chakra, Sila.
5. *Jain Ethics :* Triratnas.
6. *Sikh Ethics :* Hukum, Nirbhau, Nirvair, Parupkar, Sewa and Sahaj.

Unit-III

7. *Gandhi :* Ahimsa, Satyagraha and Antyodaya.
8. *Gita :* Anasakta Karma, Lok Samgraha, Sva-Dharma.

Unit-IV

9. *Yoga and Stress Management :* Definition of Yoga, Ashtanga Yoga.
10. *Yoga :* Yama, Niyama, Asana, Pranayama, Pratyahara, Dharna, Dhyan, Samadhi.

Essential Readings :

1. Avtar Singh : *Ethics of the Sikhs,* Punjabi University, Patiala, 1970.
2. I. C. Sharma : *Ethical Philosophies of India,* George Allen and Unwin, London, 1965.
3. S. K. Maitra : *The Ethics of the Hindus,* Asian Publication Services, New Delhi, 1956.
4. Vivekananda : *Raj Yoga,* Advait Ashram, 1896.

Further Readings :

1. Surama Dasgupta : Development of Moral Philosophy in India, Munshi Ram Manoharlal Pub., New Delhi, 1994.
2. Four chapters on Freedom, Commentary on the Yoga Sutras of Patanjali, Swami Satyananda Saraswati Yoga Publications Trust, Munger, Bihar-Chapter II Section 29 onwards, Chapter III—Sections 1 to 6.

Paper–II : WESTERN ETHICS

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours
Lectures	:	75

Aims and Objectives :

This paper gives an insight into the nature of Ethics, moral notions and basic moral theories as propounded by Western Ethical philosophers. This paper also deals with problems of applied ethics.

INSTRUCTIONS FOR THE PAPER-SETTER AND CANDIDATES :

(i) *The syllabus has been divided into four units.*
There shall be **9** questions in all. The first question is **compulsory** and shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidates are required to attempt any 9 short answer type questions carrying 18 marks i.e. 2 marks of each. Rest of the paper shall contain **4** units. Each unit shall have **two** essay type questions, and the candidate shall be given internal choice of attempting one question from each Unit – 4 in all. Each question will carry 18 marks.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.
The paper-setter must put note (ii) in the question paper.

Unit-I

1. *The Nature of Ethics :* Virtues and Goodness, its concerns, Relation of Ethics with Aesthetics, Psychology and Religion.
2. The notion of Truth, Beauty & Goodness, Reflective Customary Morality.
3. *Object of Moral Judgement :* Actions, Intentions, Situations and Character.

Unit-II

4. *Virtue Ethics :* Aristotle.
5. *Categorical Ethics :* Kant.
6. Concept of Love, sin and freedom in Christian Tradition.

Unit-III

7. Teleological Ethics :
   (i) Hedonism
   (ii) Utilitarianism
      (a) Mill
      (b) Bentham.

Unit-IV

8. Moral Rights of Foetus and Female Foeticide.
9. Sexual harassment and Gender Justice.
10. *Environmental Ethics :* Pollution, Land Degradation and Organic Farming and Global Warming.

Essential Readings :

1. Aristotle : *Nicomachean Ethics,* Dordrecht : D. Reidel, Penguin Harmondsworth, 2001.
2. Peter Singer : *Practical Ethics* (Second Edition), Cambridge University Press, 1995.
3. Ved Parkash Verma : *Neeti Shastra* (in Hindi), Allied Publishers (New Delhi, 1987).
4. William Lillie : *An Introduction to Ethics* (London, Matheun, 1957).

Further Readings :

1. J.D. Mcbbott : *An Introduction to Ethics,* London : Hutchinson, 1966.
2. John Hospers : *Human Conduct : An Introduction to the Problems of Ethics*, N.Y., Harcourt, 1961.
3. Rosaline Hursthome : *Virtue Ethics,* New York, Harcourt, 1961.
4. W. Frankena : *Ethics,* Englewood Cliffs, Prentice Hall, New Delhi, India, 1994.

PSYCHOLOGY

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Objectives :

(i) Experimental Psychology course will emphasize on helping the students to gain knowledge about nervous system, sensation and perception. This course will also give an insight into psychophysics, statistics and cognitive aspects of psychology.

(ii) Pedagogy of the Course Work.
80% Lectures (including expert lectures).
20 % assignments, discussion and seminars.

Paper–A : EXPERIMENTAL PSYCHOLOGY

Max. Marks	:	80
Theory	:	70 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

(i) *The syllabus has been divided into four units.*
There shall be **9** questions in all. The first question shall be short answer type containing 12 short answer type questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 7 short answer type questions i.e. 2 marks of each. It shall carry 14 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all. Each question will carry 14 marks.

UNIT-I : *The Nervous System :* Structure and Functioning of the Neuron; Divisions of Nervous System—Central Nervous System and Peripheral Nervous System; Structure and Functioning of the Brain.

UNIT-II : *Sensations :* Concept and types. Vision—The Visual Stimulus, Structure and Functioning of the Eye, Theories of Colour Vision; Audition The Auditory Stimulus, Structure and Functioning of the Ear, Theories of Hearing.

UNIT-III : *Thinking :* Concept Attainment Reasoning; problem Solving; Language in Thinking.

UNIT-IV : *Perception :* Attention as Selective Perception; Perception of Form, Space and Movement.

Paper–B : EXPERIMENTAL PSYCHOLOGY

Max. Marks	:	80
Theory	:	70 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

(i) *The syllabus has been divided into four units.*
There shall be **9** questions in all. The first question shall be short answer type containing 12 short answer type questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 7 short answer type questions i.e. 2 marks of each. It shall carry 14 marks and shall be **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all. Each question will carry 14 marks.

UNIT-I : *Learning :* Concept, Classical Conditioning, Operant Conditioning, and Observational Learning.

UNIT-II : *Memory :* Sensory, Short Term and Long Term Memory. Levels of Processing; Measures of Memory; Forgetting due to Decay and Interference.

UNIT-III : *Psychophysics :* Thresholds; Psychophysical Methods of Limits, Constant Stimulus and Average Error; Brief introduction to Psychophysical Relationship.

UNIT-IV : *Statistics :* Normal Probability Distribution:- Properties and Applications; Hypothesis testing—Type I and Type II Errors; Chi Square Test Application to One Way and Two Way Classifications.

Note : The use of non-programmable calculators and statistical tables is allowed in the examination.

PSYCHOLOGY PRACTICALS

Max. Marks : 40
Time : 3 Hours

Eight practicals have to be performed :

1. Division of Attention.
2. Span of Attention.

3. Two Point Threshold.
4. Muller Lyer Illusion.
5. Differential Limen.
6. Effect of Mental Set on Perception.
7. Maze Learning.
8. Paired Associate Learning.
9. Comparing Recall and Recognition as Measures of Memory.
10. Retroactive Interference.
11. Bilateral Transfer of Training.
12. Problem Solving.

Books Recommended :

Essential Books :

1. Ciccarelli, S.K. and Meyer, G.E. (2008) : *Psychology :* South Asian Edition. Pearson Education.
2. Feldman, R.S. (1996) : *Understanding Psychology.* New Delhi : Tata McGraw Hill.
3. Garrett, H. E. (1966) : *Statistics in Psychology and Education.* New Delhi : Vakils, Feffer, and Simons.
4. Morgan, C.T., King, R.A., Weisz, J.R. and Schopler, J. (1987) : *Introduction to Psychology.* Singapore : McGraw Hill.
5. Woodworth, R.S. and Schlosberg, H. (1954) : *Experimental Psychology.* New Delhi : Oxford and IBH.

Reference Books :

1. Baron, R.A. (2002) : *Psychology.* New Delhi : Pearson Education.
2. D'Amato, M.R. (1970) : *Experimental Psychology.* New Delhi : Tata McGraw Hill.
3. Petdman, R.S. (1998) : *Understanding Psychology.* New Delhi : Tata McGraw Hill
4. Gazzaniga, M.S. (2006) : *Psychological Science.* New York: W.W. Norton.
5. Guilford, J.P., and Fruchter, B. (1981) : *Fundamental Statistics in Psychology and Education.* Singapore : McGraw Hill.

GEOGRAPHY

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A : WORLD RESOURCES AND ENVIRONMENT

Max. Marks	:	75
Theory	:	65 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Objectives :

- To understand concepts of resources and their interface with environment;
- To examine use and misuse of various resources and to anlayse future prospects;
- To study various methods and approaches of conservation and management of natural resources;
- To understand the quantitative and qualitative aspects of human resources in spatial perspectives and the associated environmental problems.

Course Content :

UNIT-I

Meaning, nature and components of environment. Nature and definition of resources. Resource—environment interface. (6 lectures)

Classification of Resources : Renewable and nonrenewable : Biotic (forests, wild-life, live-stock, fisheries, agricultural crops) and abiotic (land, water, minerals). (6 lectures)

UNIT-II

Distribution, availability, utilization and conservation of water; minerals and energy resources, their economic and environmental significance and sustainability. (15 lectures)

Types and distribution of Forests, Fisheries and major Soils : Their economic and environmental significance and conservation; problems of soil erosion and soil conservation. (15 lectures)

UNIT-III

Human Resources : Quantitative and qualitative aspects of population; numbers, distribution, density, growth, literacy and urbanization. (10 lectures)

Population-Resource Relationship, Population – resource regions of the world. (4 lectures)

UNIT-IV

Environment : Natural and Human. Man-environment relationships—determinism, possibilism, ecology. (10 lectures)

Biodiversity : Meaning and importance. (4 lectures)

Environmental Issues : Pollution; population explosion; food security; deforestation; conservation of wild life. (5 lectures)

Note :

1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The students shall attempt any 10 parts. The answer of each part should not exceed 25 words. Each part will carry 1.5 marks (Total 15 marks.)

2. The whole syllabus will be divided into 4 units. Eight questions will be set out of the whole syllabus, 2 from each unit. The students will be required to attempt one question from each unit. Each question will carry 12.5 marks. These will be in addition to the compulsory question at serial number 1.

3. Special credit will be given to suitable use of maps and diagrams. Use of unmarked stencils and colours are allowed.

4. Internal assessment will include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

5. For USOL/reappear/improvement candidate(s), who have not been assessed earlier for Internal Assessment, the question paper(s) in their case shall be of Maximum Marks allotted to the paper(s) concerned.
   The paper-setter must put note (5) in the question paper.

Books Recommended :

Essential Readings :

1. Chandna, R.C. : *A Geography of Population,* Kalyani Publishers, Ludhiana, 2009.

2. Chandna, R.C. : *Environmental Awareness,* Kalyani Publishers, New Delhi , 1998.

3. Hartshorne, Truman A. and John W. Alexander : *Economic Geography,* Prentice Hall of India, New Delhi, 1994, 3rd Edition.

4. Singh, Jagdish : *Geography of Resources,* Gyanodaya Prakashan, Gorakhpur, 2001.

Further Readings :

1. Agarwal, A. et. al : *The Citizen's Fifth Report,* Centre for Science and Environment, New Delhi, 1999.
2. Kates, R.W. & Burton, I. (Eds.) : *Geography, Resources and Environment,* Vols. I & II, University of Chicago Press, Chicago, 1986.
3. Zelinsky, Wilbur : *A Prologue to Population Geography*, Prentice Hall, New Jersey, 1966.
4. Zimmerman, E.W. : *World Resources and Industries,* Harper, New York.

Pedagogy :

- Maps can be shown related to distribution of various resources in the World and India
- Diagrams and flow charts related to institutional setup and policy models may be prepared and shown to students.
- Impact of problems associated to environment may be comprehended through field visits.

Paper-B : GEOGRAPHY OF INDIA – With special reference to Punjab

Max. Marks	:	75
Theory	:	65 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Objectives :

To foster compreshensive understanding of physical, human and economic resource base of India as well as Punjab.

Course Content :

UNIT-I

India in the context of Asia and the World (5 lectures)

Land : Relief, drainage, climate, natural vegetation, soils, minerals (Iron ore, manganese, mica, copper, gold) and power resources. (5 lectures)

People : Numbers, distribution and density, growth, urbanization and religious composition. (5 lectures)

UNIT-II

Economy (I) : (i) Agriculture - Characteristics and Problems of Indian Agriculture : Land use pattern, irrigation, major crops (rice, wheat, maize, cotton, jute, tea and groundnut) (15 lectures)

(ii) Transport - Rail, road, airways and waterways.

UNIT-III

Economy (II) : (i) Distribution and localization factors of major industries (iron and steel, cotton textiles, sugar, fertilizers, automobile). (15 lectures)

(ii) International trade.

UNIT-IV

Location, evolution and administrative divisions of Punjab (15 lectures)

Land : Physiography and drainage, climate, vegetation and soils.

People : Number, distribution, density, growth, religious composition and urbanization.

Economy : Agriculture, Industry and Transport.

Note : 1. A compulsory question containing 15 short answer type questions shall be set covering the whole syllabus. The students shall attempt any 10 parts. The answer of each part should not exceed 25 words. Each part will carry 1.5 marks (Total 15 marks.)

2. The whole syllabus will be divided into 4 units. Eight questions will be set out of the whole syllabus, 2 from each unit. The students will be required to attempt one question from each unit. Each question will carry 12.5 marks. These will be in addition to the compulsory question at serial number 1.

3. Special credit will be given to suitable use of maps and diagrams. Use of unmarked stencils and colours are allowed.

4. Internal assessment will include written assignments, snap tests, participation in discussion in the class, term papers, attendance etc.

5. For USOL/reappear/improvement candidate(s), who have not been assessed earlier for Internal Assessment, the question paper(s) in their case shall be of Maximum Marks allotted to the paper(s) concerned.

The paper-setter must put note (5) in the question paper.

Books Recommended :

Essential Readings :

1. Deshpande, C.D. : *India : A Regional Interpretation,* Northern Book Centre, New Delhi, 1992.
2. Johnson, B.L.C. : *India : Resources and Development,* Arnold Heinemann, London, 1980.
3. Johnson, B.L.C. : *South Asia,* Heinemann, London, 1981.
4. Mankoo, Darshan Singh : *Geography of Punjab,* Kalyani Publications, Ludhiana, 2009.
5. Mavi, H.S. and DS Tiwana : *Geography of Punjab,* National Book Trust, Delhi, 1993.
6. Spate, O.H.K. and ATA Learmonth : *India & Pakistan* : *Land, People and Economy,* Methuen, London, 1967.

Further Readings :

1. Khullar, D.R. : *India : A Comprehensive Geography,* 2007, Kalyani Publishers, New Delhi.
2. Singh, Gopal : *A Geography of India,* Atma Ram & Sons, New Delhi.
3. Singh, Jagdish : *India : A Comprehensive Systematic Geography,* Gyanodaya Prakashan, Gorakhpur, 2nd Edition, 2003.
4. Singh, R.L. (Ed.) : *India : A Regional Geography,* National Geographical Society of India, Varanasi, 1971.
5. Sharma, T.C. and O. Coutinho : *Economic and Commercial Geography of India,* Vikas, New Delhi 2005.
6. Tirtha, Ranjit & Gopal Krishan : *Emerging India,* Conpub Ann Arbour Publishers, Michigan, 2006.

Pedagogy :

Teaching should involve usage of visual aids especially maps. Students should be encouraged to use an atlas in the classrooms. Audio-video shows on different parts of India and Punjab should be arranged if possible.

Paper-C : CARTOGRAPHY

Max. Marks : 50

Time : 3 Hours

Distribution of Marks :

Written paper of 3 hours duration	:	30 marks
Viva and Practical record (10 + 10)	:	20 marks

Objectives :

- To apprise the students with symbolization of different types of geographical data and depiction of various spatial data.
- To provide training in application of various graphical methods of depicting geographic data.
- To train the students to interpret the topographical sheets at different scales.

Course Content :

UNIT-I

Symbolization of Geographical Data :

(a) *Point Symbols :* Dot, circle, sphere.

(b) *Line Symbols :* Isopleths and flow lines.

(c) *Areas Symbols :* Choropleth.

(5 lectures,10 lab. sessions)

UNIT-II

Construction and Significance of the following :

(a) *Columnar Diagrams :* Simple, percentage, superimposed, composite.

(b) *Graphs :* Line graphs, climograph, hythergraph, ergograph, wind rose.

(6 lectures, 12 lab. sessions)

UNIT-III

Cartographic Representation of :

(a) Population data (distribution, density, growth, migration and literacy).

(b) Agricultural data (land utilization, distribution of crops, percentage of cropped area and irrigated area).

(c) Industrial data (distribution, employment and production).

(9 lectures, 30 lab. sessions)

UNIT-IV

Topographical Maps : Significance of topographical maps in geographical studies.

Study and Interpretation of Topographical Maps of India including OSM and DSM (two sheets : one representing a hilly/mountainous tract and the other a plain tract).

Profiles : Longitudinal and Transverse, profiles of rivers.

Introduction to Elementary concepts of GIS.

(6 lectures, 12 lab. sessions)

Note :

1. The written and practical examination including viva-voce shall be conducted at the respective college itself except for USOL students. However, the format of the questions paper shall be uniform. A separate paper for 30 marks shall be prepared for colleges by the University from the prescribed syllabus.

2. Practical exam. at the respective colleges shall be conducted by one internal and one external examiner. The external examiner shall be appointed by the Principal of the respective colleges in consultation with the senior most teacher of geography in the college.

3. For students of USOL, a written theory paper for 30 marks shall be conducted by the University alongwith the University examination.

4. A compulsory question containing 10 short answer type questions shall be set covering the whole syllabus. The students shall attempt any 6 parts. The answer of each part should not exceed 25 words. Each part will carry 1 mark (Total 6 marks).

5. The whole syllabus has been divided into 4 units. Eight questions will be set out of the whole syllabus, i.e. 2 from each unit. The students will be required to attempt one question from each unit. These will be in addition to the compulsory question at serial number1.

6. Evaluation of Practical Record will be done at the time of viva-voce examination. A minimum of 20 sheets are to be prepared by the students. There will be no laboratory exercise at that time.

7. There will be no viva-voce examination for the candidates appearing through USOL. They will be required to submit their Practical Note Book (Practical files) with the University School of Open Learning (Department of Geography) at least 10 days before the commencement of their examination. Their Note Books (Practical files) will be evaluated by two examiners (including at least one from the USOL).

8. For the students of USOL there will be an internal assessment of 10 marks in lieu of the viva-voce examination in practical record and field report. The marks obtained by the candidate will be added to the marks awarded by the internal and external examiners evaluating the Practical Record and Field Report.

9. A fresh practical note book/field report shall be prepared by failed/improvement candidates.

10. For Practical classes, the number of students in one group shall not exceed fifteen.

11. There will be 3 hours of teaching per week for this paper.

12. For USOL/reappear/improvement candidate(s) who have not been assessed earlier for Internal Assessment, the question paper(s) in their case shall be of Maximum Marks allotted to the paper(s) concerned.

The paper setter must put note (12) in the question paper.

Books Recommended :

Essential Readings :

1. Robinson, A.H. : *Elements of Cartography,* John Wiley, New York, 1995.

2. Singh, Gopal : *Mapwork & Practical Geography*, Vikas Publishing House Pvt. Ltd., New Delhi, 1995.

3. Singh, L.R. : *Practical Geography,* Prayag Pustak Bhavan, Allahabad, 2006.

Further Readings :

1. Birch, T.W. : *Maps : Topographical & Statistical*, *Clarendon Press,* Oxford, 1949.

2. Garnett, A. : *Geographical Interpretation of Topographical Maps*, George Harrap & Co., London, 1953.

3. Khullar, D.R. : *Essentials of Practical Geography,* New Academic Publishing Co., Jallandhar, 2009.

4. Monkhouse, F.J. : *Maps and Diagrams*, Methuen & Co., London, 1994 reprint.

Pedagogy :

The students need to be trained in the use of symbols to depict various geographic data.

They should learn to comprehend the topographical maps by deriving slopes and drawing profiles and should be able to interpret the maps.

GANDHIAN STUDIES

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

PAPER-A : SOCIAL THOUGHT OF MAHATMA GANDHI

Max. Marks	:	100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Course Objectives :

The paper is designed to acquaint the students with the Social Thought of Mahatma Gandhi.

Pedagogy of the Course Work :

90% Lectures (including expert lectures)
10% Unit tests, snap tests, assignments, attendance and class room participation

Note :

1. The syllabus has been divided into four (4) units.
2. There shall be **9** questions in all.
3. The first question shall be short answer type containing 15 short answer type questions spread over the whole syllabus and each to be answered in about 25-30 words. The candidate is required to attempt any **9** short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question.
4. Rest of the paper shall contain four (4) units and each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each unit – 4 in all. All questions shall carry 18 marks.
5. For the private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

The paper setter must put note (5) in the question paper.

Unit-I

1. Foundations and Characteristics of Gandhi's Social Thought
2. Truth
3. Non-violence

Unit-II

4. Human Rights
5. Ideal Society, State and Rama Rajya

Unit-III

6. Education
7. Theory of Civilization
8. Morality and Religion

Unit-IV

9. Drive against Untouchability :
   (a) Caste System
   (b) Communal Award
   (c) Poona Pact
10. Women Empowerment

Essential Readings :

1. Bandyopadhyaya, J. : *Social and Political Thought of Mahatma Gandhi,* Bombay : Allied Publisher, 1969.
2. Bharathi, K.S. : *Social and Political Thought of Mahatma Gandhi,* New Delhi : Concept, 1995.
3. Varma, V.P. : *Political Philosophy of Mahatma Gandhi & Sarvodaya,* Agra : Laxmi Narain Aggarwal, 1998.
4. Mishra, A.D. : *Reading Gandhi,* New Delhi : Dorling Kindersley, Pearson, 2012

Further Readings :

1. Suda, J.P. : *Main Currents of Social and Political Thought of Modern India,* Vol. II, Meerut : K. Nath & Co., 1973.
2. Unnithan, T.K.N. : *Gandhi and Social Change,* New Delhi : Arnold Heinemann India, 1973.

PAPER-B : POLITICAL THOUGHT OF MAHATMA GANDHI

Max. Marks		: 100
Theory	:	90 marks
Internal Assessment	:	10 marks
Time	:	3 Hours

Course Objectives :

The paper is designed to acquaint the students with the Political Thought of Mahatma Gandhi.

Pedagogy of the Course Work :

90% Lectures (including expert lectures)

10% Unit tests, snap tests, assignments, attendance and class room participation

Note :

1. The syllabus has been divided into four (4) units.
2. There shall be **9** questions in all.
3. The first question shall be short answer type containing 15 short answer type questions spread over the whole syllabus and each to be answered in about 25-30 words. The candidate is required to attempt any **9** short answer type questions i.e. 2 marks of each. It shall carry 18 marks and shall be **compulsory** question.
4. Rest of the paper shall contain four (4) units and each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each unit – 4 in all. All questions shall carry 18 marks.
5. For the private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

   The paper setter must put note (5) in the question paper.

Unit-I

1. Influences on Gandhi's Political Thought
2. Theory of Ends & Means
3. Theory of State : Ideal and Practical

Unit-II

4. Panchayat Raj and Gram Swaraj
5. Freedom & Equality
6. Rights and Duties

Unit-III

7. Religion and Politics

8. Satyagraha
   (a) Meaning
   (b) Essentials
   (c) Forms & Significance
   (d) Satyagraha & Passive Resistance
   (e) Relevance of Satyagraha in the Modern State

Unit-IV

9. Power and Authority
10. Assessment and Contemporary Relevance

Essential Readings

1. Bhattacharyya, Buddhadeva : *Evolution of Political Thought of Gandhi,* Calcutta: Calcutta Book House, 1969.

2. Dhawan, G.N. : *Political Philosophy of Mahatma Gandhi,* Ahmedabad: Navajivan Publishing House, 1990, 4th Edition.

3. Varma, V.P. : *The Political Philosophy of Mahatma Gandhi and Sarvodaya,* Agra: Lakshmi Narain Aggarwal, 1981.

10. Mishra, A.D. : *Reading Gandhi,* New Delhi : Dorling Kindersley, Pearson, 2012.

Further Readings

1. Iyer, Raghavan N. : *The Moral & Political Thought of Mahatma Gandhi,* London: Oxford University Press, 1973.

2. Mishra, A.D. (ed.) : *Perspectives on Human Rights,* New Delhi: Radha Publications, 2002.

JOURNALISM & MASS COMMUNICATION

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Objectives :

(i) The course will introduce students to the basic techniques of sourcing news stories and features; train them in the art and science of writing and presenting in the areas of print, radio and the television medium.

(ii) Pedagogy of the Course Work :
80 % lectures (including expert lectures).
20 % assignments, discussion and seminars.

PAPER-A : PRINT JOURNALISM

Max. Marks	:	80
Theory	:	70 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

There shall be **9** questions in all. The first question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 7 short answer type questions i.e. of 2 marks each. It shall carry 14 marks and is a **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all. Each question will carry 14 marks.

Unit-I

Definition, nature and types of news and features; sourcing news and features.

Unit-II

Writing techniques and styles of news reports and features.

Unit-III

Role and responsibility of a sub-editor; basic editing techniques for copy and pictures; importance and writing of headlines.

Unit-IV

Introduction to typography and design principles; page make up using publishing software.

PAPER-B : RADIO & TV JOURNALISM

Max. Marks	:	80
Theory	:	70 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

There shall be **9** questions in all. The first question shall be short answer type containing 12 short questions spread over the whole syllabus and each to be answered in about 25 to 30 words. The candidate is required to attempt any 7 short answer type questions i.e. of 2 marks each. It shall carry 14 marks and is a **compulsory** question. Rest of the paper shall contain **4** units. Each unit shall have **two** questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all. Each question will carry 14 marks.

Unit-I

Reporting and writing for Radio and TV.

Unit-II

Basic introduction to audio/video production & its stages (pre-production, production, post production)

Unit-III

Pronunciation and modulation in Radio/TV reporting : Pitch, volume, tempo, vitality.

Unit-IV

Basic camera shots and language of camera movements.

PRACTICALS

Max. Marks : 40

1.	Portfolio of all issues of a laboratory quarterly publication.	:	10 Marks
2.	Compilation of 5-minute weekly TV news bulletins.	:	10 Marks
3.	Compilation of 10-minute weekly radio bulletins.	:	10 Marks
4.	Performance report of internship in the media of print, radio or TV.	:	10 Marks

Books Recommended :

Essential Readings :

1. Rivers, William, L. : *Freelancer and Staff Writer : Newspaper Features and Magazine Articles,* Wadsworth Publishing Co., 1991.
2. Sharma, S.R. : *Elements of Modern Journalism,* S.S. Publishers, New Delhi, 1999.
3. Chauhan, Swati and Chandra, Navin : *Foundation of News and Journalism,* Kanishka Publishers, New Delhi, 2000.
4. Maur, Susan : *Writing Your Own Scripts and Speeches,* Mc Graw Hill, London, 1991.
5. Ahuja, B.N. and Chhabra, S.S. : *Principles and Techniques of Journalism,* Surjeet Publications, New Delhi, 1995.
6. Blundell, William : *The Art and Craft of Feature Writing,* Plume Inc., 1998.
7. Scannell, Paddy : *Broadcast Talk,* Sage Publications, London, 1991.
8. Hyde, Stuart : *Television and Radio Announcing,* Kanishka Publishers, New Delhi, 1998.
9. Jain, S.P. : *Art of Broadcasting,* Intellectual Publishing House, New Delhi, 1985.
10. Mufti Maqbool : *Basics of Teaching of Arts and Audio Visuals,* Dilpreet Publishing House, New Delhi, 1996.
11. Look, Marshall, J. : *Leads and Conclusions,* Writers' Digest Books, 1995.
12. Mencher, Melvin : *Basic News Writing,* Brown Publishers, Iowa, 1989.
13. Bhatt, S.C. : *Broadcast Journalism : Basic Principles,* Har Anand Publications, New Delhi, 1993.
14. Sudarshan, K.N. : *Electronic Media,* Indian Publishers Distributors, 1998.
15. Jacobi, Peter B. : *The Magazine Article : How to Think it, Plan it, Write it,* Indiana University Press, 1997.
16. Ravindran, R.K. : *Handbook of Reporting and Editing,* Anmol Publications, N.D., 1999.
17. Sharma, S.R. : *Information Technology, Mass Media and Management Systems,* Book Enclave, Jaipur, 1997.

18. Joseph, M.K. : *Textbook of Editing and Reporting,* Dominant Publishers, New Delhi, 2000.

19. George, T.J.S. : *News Editing,* IIMC, New Delhi, 1980.

20. Krause, Jim : *Ideas Index : Graphic Effects and Typographic Treatment,* North Light Books, N.Y., 2000.

21. Moen, Daryl R. : *Newspaper : Layout and Design,* Iowa State University Press, 1989.

22. Bringhurst, Robert : *The Elements of Typographic Style,* Hartley and Marks, 1997.

23. Wong, Wucius : *Principles of Color Design,* John Wiley and Sons, 1997.

24. Hoy, Frank : *Photojournalism : The Visual Approach,* Prentice Hall, New Jersey, 1986.

25. Hurlburt, Allen : *Publication Design,* Van Norstrand-Reinhold and Co., 1977.

26. Lewis, John : *Typography,* Barrie and Jenikins, 1978.

27. Maghdam, Dinah : *Computers in Newspaper Publishing,* Marcel-Dekker Inc., 1978.

28. Worlock, Peter : *The Desktop Publishing Book,* Heinmann, 1983.

29. Davis, Bront : *Teach Yourself Photoshop,* Hayden Books, 1998.

30. Trewin, J. : *Presenting on TV and Radio,* UniLib Book Distributors, 2004.

31. Keith, M.C. : *The Radio Station,* UniLib Book Distributors, 2004.

32. Chapman, N. : *Digital Media Tools,* UniLib Book Distributors, 2004.

33. Bruce, G. : *Practical Recording Techniques,* UniLib Book Distributors, 2004.

34. Stovall, J. : *Writing for the Mass Media,* Pearson Education, 2007.

35. Lorenz, A. and Vivian, J. : *News : Reporting and Writing,* Pearson Education, 1996.

36. Shook, F. : *Television Field Production and Reporting,* Pearson Education, 2006.

Reference Books :

1. Kennedy, A. : *Beyond the Inverted Pyramid,* Bradford Books, 1992.
2. Thomas Sunny : *Writing for the Media,* Vision Books Ltd., N.D., 1997.
3. Singh, Chandrakant : *Before the Headlines : A Handbook of Television,* MacMillan India, New Delhi, 1999.
4. Shrivastava, K.M. : *Radio and TV Journalism,* Sterling Publishers, New Delhi, 1989.
5. Parthasarthy, Rangaswami : *Here is the News : Reporting for the Media,* Sterling Publishers, 1994.
6. Shaw, Eva : *The Successful Writer's Guide to Publishing Magazine Articles,* Rodgers and Nelson Publishing Co., 1998.
7. Wells, Gorden : *The Craft of Writing Articles,* Allison and Burby Inc., 1997.
8. Yudkin, Marcia : *Writing Articles About the World Around You,* Writer's Digest Books, 1998.
9. Wray, Cheryl Sloan : *Writing for Magazines : A Beginners Guide,* NTC Publishing Group, 1996.
10. Hay, Vicky : *The Essential Feature : Writing for Newspapers and Magazines,* Columbia University Press, 1991.
11. Hennessey, Brendan : *Writing Feature Articles : A Practical Guide to Methods and Markets*, Focal Press, 1997.
12. Wilber, Rick : *Magazine Feature Writing,* Bedford Books, 1994.
13. Campbell, Lawrence : *A Guide to Radio-TV Writing,* Iowa State University Press, Iowa, 1950.
14. Cohen, Akiba : *The Television News Interview,* Sage, London, 1987.
15. Williams, Robin : *The Non Designers Design Book,* Peachpitt Press, 1994.
16. Fink, Anne : *Faces on the Edge : Type Design in the Digital Age,* Peachpitt Press, 1998.
17. Allen, John : *Newspaper Design,* Harper Bro., 1947.

18.	Craig, James	:	*Production for the Graphic Designer,* Watson-Gutpill Publications, N.Y., 1979.
19.	Arnold, Edmund C.	:	*Functional Newspaper Design,* Harper and Row, 1956.
20.	Davis, Alec	:	*Graphics : Design into Production,* Faber and Faber, 1973.
21.	Sutton, Albert A.	:	*Design and Make-up of the Newspaper,* Prentice Hall, 1970.
22.	Evans, Harold	:	*Newspaper Design,* Holt, Rinehart & Winston, 1974. Universitty of Michigam ISBN No. 0030913489

POLICE ADMINISTRATION

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Outline of tests, syllabi and courses of reading

Paper-A : POLICE PERSONNEL ADMINISTRATION

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

(A) Course Objectives :

The objective of this course is to give inputs to the students on the various aspects of personnel relevant to police administration. Effort is made to impart knowledge to the candidates about the meaning, nature, scope and significance of Police Personnel Administration in India. In addition, job analysis, job description, pay policy, compensation and fringe benefits have been discussed. Further, the recruitment, training and promotion system in Police Service in India has been included in the syllabus. The students are also taught about the performance appraisal, police ethics, corruption in police, code of conduct and discipline along with doctrine of pleasure.

(B) Pedagogy of the Course Work :

90 per cent of the Course Content would be delivered through Lecture Method and rest 10 per cent would comprise of two internal examinations and attendance.

(C) Instructions for Paper Setters and Candidates :

- There shall be 9 questions in all.
- The first question shall be **compulsory** and be short answer type containing 12 short questions spread over the whole syllabus and to be answered in about 25 to 30 words, each. The candidate is required to attempt any 9 short answer type questions carrying 2 marks each (9×2=18 marks).
- Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall have two questions and the candidates shall attempt one question from each unit i.e. four questions in all. Each question will carry 18 marks (4×18=72 marks).

Course Content :

Unit-I

Meaning, Nature and Scope of Police Personnel Administration; Significance of Police Personnel Administration in India; Career System in Police Administration.

Unit-II

Job Analysis and Job Description of Police Personnel – Concept and Significance; Pay Policy, Compensation and Fringe Benefits.

Unit-III

Recruitment : Meaning and Methods; Recruitment of Police Personnel in India; Training : Meaning and Types; Training of IPS Officers and State Police Services in India; Promotion : Meaning and Principles; Promotion of Police Personnel in India.

Unit-IV

Performance Appraisal; Police Ethics and Code of Conduct; Corruption – Meaning and Causes; Corruption in Police Force; Disciplinary Actions; Doctrine of Pleasure – Safeguards against Arbitrary Punishment.

Essential Readings

1. Administrative Reforms Commission : *Report on Personnel Administration,* 11th Report, Government of India, New Delhi.
2. Goel, S.L. : *Public Personnel Administration,* Deep & Deep Publication Pvt. Ltd., New Delhi, 2006.
3. Mamoria, C.B. : *Personnel Management,* Himalaya Publishing House, Mumbai, 2007.
4. Zehra, Kaneez : *Training of I.P.S.,* Ashish Publishing House, New Delhi 1993.
5. Muthuswamy, P. : *Swamy's Manual on Disciplinary Proceedings,* Swamy Publishers (P.) Ltd., Madras, 1986.

Further Readings :

1. Saxena, A.P. : Training and Development in Government, The Indian Institute of Public Administration, New Delhi, 1974.
2. Singh, Sahib Bhayana, Singh, Swinder and Dogra, R.K. : *Public Personnel and Financial Administration,* New Academics, Jalandhar,1990.
3. Ghosh, P. : *Personnel Administration in India,* Sudha Publications Private Limited, New Delhi, 1973.

Paper-B : LAW AND ORDER ADMINISTRATION

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

(A) Course Objectives :

The principal objective of the syllabi is to acquaint the students with the basics of Law and Order Administration in India. Specifically, the students will be imparted the knowledge about the meaning, nature, scope and significance of Law and Order Administration in India along with some basic issues and emerging patterns. The endeavour is to discuss in detail the role and significance of Primary and Auxiliary Agencies of Law and Order Administration. Further, the role of Law and Order Administration in crowd management, communal riots, agrarian and industrial conflicts has also been discussed. The students are also taught about the challenges before Law and Order Administration. In addition, the concept of National Security along with challenges to it, in contemporary terms have been included in the syllabus.

(B) Pedagogy of the Course Work :

90 per cent of the Course Content would be delivered through Lecture Method and rest 10 per cent would comprise of two internal examinations and attendance.

(C) Instructions for Paper Setters and Candidates :

- There shall be 9 questions in all.
- The first question shall be **compulsory** and be short answer type containing 12 short questions spread over the whole syllabus and to be answered in about 25 to 30 words each. The candidate is required to attempt any 9 short answer type questions carrying 2 marks each (9×2 = 18 marks).
- Rest of the paper shall contain 4 units. Each units shall have two questions and the candidates shall be given internal choice i.e. the candidates shall have two questions and the candidates shall attempt one question from each unit i.e. four questions in all. Each question will carry 18 marks (4×18=72 marks)

Course Content :

Unit-I

Meaning, Nature and Scope of Law and Order Administration; Significance of Law and Order Administration in India; Law and Order Administration – Some Basic Issues and Emerging Patterns.

Unit-II

Role, Significance and interrelationships of Law and Order Administration Agencies at the District level : The Police; Civil Administration; Court of Law; Prisons; and Health.

Unit-III

Role of Law and Order Administration in Crowd Management, Communal Riots, and Industrial & Agrarian Conflicts. Challenges before Law and Order Administration.

Unit-IV

National Security : Concept and Significance. Challenges to National Security in India – Contemporary Issues.

Essential Readings :

1. Bailey, David, H. : *The Police and Political Development in India,* Princeton University Press, New Jersey, 1969.
2. Sharma, K.K. : *Law and Order Administration,* National Book Organization, New Delhi, 1985.
3. Srivastava, Aparna : *Role of Police in a Changing Society,* A.P.H. Publishing Corporation, New Delhi, 1999.
4. Ghosh, S.K. and Rustamji, K.F. : *Encyclopaedia of Police in India,* Volume I, Ashish Publishing House, New Delhi, 1993.
5. Bharti, Dalbir : *Police Evam Log: Donon ke Adhikar Va Zimmedarian,* A.P.H. Publishing Corporation, New Delhi, 2007.
6. Singh, Yogendra : *Social Change in India: Crisis and Resilience,* Har-Anand Publications Pvt. Ltd., New Delhi, 2007.

Further Readings :

1. Mayhall, Pamela D. et.al. : *Police-Community Relations and the Administration of Justice*, Prentice-Hall Inc., New Jersey, 1995.
2. Ghosh, S.K. : *Keeping the Peace : For Whom the Bell Tolls* (Police Then and Now), Ashish Publishing House, New Delhi, 1989.
3. Vadackumchery, James & Kattakayam, John : *Human Behaviour and Law Enforcement,* Ashish Publishing House, New Delhi, 1995.
4. Owens, R. Glynn and Ashcraft, J. Barrie : *Violence: A Guide for the Caring Professions,* Croom Helm, New Hampshire, USA, 1985.

WOMEN'S STUDIES

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A : FEMINIST THEORY

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objective :
This course offers the prominent theories that feminists have formulated to explain the matrix of gender inequality from the nineteenth century to the present. It further seeks to introduce the student to the key debates within feminist theories as well as the shifts in feminism.

Note : The students are expected to have a general understanding of the various strands of feminism and not of any specific feminist thinker in this course.

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

In this paper, the candidate will be assessed for 90 marks on the basis of a written examination and 10 marks will be for internal assessment.

There shall be 9 questions in all. The first question shall be **compulsory** containing 12 short questions spread over the whole syllabus to be answered in about 25 to 30 words each. The candidate is required to attempt any 09 short answer type questions carrying 2 marks each (18 marks). Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidate shall attempt one question from each unit - 4 in all. Each question will carry 18 marks.

Course Contents :

Unit-I : Feminism

(a) Definition

(b) Emergence of Feminism with special focus on

- Women in French Revolution,
- Suffrage Movement in the U.S.A. (with reference to Seneca Falls Convention and Declaration of Sentiments),
- Suffrage Movement in the U.K.

(c) Three waves of Feminism.

Unit-II : Strands of Feminism-I

(a) Liberal Feminism,

(b) Socialist Feminism,

(c) Marxist Feminism.

Unit-III : Strands of Feminism-II

(a) Radical Feminism,

(b) Post modern Feminism,

(c) Psycho-analytic Feminism (Freudian).

Unit-IV : Strands of Feminism-III

(a) Black Feminism,

(b) Cultural Feminism,

(c) Eco-Feminism.

Essential Readings :

1. Agarwal, Bina : *The Gender and Environment Debate: Lessons from India, Feminist Studies,* Vol. 18, No.1, Spring 1992, pp 119-158.

2. Anderson, Margaret : *Thinking About Women*, Macmillan, New York, 1993.

3. Beasley, Chris : *What is Feminism?* Sage, London, 1999.

4. Bell Hooks : *Ain't I a Woman: Black Women and Feminism,* Pluto Press, London, 1982.

5. Chatterjee, Mohini : *Feminism and Gender Equality,* Aavishkar, Jaipur, 2005.

6. Collins, Patricia Hill : *Black Feminist Thought: Knowledge, Consciousness and the Politics of Empowerment*, Routledge, New York, 1999.

7. Desai, Leela : *Issues in Feminism,* Pointer Pub., Jaipur, 2004.

8. Molyneux, Maxinne and Steiberg, Deborah Lynn : *Mies and Shiva's Ecofeminism: A New Testament, Feminist Review*, No. 49, Spring 1995, pp 86-107.

Further Readings :

1. Banks, Olive : *Faces of Feminism: A Study of Feminism as a Social Movement,* St. Martin's Press, New York, 1981.

2. Freeman, Jo (ed.) : *Women: A Feminist Perspective,* Palo Alto, California : Mayfield, 1975.

3.	Kosambi, Meera	:	*Crossing Thresholds: Feminist Essays in Social History,* Permanent Black, New Delhi, 2007.
4.	Mishra, Binod (ed.)	:	*Critical Responses to Feminism,* Sapru & Sons, New Delhi, 2006.
5.	Shukla, Bhaskar A.	:	*Feminism : From Mary Wollstonecraft to Betty Friedan,* Sapru & Sons, New Delhi, 2007.
6.	Spender, Dale	:	*Feminism Theorists: Three Centuries of Women's Intellectual Traditions,* The Women's Press, London, 1982.

Paper-B : STATUS OF WOMEN IN INDIA-II

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

Objective :

This course aims to sensitize the student with the status of women in contemporary India, with a special focus upon the factual situation apart from the major issues confronting Indian women.

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

In this paper, the candidate will be assessed for 90 marks on the basis of a written examination and 10 marks will be for internal assessment.

There shall be 9 questions in all. The first question shall be **compulsory** containing 12 short questions spread over whole syllabus to be answered in about 25 to 30 words each. The candidate is required to attempt any 09 short answer type questions carrying 2 marks each (18 marks). Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidate shall attempt one question from each unit - 4 in all. Each question will carry 18 marks.

Unit-I : Women and Family :

(a) Origin of the Family,

(b) Types of Family,

(c) Gender Division of labour in family,

(d) Female headed households,

(e) Changing role of Women in the family.

Unit-II : Women and Work :

(a) Changing definition of work,

(b) Gender differential in Work Participation Rates,

(c) Obstacles to Women's Workforce Participation,

(d) Sexual Harassment at the Workplace.

Unit-III : Women, Religion and Caste :

(a) *Religion :* Women's Status in major Indian religions – Hinduism, Islam, Sikhism, Christianity.

(b) *Caste :* Introduction to caste system in India; Issues of dalit women.

Unit-IV : Violence against Women :

(a) Violence against Women : Definitions

(b) Types of violence :
(i) Incidence Based Violence :
- Domestic Violence
- Rape and Dowry

(ii) System Based Violence :
- Caste and Communal Violence
- Cultural Violence

Essential Readings :

1. Chatterjee, Meera : *A Report on Indian Women from Birth to Twenty,* NIPCCD, New Delhi, 1999.

2. Gopalan, Sarala and Shiva, Mira : *National Profile on Women, Health and Development,* VHAI & WHO, 2000.

3. Gopalan, Sarala : *Towards Equality - The Unfinished Agenda : Status of Women in India -2001,* National Commission for Women, New Delhi, 2002.

4. Menon-Sen, Kalayani and Shiva Kumar, A.K. : *Women in India ! How Free?, How Equal?* Report Commissioned by the Office of the United Nations Resident Coordinator in India, New Delhi, 2001.

5. *National Human Development Report, 2001,* Planning Commission, Govt. of India, March, 2002.

Further Readings :

1. Anand, Meena : *Dalit Women : Fear and Discrimination,* Isha Books, Delhi, 2005.
2. Bhattacharya, Rinki (ed.) : *Behind Closed Doors : Domestic Violence in India,* Sage, New Delhi, 2004.
3. Chanana, Karuna : *Socialization, Education and Women : Explorations in Gender-Identity,* Orient Longman, New Delhi, 1988
4. Choudhary, Prem : *Contentious Marriages: Eloping Couples: Gender, Caste and Patriarchy in Northern India,* OUP, New Delhi, 2007.
5. Desai, Neera and Krishnaraj, Maithreyi (ed.) : *Women and Society in India,* Ajanta Publication, New Delhi, 1987.
6. Lipman, Jean Blumen : *Gender-Roles and Power,* New Jersey, Prentice Hall Inc., 1984
7. Mohanti, Bedabati : *Violence Against Women : An Analysis of Contemporary Realities,* Kanishka Publications, New Delhi, 2005.
8. Mohini Giri, V. : *Deprived Devis : Women's Unequal Status in Society,* Gyan, New Delhi, 2006.
9. Nagar, N.S. : *Changing Status of Indian Women,* Vista International Pub., New Delhi, 2008.
10. Oakely, Ann : *Sex, Gender and Society,* Harper & Row Pub. London 1972.
11. Patel, Tulsi : *The Family in India: Structure and Practice,* Sage Publication, New Delhi, 2005
12. Rajawat, Mamta : *Dalit Women:Issues and Perspectives,* Anmol Pub., New Delhi, 2005.
13. Saradamoni K. (ed) : *Finding the Household:* Conceptual and Methodological Issues, Sage Publication, New Delhi, 2005.
14. Sen, Sujata : *Gender Studies,* Pearson, New Delhi, 2012.
15. Uberoi, Patricia (ed.) : *Family, Kinship and Marriage in India,* Oxford University Press, Delhi 1993.
16. Vardhan, Ranjay : *Female Headed Households in Patriarchal Society: A Sociological Study,* Indian Publishers Distributors, Delhi, 1999.

Readings in Hindi :

1. Kumar, Raj (ed.) : *Bhartiya Mahila,* Arjun Publishing, New Delhi, 2003.
2. Saraswat, Swapnil : *Mahila Vikas Ek Paridrishya,* Nayan Pub., New Delhi, 2005.
3. Sharma, Pragya : *Mahila Vikas Aur Sashaktikaran,* Aavishkar Pub., Jaipur, 2001.
4. Sharma, Pragya : *Bharatiya Samaj Mein Nari,* Aavishkar Pub., Jaipur, 2001.
5. Vohra, Asha Rani : *Adhunik Samaj Mai Istri,* Ji Navraj Prakashan, Delhi, 2005.

HUMAN RIGHTS & DUTIES

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A : HUMAN RIGHTS & DUTIES IN INDIA

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES:

1. For written paper, the students will be required to attempt five questions in all. Question No.1 will be compulsory comprising of 12 short answer type questions of 2 marks each and will cover the entire syllabus. The students are required to attempt nine short answer type questions out of 12 i.e. $9 \times 2 = 18$ marks.

   In addition to it, Question Nos. II to IX will consist of long answer (essay type) questions, two questions from each unit with internal choice carrying 18 marks of each i.e. $4 \times 18 = 72$ marks.

2. On an average, 15 hours are to be devoted to each unit.

3. For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

   The paper-setter must put note (3) in the question paper.

Objective :

This paper deals with the growth of human rights phenomenon in India focusing on its history, normative agenda and implementation mechanisms.

Unit-I : Evolutionary Phase: Evolution of the concept of Human Rights & Duties in India; Values of composite culture of India.

Unit-II : Contemporary Phase: Human Rights & Duties in Contemporary India.

Unit-III : Normative framework: Constitutional Values such as The Preamble, Fundamental Rights, Directive Principles, Fundamental Duties.

Unit-IV : Implementation of Human Rights: Constitutional Machinery; Statutary Machinery: National Human Rights/State Human Rights Commissions.

Suggested Readings :

1. Alston, Philip (ed.), 2004, *The Dictionary of Human Rights,* London: Europa Publications.
2. Basu, D.D., 2009, *Introduction to the Constitution of India,* Nagpur, New Delhi, NexisLexis Butterworths.
3. Brownlie, Ian and Guy S. Goodwin-Gill (eds.), 2006, *Basic Documents on Human Rights,* Oxford: Oxford University Press.
4. Brundland Commission, 1986 (Brundland Report), *Our Common Future,* Oxford : Oxford University Press.
5. Das, A.K., 2007, *Human Rights in India,* New Delhi : Sarup Book Publishers.
6. Das, A.K., Mohanty, P.K., 2007, *Human Rights in India,* New Delhi : Sarup and Sons.
7. Gandhi, P.R. (ed.), 2001, *Blackstone International Human Rights Documents,* New Delhi: Universal Law Publishing Co.
8. Kapoor, S.K., 2005, *Human Rights under International Law and Indian Law,* Allahabad : Central Law Agency.
9. Nirmal, C.J., 2009, *Human Rights in India,* New Delhi : Oxford University Press.
10. Robertson, David, 2004, *A Dictionary of Human Rights,* London: Europa Publications.
11. Rao, Ashwini, 2010, *Status of Human Rights in India,* New Delhi : Pacific Publications.
12. Steiner, Henry J. et al (eds.), 1996, *International Human Rights in Context,* Chapter 10, Oxford: Clarendon Press.
13. Sehgal, B.P.S., 2008, *Human Rights in India : Problems and Perspectives*, Delhi: Deep & Deep.

Paper B : EMERGING ISSUES OF HUMAN RIGHTS IN INDIA

Max. Marks	:	100
Theory	:	90 Marks
Internal Assessment	:	10 Marks
Time	:	3 Hours

INSTRUCTIONS FOR THE PAPER-SETTER AND THE CANDIDATES :

1. For written paper, the students will be required to attempt five questions in all. Question No.1 will be compulsory comprising of 12 short answer type questions of 2 marks each and will cover the entire syllabus. The students are required to attempt nine short answer type questions out of 12 i.e. $9 \times 2 = 18$ marks.

In addition to it, Question Nos. II to IX will consist of long answer (essay type) questions, two questions from each unit with internal choice carrying 18 marks of each i.e. 4×18 = 72 marks.

2. On an average, 15 hours are to be devoted for each unit.

3. For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment.

The paper-setter must put note (3) in the question paper.

Objective :

This paper focuses on some selected aspects of society where human rights tend to play a crucial role.

Unit-I : Human Rights & Working Class : Bonded labour, Unorganized labour, Problems of working women.

Unit-II : Human Rights & Criminal Justice : Rights of the accused, Rights of the victim.

Unit-III : Social Problems : Poverty, Illiteracy, Unemployment.

Unit-IV : Social Auditing : Media and Civil Society.

Suggested Readings :

1. Agarwal, Amita, 2004, *Human Rights for Survival of Civilization*, New Delhi: Kalinga Publications.

2. Baxi, Upendra (ed.), 1987, *The Right to be Human,* New Delhi: Lancer International.

3. Baxi, Upendra, 2002, *The Future of Human Rights,* New Delhi : Oxford University Press.

4. Lyer, V.R. Krishna, *Human Rights – A Judge's Miscellany,* New Delhi: B.R. Publishing Corporation.

5. Kannabiran, Kalpana, 2006, *Crafting Human Rights Cultures,* NALSAR, University of Law.

6. Kaushal Rachna, 2000, *Women and Human Rights in India,* New Delhi : Daya.

7. Mishra, R.C., 2005, *Human Rights and Policing,* New Delhi: Authors Press.

8. Papolam, T.S., A.N. Sharma, 1999, *Gender and Employment in India,* New Delhi : Vikas.

9. Sanajaoba, N., 2000, *Human Rights in the New Millennium,* New Delhi : Manas Publications.

10. Sen Shankar, 1998, *Human Rights in a Developing Society,* New Delhi : A.P.H. Publishing.

11. Shastri, Satish C. (ed.), 2006, *Human Rights, Development and Environmental Law: An Anthology,* New Delhi: Bharat Law House.

12. Vibhute, K.I., 2004, *Criminal Justice : A Human Rights Perspective of the Criminal Justice Process in India,* Lucknow: Eastern Book Company.

ENVIRONMENT CONSERVATION

B.A. /B.Sc.(GENERAL) SECOND YEAR EXAMINATION, 2015

Paper	*Name*	*Time in Hrs*	*Theory Marks*	*Internal Assessment Marks*	*Total Marks*
A	Air Pollution and Climate change	3	65	10	75
B	Soil waste and Disaster Management	3	65	10	75
Practical					
A	Air Pollution and Climate change	3	50		50
B	Soil waste and Disaster Management				
	Total Marks				**200**

Note: The practical will include survey and its project reports carrying 15 marks, 35 marks will be allotted to laboratory practicals. The number of hours for theory and practical per week shall be 5 hours and 4 hours respectively.

There will be 9 questions in all, two each from section I to 1V. Question No. I (17 marks) will be short answer type & will cover the whole syllabus. Candidates will attempt five questions in all, selecting one each (12 marks) from section I to IV and the first compulsory question.

Paper-A **AIR POLLUTION AND CLIMATE CHANGE**

Section-I

Atmosphere: Composition of atmosphere, Air Pollution: Sources, Classification (Natural and anthropogenic sources, primary and secondary pollutants), Smoke, Smog, Particulate Matter and Acid rain. Effects of air pollution on human health, plants, animals and material. Air (Prevention and control of Pollution) Act, 1981.

Section-II

Green House Effect: Definition, Green house gases, Sources of GHE, Global Warming.

Stratospheric ozone layer depletion. Role of agriculture to green house effect (paddy and livestock, biomass burning) in GHE.

Section-III

Noise pollution: Definition, Sources, Noise standards, Silence Zones, Consequences of noise pollution.

Radioactive Pollution: Definition**,** Sources, effect on human health.

Mineral Resources: Types and Importance of minerals; Mineral extraction and environmental problems (limestone mining in Moussourie hills, coal mining in Jharkhand).

Section-IV

Control: Measures and techniques to control air pollution in vehicles and industries, Green building concept, CDM (Clean Development Mechanism), Carbon sequestration.

Control of noise pollution.

PRACTICAL

PAPER-A

To monitor the level of Dust fall, SPM in surrounding atmosphere.

To monitor the level of NOx and SO2 in surrounding atmosphere.

To prepare a survey report on different sources causing air pollution in local area.

To prepare a survey report on noise pollution in your locality.

To visit, collect data and prepare report regarding air and water pollution from sugar mill, thermal plant, cement factory in the adjoining area like Nawanshahr, Ropar, and Hoshiarpur.

Paper-B SOLID WASTE AND DISASTER MANAGEMENT

Section-I

Solid waste: Sources, nature and characteristics; categories of solid waste, Biodegradable and non biodegradable waste; Plastic nuisance. Solid waste disposal: Sanitary landfills, Composting, optimum conditions for composting, Pyrolysis, Incineration and Refuse Derived Fuels.
Fly Ash utilization, MSW Handling Rules, 2000.

Section-II

Disaster management: Natural disasters: floods, earthquakes, Tsunami, land slides, drought, forest fire, and Tropical cyclones.

Disaster management: pre-disaster phase, actual disaster phase, post- disaster phase.

Section-III

Bioenergy and waste water treatment: Biogas plant: Construction and working of biogas plants, its advantages and disadvantages. Vermi-culture and Vermi-composting, Biofertilisers, Biofuels.

Waste water treatment (STP and ETP), A brief account of primary, secondary and tertiary treatment.

Section-IV

Organic farming: Definition, Need of Organic Farming, Methods: Soil management, weed management, control of pests. Its advantages.
Organic Food, its benefits.

Waste lands: Definition, Categories of waste lands, Desertification, ways of reclamation.

PRACTICAL

Paper-B

A visit to Composting/Vermicomposting Unit.

To prepare a survey report of municipal waste of your city (Generation to disposal).

Classify the waste of your municipality waste (Source based and Waste based)

To visit any land fill site and prepare a tour report.

To prepare a list of herbicides / pesticides / insecticides commonly used in this Area

To make a list of various botanical pesticides available in the market

Assignments/project as assigned by the teacher

Books Recommended

1. Critchfield, Howard J. General Climatology, Second Edition, Prentice-Hall, Inc., 1982.
2. Houghton, J. Global Warming, The Complete Briefing, 3rd Edition, Cambridge Univ. Press, 2006.
3. Hill, M. K. Understanding Environmental Pollution, Cambridge Univ. Press, 2008.
4. Malsin, M. Global Warming, A Very Short Introduction, Oxford Univ. Press, 2009
5. Tchobanogloas, G. Integrated Solid Waste Management: Engineering, Principle and Management. McGraw Hill, USA. 1993.
6. Singh, H.M. Solid Waste Management. Alfa Publications, Delhi. 2010.
7. Bhatnagar R.K. & Palta R.K. Earthworm Vermin Culture & Vermi Composting, Kalyani, Pub.2007.
8. Ranganathan L.S Vermi biotechnology from Soil Health to Human Health, Updesh purodit pub. Jodhpur, 2006
9. Aggarwal, A. K. Water Pollution, APH Publications, New Delhi, 2008.

American Public Health Association (2003) Standard Methods for Examination of Water and Wastewater, APHA, New York.

10. Sengupta, B., John, S. & Abraham, T. — Forth Estate, A Handbook on Air, Water and Land Pollution. IDMA Foundation for Sustainable Development and BR Publications Corporation, Delhi. 2006.

11. Aggarwal, A. K. — Noise Pollution. APH Publications, New Delhi, 2008.

12. Veeresh, G. K. — Organic Farming. Foundation Books, Delhi, 2006.

13. Gehlot, D. — Organic Farming: Components and Management, Agrobios, Jodhpur, 2010.

14. Singh, K. K. & Aleyer, L. — Disaster management: Manmade Disasters. APH Publications. 2010.

15. Dhunna, M. — Disaster Management, Vayu Education of India, Delhi, 2009.

16. Sharma & Sharma — Natural Disasters, APH Puiblications, 2009.

17. Dara, S.S. — A Textbook of Environmental Chemistry. 2nd Edition. S. Chand & Company Ltd., Ram Nagar, New Delhi, 2010.

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HOME SCIENCE

B.A. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper	**Theory**	**Practical**
	Period/Week	
Clothing	3/Week	4/Week
Textiles	4/Week	3/Week

SCHEME OF EXAMINATION

Name of Paper	*No. of Papers*	*Time in hours*	*Marks*	*Internal Assessment*	*Total*
Clothing	1	3 hrs.	40	10	100
Textiles	1	3 hrs.	50		
Clothing Practical		3 hrs.	45	15	60
Laundry Practical		1 hr.	30	10	40
			Total of Theory & Practical :		**200**

Note :
1. Internal assessment should be based on : Assignments/test/seminars and attendance for Theory & Practical.
2. Practical examinations will be held before the Final Theory Exams.
3. Practical group should have 12-15 students.

Paper A : CLOTHING (THEORY)

Maximum Marks : 50
Paper : 40
Internal Assessment: 10
Time : 3 Hours/Week

INSTRUCTIONS FOR THE PAPER SETTER :

The question paper will consist of five sections : A, B, C, D & E. Sections A, B, C & D will have two questions from the respective sections of the syllabus & will carry 8 marks each. Section E will consist of 8 short answer type questions covering the entire syllabus uniformly carrying one mark each.

INSTRUCTIONS FOR THE CANDIDATES :

Candidates are required to attempt one question each from the Sections A, B, C & D of the question paper and the entire section E.

SECTION-A

1. Equipments & supplies used for Clothing-Construction on their use & care
2. Sewing Machine :
   (a) Parts of Sewing Machine and its accessories.
   (b) Common defects in sewing machine & their remedies.
3. Care of Sewing Machines.

SECTION-B

1. Recording of body measurements.
2. Care to be taken while taking body measurements.
3. Preparation of fabric for construction, shrinking and straightening.
4. Drafting
   I. Tools of Drafting
   II. Steps involved in drafting from measurement to garment construction i.e.
   - Measurements
   - Making a draft
   - Placement of draft on fabric
   - Cutting
   - Stitching
   - Finishing

   III. Points to be kept in mind while making a draft.
   IV. Terms used in drafting
   - Seam allowance
   - Ease
   - Selvedge, Dart, Pattern

   Grain line, stay stitching.

SECTION-C

1. Elements of art such as colour, form, texture, shape and line.
2. Principles of design such as Harmony, Balance, Rhythm, Emphasis & Proportion in relation to clothing.

SECTION-D

1. Introduction to Fashion, Fad and Style.
2. Selection of suitable clothes for following age groups :
   (a) Infants
   (b) Toddlers
   (c) School going children
   (d) Adolescents
   (e) Adults
   (f) Elderly
3. Care & storage of garments of cotton, wool, silk.

Paper-B : TEXTILES (THEORY)

Maximum Marks : 50
Time : 3 Hours/Week

INSTRUCTIONS FOR THE PAPER SETTER :

The question paper will consist of five sections A, B, C, D & E. Sections A, B, C & D will have two questions from the respective sections of the syllabus & will carry 10 marks each. Section E will consist of 10 short type/objective type questions covering the entire syllabus uniformly carrying one mark each.

INSTRUCTIONS FOR THE CANDIDATES :

Candidates are required to attempt one question from each of the Sections A, B, C & D of the question paper and entire section E.

SECTION-A

1. Classification of textile fibres and manufacture (In Brief).
2. Properties of different fibres :
   (a) Cotton
   (b) Linen
   (c) Silk
   (d) Wool
   (e) Nylon
   (f) Polyester
   (g) Rayon—Viscose & Acetate.

SECTION-B

1. Different types of yarns
   - Simple
   - Novelty and Bulk yarns in brief.
2. Fabric Construction—A brief study of basic weaves
   (a) Simple weaves
      - Rib
      - Basket
   (b) Twill weaves—
      - Even Twill weave
      - Uneven Twill weave
   (c) Satin weave
   (d) Sateen weave
   (e) Knitting, Felting, Bonding.

SECTION-C

1. Finishing—
2. Type of finishes
3. Importance of finishes
4. Finishing—Sizing, desizing, calendering, sanforising, mercerization, crease resistant, water proof & water repellent, flame resistant & flame proof.

SECTION-D

1. Application of colour on fabrics.
   (a) Dyeing—Simple dyeing of cotton.
   (b) Resist—Tie, Dye and Batik
2. Printing :
   (a) Block Printing
   (b) Screen Printing
   (c) Roller Printing
3. Methods of Laundry/Washing.
4. Types of Starches and Blues.

PRACTICAL

CLOTHING		LAUNDRY	
Maximum Marks	: 60	Maximum Marks	: 40
Paper	: 45	Paper	: 30
Internal Assessment:	15	Internal Assessment:	10
Time : 3 Hours/Week		Time : 1 Hours/Week	

A. CLOTHING

1. *Make samples of the following :*
   (a) Tacking, running stitch, hemming, Back Stitch, Button hole stitch, Fastners.
   (b) Seams—Flat seam, Counter hem, Mantua maker, Run & Fell seam, French seam.
   (c) Processes—Continues wrap, two piece placket opening, pleats gathers into a band, tucks.
   (d) Joining- biased strip, shaped facing on V neck line piping on round neck.

2. Embroidery—any '10' fancy stitches from following.
   Stem stitch, chain stitch, feather stitch, lasy daisy bullion, cross stitch, satin stitch, herringbone, French knot, web & button hole.

3. *Drafting of the following:*
   (i) Child's bodice block
   (ii) Sleeve—Plain sleeve, Puff sleeve
   (iii) Collars—Flat and raised peterpan collar, cape collar, baby collar.
   (iv) Jangia, Bloomer.
   (v) Drafting of- Adults bodice block, Plain sleeve
   (vi) Petticoat
   (vii) Salwar
   (viii) Kameez

4. Construction of following garments :
   (i) Children : Bloomer, Jangia, Child's gathered frock with any sleeve and collar.
   (ii) Ladies : Saree, Petticoat, Blouse, Salwar Kameez.

5. An article to be prepared by Block Printing method

Reference Books :

1. *Singer Sewing Book,* Mary Brocks Bichen, McGraw Hill Book Company, Inc., New York, Toronto, London, 1953.

2. *Basic Processes & Clothing Construction,* Sherie Doongaji & Raushni Deshpande, Raaj Prakashan, New Delhi (Fourth Revised edition).

3. *Pattern Drafting Vol. III, Dress Making,* Kamakura—Shobo Publishing Co. Ltd., Tokyo, Japan, 1972 (Fifth Edition).

4. *A Manual of Children Clothing,* Savitri Pandit, Orient Longmans Ltd., Bombay, 1967.

5. *Practical Dress Design, Principles of Fitting and Pattern Making,* Mabel D. Erwin, The Macmillan Company, New York, Ninth Printing, 1966.

6. *Indian Embroidery, Its Variegated Charms, Savitri Pandit,* Faculty of Home Sciences, Baroda, 1976, First Edition.

7. *Singer Sewing Book,* Gladys Cunningham, Pub. Golden Press, New York.

8. *The Basic Book of Sewing,* Eve Harlow Pub. Octopids Books Ltd., London.

9. *Simple Dress Making 2,* Maureen Goldsworthy Pub. Mills & Boon Ltd., London.

10. *Encyclopaedia of Dress Making,* Pub. Marshall Caven Dish Books Ltd., London.

B. LAUNDRY Time : 1 hours/week

1. Testing of cotton, wool & silk, nylon by microscope & by burning test.

2. Stain Removal : Rust, Coffee, Tea, Paint, Nail Polish, Lipstick, Perfume, Blood, Boot Polish, Ink (Ball Pen), Curry and Juice.

3. Spot cleaning of woollen garment.

4. Simple house hold dyeing of cotton fabric 12'' × 12''.

5. Preparation of an article of Tie and Dye.

6. Laundry & finishing of following garments :

   (a) stiffening, blueing & ironing

   (b) Cotton : Salwar, Kameez, Saree.

   (c) Synthetic : Gathered Frock/Pleated Skirt.

   (d) Silk : Blouse

   (e) Woollen : Cardigan/Pullovers.

Projects:

- Market survey of laundry reagents detergents, detergent cakes, bleaches, softeners, washing soap and blues.
- Visit to Block Printing unit and submitting a brief report and teaching the community

Reference Books :

1. *Textile Fabrics and their Selection,* Isable B. Wingate Prentice Hall, Inc., Englewood Cliffs, N.J., 1970.

2. *Fundamentals of Textiles and their Care,* Susheela Dantyagi, Orient Longmans, Bombay, Calcutta, Madras, New Delhi, Reprinted, 1968.

3. *Household Textiles and Laundry Work,* Durga Deulkar, Atma Ram & Sons, New Delhi, 1973.

4. *Indonesian Batik & Ikat,* Former Bedrich, Butterworth, London, 1982.

5. *Ideas for Fabric Printing and Dyeing Gooch,* Peter H. Charles Cribner's, New York, 1974.

6. *Encyclopaedia of Textiles,* Fibres and Nonwoollen Fabrics, Grayson Martin, John Wiley, New York, 1984.

7. *Household Textiles & Laundry Work,* Durga Deulkar, Pub. Atma Ram & Sons.

8. *Batik Kala, Sarla Sudershan,* Pub. Pustak Mahal Khari Baoli, Delhi.

9. *Tie-Dyed Textiles of India—Veronica Murphy & Rosemary Crill,* Pub. Mapiu Publishing Pvt. Ltd. Ahmedabad.

AGRICULTURE (ELECTIVE)

B.A./B.Sc. (GENERAL) SECOND YEAR EXAMINATIONS, 2015

Paper - A

Max. Marks	:	75 Marks
Theory	:	70 Marks
Internal Assessment	:	5 Marks
Time	:	3 Hours

Period per week : 1. THEORY – Three of 45 minutes duration.
2. PRACTICAL – One of three hours duration.

Instructions for the paper setter:

1. Question paper shall have five sections.
2. Section A shall comprise 10 short answer type questions covering whole syllabus and will be compulsory. Each question will carry 1 (one) mark.
3. Section B, C, D and E shall have two questions each from respective sections, out of which one from each Unit is to be permitted. Each question will be of 15 (fifteen) marks.
4. Total five questions are to be attempted.

Unit-I

Important parts of the tractor, combine, thresher and their maintenance.
Study of various agricultural tools and implements.

Unit-II

Maintenance of grassy lawns and flower beds
Raising of winter and summer ornamental flowers (rose, gladioli, dahlias, dianthus, foliage plants).
Fundamentals of land scape of an educational institute, factory, panchayat lands and office buildings.

Unit-III

Introduction to medicinal, aromatic and spice plants
Growing of non-conventional plants
Basic idea of G. M. crops
Fundamentals of land measurements and land revenue

Unit-IV

Agricultural aid on works :
Dairy
Poultry
Fishery
Mushroom Cultivation
Animal Husbandry
Bee Keeping

PRACTICAL

Max. Marks : 25 Marks
Practical : 20 Marks
Internal Assessment : 5 Marks

Visit to Poultry Farms, Fish Ponds, Apiary, Dairy Farms
Raising various ornamental flowers of different seasons
Maintenance of grassy lawns
Study parts of tractor and combine

Books :

1. Agriculture Engineering Vol.I and Vol. II, A. M. Michae, T.P. Ojha, 2006, Jain Brothers, New Delhi.
2. Mushroom Growing, 1997, H. S. Garcha, P. A. U., Ludhiana.
3. Hand Book of Agriculture, 2008, ICAR, New Delhi.
4. Farm Machinery, An Approach, S.C. Jain and Grace Philips, 2006, Standard Publishers, Distributors, New Delhi.
5. Fundamentals of Dairy Chemistry, Wong, Jenness, Keeney and Marth, 1998, CBS Publishers and Distributors, New Delhi.
6. Milk and Milk Products, Clarence Henry Eckles, Willes Barnes Combs and Harold Macy, Delhi, 2006, Tata McGraw Hill Publishing Company Limited, New Delhi.
7. Introduction to Plant Biotechnology, H. S. Chawla, 2008, Oxford and IBH Publishing Company Pvt. Ltd., New Delhi 2.
8. Plant Biotechnology, B. D. Singh, 2009, Kalyani Publishers, Ludhiana.
9. Aquarium and Fish Keeping, C. B. L. Srivastva, 2006, Kitab Mahal Allahabad.
10. Medicinal Plants of India, R. Zafar, 2006, CBS Publishers and Distributors, New Delhi.
11. Spices and Condiments, J. S. Pruthi, 2008, National Book Trust, New Delhi.

Paper - B

Max. Marks : 75 Marks
Theory : 70 Marks
Internal Assessment : 5 Marks
Time : 3 Hours

Period per week : 1. THEORY – Three of 45 minutes duration.
2. PRACTICAL – One of three hours duration.

Instructions for the paper setter:

1 Question paper shall have five sections.

2 Section A shall comprise 10 short answer type questions covering whole syllabus and will be compulsory. Each question will carry 1 (one) mark.

3 Section B, C, D and E shall have two questions each from respective sections, out of which one from each Unit is to be permitted. Each question will be of 15 (fifteen) marks.

4 Total five questions are to be attempted.

Unit – I

Tillage, Fundamentals and Principles of Tillage, Zero Tillage, Tillage Implements.

Unit – II

Cultivation Practices of Potato, Tomato; Bhindi, Cabbage; Onion and Asparagus

Unit – III

What are Timbers and Lumbers?
Raising of Dalbergia; Tectona; Poplar; and Eucalyptus

Unit- IV

Statistical Methods for Agricultural workers including Mean, Mode, Median Chi-Square, Standard Deviation.

PRACTICAL

Max. Marks	:	25 Marks
Practical	:	20 Marks
Internal Assessment	:	5 Marks

Visit to Agricultural Farms for Fundamental Aspects of Tillage
Raising Forest Nursery
Cultivation of Seasonal Vegetables
Numerical Problems of Statistics

Books :

1. Principles of Agronomy**,** T. Yellamanda Reddy and G. H. Sankara Reddy, 2008, Kalyani Publishers, Ludhiana.

2. Statistical Methods for Agricultural Workers, V.G. Panse and P.V. Sukhatme, 1997, ICAR, New Delhi.

3. Modern Techniques for Raising Field Crops, Chidda Singh, Prem Singh and Rajbir Singh, 2009, Oxford and IBH Publishing Co. Pvt. Ltd., New Delhi.

4. Vegetable Crops of India, P.C. Das, 1998, Kalyani Publishers, Ludhiana.

5. Potato Cultivation, 1998, Nandpuri and Kumar, P.A.U., Ludhiana.

MATHEMATICS

B.A./B.Sc. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-I : ADVANCED CALCULUS

Max. Marks	:	67
Theory	:	60 marks
Internal Assessment	:	7 marks
Time	:	3 Hours

Note : 1. The syllabus has been split into two sections—Section A and Section B. Four questions will be set from each section.

2. A student will attempt five questions in all selecting at least two questions from each section. Each question will be of 12 marks.

3. The teaching time shall be five periods (45 minutes each) per paper per week including tutorials.

SECTION–A

Definition of a sequence and subsequence. Theorems on limits of sequences. Bounded and monotonic sequences. Cauchy's convergence criterion. Series of non-negative terms. Comparison Tests. Cauchy's Integral Test. Ratio Tests. Cauchy's Root Test. Raabe's Logarithmic., De Morgan and Bertrand's tests. Gauss Test. Alternating series. Leibnitz's theorem. Absolute and conditional convergence.

Vector differentiation, Gradient, divergence and curl.

Continuity. Sequential continuity. Uniform continuity, Proof of chain rule of Differentiability.

SECTION-B

Limit and continuity of functions of two and three variables. Partial differentiation. Change of variables. Partial derivation and differentiability of real-valued functions of two and three variables. Schwarz and Young's theorem. Statements of Inverse and implicit function theorems and applications.

Euler's theorem on homogeneous functions. Taylor's theorem for functions of two and three variables. Jacobians. Envelopes. Evolutes. Maxima, minima and saddle points of functions of two and three variables. Lagrange's multiplier method.

References :

1.	T.M. Apostol	:	*Mathematical Analysis,* Narosa Publishing House, New Delhi, 1985.
2.	R.R. Goldberg	:	*Real Analysis,* Oxford & I.B.H. Publishing Co., New Delhi, 1970.
3.	P. K. Jain and S. K. Kaushik	:	*An Introduction to Real Analysis,* S. Chand & Co., New Delhi, 2000.
4.	Gorakh Prasad	:	*Differential Calculus,* Pothishala Pvt. Ltd., Allahabad.
5.	Murray R. Spiegel	:	*Theory and Problems of Advanced Calculus,* Schaum Publishing Co., New York.
6.	S.C. Malik	:	*Mathematical Analysis,* Wiley Eastern Ltd., New Delhi.
7.	Chandrika Prasad	:	*Text Book on Algebra and Theory of Equations,* Pothishala Pvt. Ltd., Allahabad.
8.	N. Piskunov	:	*Differential and Integral Calculus,* Peace Publishers, Moscow.
9.	Shanti Narayan	:	*A Course of Mathematical Analysis,* S. Chand and Company, New Delhi.
10.	Murray, R. Spiegel	:	*Vector Analysis,* Schaum Publishing Company, New York.
11.	Shanti Narayan	:	*A Text Book of Vector Calculus,* S. Chand & Co., New Delhi.

Paper-II : DIFFERENTIAL EQUATIONS

Max. Marks	:	67
Theory	:	60 marks
Internal Assessment	:	7 marks
Time	:	3 Hours

Note : 1. The syllabus has been split into two sections—Section A and Section B. Four questions will be set from each section.

2. A student will attempt five questions in all selecting at least two questions from each section. Each question will be of 12 marks.

3. The teaching time shall be five periods (45 minutes each) per paper per week including tutorials.

SECTION–A

Exact differential equations. First order higher degree equations solvable for x, y, p. Clairaut's form and singular solutions. Geometrical meaning of a differential equation. Orthogonal trajectories. Linear differential equations with constant coefficients. Homogeneous linear ordinary differential equations. Linear differential equations of second order. Transformation of the equation by changing the dependent variable/the independent variable. Method of variation of parameters.

SECTION-B

Series solution of differential equations—Power Series method, Bessel and Legendre equations. Bessel functions of First and Second kind and their properties. Legendre function. Generating Function. Recurrence relation and orthogonality of Bessel and Legendre function.

Laplace Transformation—Linearity of the Laplace transformation. Existence theorem for Laplace transformations, Shifting theorems, Laplace transforms of derivatives and integrals, Multiplacation of t^n, Division by t.

Inverse Laplace transforms- Linearity property, Shifting theorems, Inverse Laplace transforms of derivatives and integrals, Convolution theorem, Solution of system of ordinary differential equations using Laplace transforms.

References :

1. Erwin Kreyszig : *Advanced Engineering Mathematics,* John Wiley & Sons Inc., New York, 1999.
2. D.A. Murray : *Introductory Course on Differential Equations,* Orient Longman (India), 1967.
3. A.R. Forsyth : *A Treatise on Differential Equations,* Macmillan and Co. Ltd., London.
4. Bill, R.J.T : *Elementary Treatise on Coordinate Geonmetry of Three Dimensions,* Macmillan India Ltd., 1994.
5. Sneddon, I.N. : *Element of Partial Differential Equations,* McGraw Hill, 1957.
6. Murry & Sphegiel : *Schaum Series, Laplace Transforms.*
7. Ross, S.L. : Differential Equations, John Willey & Sons, 2004.

Paper-III : MECHANICS

Max. Marks	:	66
Theory	:	60 marks
Internal Assessment	:	6 marks
Time	:	3 Hours

Note :
1. The syllabus has been split into two sections—Section A and Section B. Four questions will be set from each section.
2. A student will attempt five questions in all selecting at least two questions from each section. Each question will be of 12 marks.
3. The teaching time shall be five periods (45 minutes each) per paper per week including tutorials.

SECTION-A

Statics :

Basic notion, Newton's laws of motion, system of two forces, parallelogram law of forces, resultant of two collinear forces, resolution of forces, moment of a force, couple, theorems on moment of a couple, coplanar forces, resultant of three coplanar forces acting at a point, algebraic method of finding the resultant of any number of coplanar concurrent forces, theorem of resolved parts, resultant of two forces acting on a rigid body, Varignon's theorem, generalized theorem of moments.

Equilibrium of two concurrent forces, equilibrium conditions for any number of coplanar concurrent forces, Lami's theorem. $\lambda - \mu$ theorem, theorems of moments, resultant of a force and a couple.

Equilibrium conditions for coplanar non-concurrent forces.

Friction : Definition and nature of friction, laws of friction, equilibrium of a particle on a rough plane.

SECTION-B

Dynamics :

Motion of a particle with constant acceleration, acceleration of falling bodies, motion under gravity, motion of a body projected vertically upwards: Motion of two particles connected by a string, motion along a smooth inclined plane, constrained motion along a smooth inclined plane. Variable acceleration: Simple harmonic motion, elastic string.

Curvilinear motion of a particle in a plane: Definition of velocity and acceleration, projectiles, motion in a circle, motion under constraint, central force motion.

Work, power, conservative fields and the potential energy, work done against gravity, potential energy of a gravitational field.

Relative motion, relative displacement, velocity and acceleration, motion relative to a rotating frame of reference.

Linear momentum, angular momentum, conservation of angular momentum, impulsive forces, principle of impulse and momentum, motion with respect to centre of mass of a system of particles, collisions of elastic bodies, loss of energy during impact.

References :

1. S.L. Loney : *Statics,* Macmillan and Company, London.
2. R.S. Verma : *A Text Book on Statics,* Pothishala Pvt. Ltd., Allahabad.

3. S.L. Loney : *An Elementary Treatise on the Dynamics of a Particle and of Rigid Bodies,* Cambridge University Press, 1956.

4. K.R. Chaudhery, and A.C. Aggarwal : *Elements of Mechanics, Statics and Dynamics,* S. Chand and Company.

5. S.L. Loney : *The Elements of Statics and Dynamics,* 5th edition, Cambridge University Press, 1947.

6. Donald, T., Green Wood : *Principles of Dynamics,* Second Edition, Prentice Hall of India.

7. M. Ray : *A Text Book on Dynamics,* S. Chand and Company, 1989.

COMPUTER SCIENCE

B.A./B.Sc. (GENERAL) SECOND YEAR EXAMINATION, 2015

Scheme of Examination

					Exam. Hrs.	*Ext.*	*Int.*	*Max. Marks*
Paper-A	:	Theory	:	Computer Organization & Architecture	3	65	10	75
Paper-B	:	Theory	:	Object Oriented Programming (Using C++)	3	65	10	75
Paper-C	:	Practical	:	Practicals based on Paper A and Paper B	3	50	--	50

Note : Practical marks will include the appropriate weightage for proper maintenance of lab. record.

Syllabus and Courses of Reading

Paper-A : COMPUTER ORGANIZATION AND ARCHITECTURE

Objective :

The course is designed to provide inside details of computer system including processor, memory card & I/O devices and to solve problems using Assembly Language Programming.

Note : (i) The question paper will consist of four sections.

(ii) Examiner will set total of *nine* questions comprising *two* questions from each section and *one* compulsory question of short answer type covering whole syllabi.

(iii) The students are required to attempt *one* question from each section and the compulsory question.

(iv) All questions carry equal marks unless specified.

SECTION-A

1. *Representation of Information :* Number system, integer and floating point representation, character codes (ASCII, EBCDIC), error detection and correction codes.

SECTION-B

2. *Basic Building Blocks :* Boolean algebra, combinatorial logic design, flip-flops, registers, counters, ALU : arithmetic and logic unit operations, organization of control units, memory : types and organization, peripheral devices : I/O devices (video terminals and printers) and controllers, storage devices (tapes and disks), Programmed and interrupt control mechanism, I/O controllers, bus bandwidths.

SECTION-C

3. *Assembly Language Programming for INTEL 8086/8088 Machine :* Programmers model of a machine, example of a typical 16 or 32 bit processor, registers, addressing modes, instruction set, use of assembly language for specific programs for typical problems like : Table search, subroutines, symbolic and numeric manipulations and I/O.

SECTION-D

4. *System Maintenance :* Introduction to various physical components of a computer, physical inspection of IBM-PC and internal cards, Diagnostics on IBM-PC, Functional description of various modules and cards. Various types of displays and other peripherals used in IBM-PC, Installing software, and detection of viruses and protection on IBM-PC.

References :

1. Mano, M.M. : *Computer System Architecture,* 3rd Ed., Dec., 1998, PHI.
2. Tannenbaum, A.S. : *Structured Computer Organization,* 3rd Ed., Oct., 1993, PHI.
3. Gear, C.W. : *Computer Organisation and Programming,* McGraw Hill, 1975.
4. Langholz, G., Grancioni, J. & Kandel, A. : *Elements of Computer Organization,* Prentice Hall International, 1988.
5. Hayes : *Computer Architecture and Organization,* McGraw Hill, International Edition.
6. Sloan, M.E. : *Computer Hardware and Organization,* 2nd Edition, Galgotia Pvt. Ltd.

Paper–B : OBJECT ORIENTED PROGRAMMING (USING C ++)

Objective :

This course enables students to appreciate the concepts of OOP using C++ and to understand linear & non linear DS.

Note : (i) The question paper will consist of four sections.

(ii) Examiner will set total of *nine* questions comprising *two* questions from each section and *one* compulsory question of short answer type covering whole syllabi.

(iii) The students are required to attempt *one* question from each section and the compulsory question.

(iv) All questions carry equal marks unless specified.

SECTION-A

1. *Concepts of Object Oriented Programming* : Object, Class, Encapsulation, Data Hiding, Inheritance, Polymorphism. Analysis and design of system using Object Oriented Approach.

2. *Structure of a C++ Program :* Include files, Declaration of an object, Main function, I/O streams.

3. *Classes :* Class Declaration : Data Members, Member Functions, Private and Public members, data hiding and encapsulation, arrays within a class.

   Class Function Definition : Member function definition inside the class declaration and outside the class declaration, scope resolution operator, Private and Public member function, Nesting of member functions.

SECTION-B

4. *Objects :* Creating Objects, Accessing class data members, Accessing member functions, Arrays of Objects, Objects as function arguments : Pass by value, Pass by Reference, Pointers to Objects.

5. *Constructors and Destructors :* Constructors : Declaration and Definition, Default Constructors, Parameterized Constructors, Copy Constructors. *Destructors* : Definition and use.

SECTION-C

6. *Function Overloading :* Function Overloading : Declaration and definition.

7. *Inheritance – Extending Classes :* Concept of inheritance, base class, derived class, defining derived classes, visibility modes, private, public, protected; single inheritance : privately derived, publicly derived; making a protected member inheritable, access control to private and protected members by member functions of a derived class, multilevel inheritance, nesting of classes.

SECTION-D

8. *Data Structures :* Linked Lists, Queues, Stacks, Trees and Sorting algorithms.

References :

1.	Stroustrup	:	*The C ++ Programming Language,* Addison Wesley.
2.	Robert Lafore	:	*OOP in Turbo C++,* Galgotia.
3.	E. Balaguruswamy	:	*Object Oriented Programming with C++,* TMH.
4.	Herbert Schildt	:	*Schildt's Advanced Win 95 Prog. in C & C++,* TMH.
5.	Herbert Schildt	:	*C++ The Complete Reference,* TMH.

Paper-C : PRACTICAL : Practicals based on Paper-A and Paper-B.

STATISTICS

B.A./B.Sc. (GENERAL) SECOND YEAR EXAMINATION, 2015

Note : 1. A candidate shall offer this subject in B.A./B.Sc. only if he/she takes up Mathematics as a subject in B.A./B.Sc.

2. A candidate shall offer this subject in B.A./B.Sc. 2nd year only if he/she had taken up the corresponding subject in B.A./B.Sc. 1st year.

3. There are three papers code named 201, 202 and 203 in the subject of Statistics in B.A./B.Sc. 2nd Year. These are to be taught simultaneously throughout the year.

4. 4 lectures (45 minutes each) per paper per week amounting in all to 12 lectures for three papers shall be allocated for the teaching.

Paper–201 : STATISTICAL INFERENCE

Max. Marks	:	75
Theory	:	65 Marks
Internal Assessment	:	10 Marks
Time allowed	:	3 Hours

Note : 1. There will be in all nine (9) questions, all of equal marks. The first question is **compulsory** and will be of short answer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set from each section. The candidate will be required to attempt five questions in all including the compulsory first question and two questions from each section.

2. Simple non-programmable calculator is allowed.

3. Statistical tables and log tables will be provided on request.

SECTION-I

Concept of a statistic and its sampling distribution. Point estimate of a parameter, concept of unbiasedness, consistency, efficiency and sufficiency (only the definitions and examples). Maximum likelihood estimation (standard distributions). Standard errors of sample mean and sample proportion. Sampling distribution of sum of independent binomial and Poisson random variables. Independence of sample mean and variance in random sampling from a normal distribution (without derivation).

Statements and sampling distributions of chi-square, t & F distributions and their relationships.

SECTION-II

Statistical Tests and Interval Estimation : Null and alternative hypothesis, two types of errors and p-value. Testing for the mean and variance of a normal distribution, testing of equality of means and variances of two univariate normal distributions, and their related confidence intervals. Testing the significance of sample correlation coefficient.

Large Sample Tests : Use of central limit theorem for testing and interval estimation of a single mean and a single proportion, difference of two means and two proportions. Fisher's Z-transformation and its uses. Chi-square test for goodness of fit and testing of independence of attributes, Yate's correction.

References :

1. Goon, A.M., Gupta, M.K., Das Gupta, B. (2005) : *Fundamentals of Statistics,* Vol. I, World Press, Calcutta.
2. Miller, I. and Miller, M. (2002) : *John E. Freund's Mathematical Statistics* (6th Edition, Low Price Edition), Prentice Hall of India.

Additional References :

1. Mood, A.M., Graybill, F.A. and Boes, D.C. (1974) : *Introduction to the Theory of Statistics,* McGraw Hill.
2. Bhat, B.R., Srivenkatramana, T. and Rao Madhava, K.S. (1997) : *Statistics : A Beginner's Text, Vol. I,* New Age International (P.) Ltd.
3. Snedecor, G.W. and Cochran, W.G. (1967) : *Statistical Methods,* Iowa State University Press.

Paper–202 : SAMPLE SURVEYS, DESIGN AND ANALYSIS OF EXPERIMENTS

Max. Marks	:	75
Theory	:	65 Marks
Internal Assessment	:	10 Marks
Time allowed	:	3 Hours

Notes : 1. There will be in all nine (9) questions, all of equal marks. The first question is **compulsory** and will be of short answer type covering the entire syllabus. Out of the remaining eight (8) questions, four (4) questions will be set from each section. The candidate will be required to attempt five questions in all including the compulsory first question and two questions from each section.

2. Simple non-programmable calculator is allowed.

3. Statistical tables and log tables will be provided on request.

SECTION-I

Concepts of population and sample, need for sampling, census and sample surveys, basic concepts in sampling. Sample selection and sample size. Sampling and non-sampling errors.

Some basic sampling methods for estimation of population mean, variance and standard error of estimates : simple random sampling (SRS) with and without replacement, stratified random sampling under various allocations and systematic sampling.

SECTION-II

Linear Models : Concepts of fixed effect, random effect and mixed effect models. Analysis of variance for one-way, two-way (with one observation per cell and with multiple but equal observations per cell) classifications under the fixed effect models.

Need for design of experiment, three fundamental principles of design, basic designs—CRD, RBD, LSD and their analysis.

References :

1. Goon, A.M., Gupta, M.K., Das Gupta, B. (2005). : *Fundamentals of Statistics,* Vol. II, World Press, Calcutta.

2. Singh, D. and Chaudhary, F.S. (1986). : *Theory and Analysis of Sample Survey Designs*.

Additional References :

1. Sukhatme, B.V. (1984) : *Sample Survey Methods and Its Applications,* Indian Society of Agricultural Statistics.

2. Murthy, M.N. (1967) : *Sampling Theory and Methods,* Statistical Publishing Society, Calcutta.

3. Sampath, S. (2000) : *Sampling Theory and Methods,* Narosa Publishing House.

4. Dass, M.N. and Giri (1986) : *Design and Analysis of Experiments,* Springer Verlag.

5. Cochran, W.G. and Cox, G.M. (1992) : *Experimental Designs,* John Wiley and Sons.

6. Dougle C. Mantgomery (2000) : Design and Analysis of Experiment John Wiley, New York.

Paper-203 : PRACTICAL

Max. Marks : 50
Time allowed : 3 Hours

(Viva-Voce : 10 marks; Record of the year : 10 marks; Annual Paper : 30 marks)

Note : The Practical Question Paper will contain five questions from the following topics. A student will be required to attempt three questions, each of 10 marks in three hours' duration.

1. Drawing random samples from standard distributions : Uniform, Exponential and Normal.

2. Tests of significance based on t, chi-square and F for one sample, two samples and paired sample problems; significance of correlation coefficient. Use of Z-Transformation for testing $p = p_0$.

3. Large sample tests for means and proportions, tests of goodness of fit and independence of attributes in contingency tables.

4. Selection of sample and determination of sample size : Simple random sampling (SRS), stratified SRS, allocation problems in stratified SRS and systematic sampling.

5. Analysis of variance for one-way and two-way classifications. Analysis of CRD, RBD, and LSD.

APPLIED STATISTICS

B.A./B.Sc. (GENERAL) SECOND YEAR EXAMINATION, 2015

Note :
1. This course shall not be opted for along with courses in B.A./B.Sc. Mathematics and/or B.A./B.Sc. Statistics.
2. The candidate opting for this course will not be eligible for admission to M.A./M.Sc. Statistics.
3. There are two papers code named Papers A and B in the subject of Applied Statistics in B.A./B.Sc., each of one credit and having a total of 100 marks. These are to be taught simultaneously throughout the year.
4. 4 to 5 lectures (40 minutes each) per paper per week amounting in all to 9 lectures for two papers shall be allocated for the teaching.

Paper-A : MATHEMATICAL METHODS-II

Max. Marks	:	100
Theory	:	90
Internal Assessment	:	10
Time allowed	:	3 Hours

Note : There will be in all nine (9) questions. The first question is **compulsory** and will be of short answer type covering the whole syllabus. This question will have 9 parts of 2 marks each. Of the remaining eight (8) questions, four (4) questions will be set from each section. The candidates will be required to attempt five (5) questions in all including the compulsory first question and two questions from each section.

SECTION-I

Coordinate Geometry of 3 Dimensions : Cartesian, spherical coordinates, equations of line, plane, sphere and ellipsoid.

Matrices and Linear Algebra : Determinants, algebra of matrices, rank of a matrix, inverse of a matrix, symmetric matrices (upto 4×4 matrices only).

SECTION-II

Matrices and Linear Algebra : Systems of Linear equations and their solutions.

Calculus of Several Variables : Functions of two variables, Partial derivatives and double integrals. Applications to evaluation of area.

Books Prescribed :

1. Ayres, F. (1999) : *First Year College Mathematics,* Schaum Series, Mc Graw Hill, New York.
2. Ayres, F. (1980) : *Matrices,* Schaum Series. Mc Graw Hill, New York.

Books Suggested for Supplementary Reading :

1. Kindle, J.H. (1990) : *Analytical Geometry,* Schaum Series, Mc Graw Hill, New York.
2. Ayres, F. (1999) : *Calculus,* Schaum Series, Mc Graw Hill, New York.
3. Spiegel, M.R. (2002) : *Advanced Calculus,* Schaum Series, Mc Graw Hill, New York.

Paper-B : STATISTICAL METHODS

Max. Marks	:	100
Theory	:	90
Internal Assessment	:	10
Time allowed	:	3 Hours

Notes : 1. There will be in all nine (9) questions. The first question is **compulsory** and will be of short answer type covering the whole syllabus. This question will have 9 parts of 2 marks each. Of the remaining eight (8) questions, four (4) questions will be set from each section. The candidates will be required to attempt five (5) questions in all including the compulsory first question and two questions from each section.

2. Simple non-programmable calculator is allowed.

3. Statistical tables and log tables will be provided on request.

SECTION-I

Compilation, classification, tabulation and diagrammatic representation of statistical data. Concepts of Statistical population, random sample and frequency curve, measures of location, dispersion, skewness and kurtosis.

Two dimensional random variable, joint probability distributions, marginal and conditional probability distributions, conditional expectation, covariance and correlation coefficient.

SECTION-II

Measures of association and contingency, correlation and linear-regression involving two variables. Bivariate normal distributions.

Statement of weak law of large numbers and central limit theorem for independent and identical random variables. Sampling distributions of means, chi-square, t and F in sampling from normal populations (without proof).

Books Recommended :

1. Goon, A.M., Gupta, M.K. and Dasgupta, B. (2005) : *Fundamentals of Statistics,* Vol. 1 to 6, 9, 10, 13 (only the relevant portion from these chapters as suggested by the body of the syllabus).

2. P.L. Meyer (1971) : Introductory Probability and Statistical Applications, Addison Wesley. American Pub. Co.

PHYSICS

B.Sc. (GENERAL) SECOND YEAR EXAMINATION, 2015

Objectives :

Principles of Communication Systems :

The objective of Principles of Communication Systems course is to develop student's understanding with communication principles, Components of superheterodyne receivers and recorders, Digital audio, understanding of fundamentals of B/W and Colour Video transmission and Television receivers.

Microprocessor Based Systems :

The objective of Microprocessor Based Systems course is to cover various aspects of 8085 microprocessor hardware and instruction architecture, interfacing with RAM and ROM, and I/O devices, I/O data transfer, 8155 programmable device, instruction set and programming applications.

Practicals

The objective of PRACTICALS course is 8085 microprocessor programming, Understanding of Television, Sequential digital circuits.

Note :

1. There will be three papers of theory and one laboratory (practical) course.
2. The number of lectures per week will be three for each theory paper and six for practicals.
3. The examination time for each theory paper will be three hours and four hours for practical.
4. Each theory paper will consist of nine questions carrying equal marks and spread over five prescribed units.
5. Eight questions in each theory paper will be set from units I to IV i.e. two questions from each unit.
6. Ninth question in each paper is compulsory and will comprise of seven small answer type questions covering the whole syllabus.
7. The numerical problems/exercises in the question paper should be 25-30%.
8. Student will attempt one question from each unit (I-IV) and any six parts of question nine.
9. The use of Non-programmable calculators will be allowed (paper setters should explicitly mention this on the question paper) in the examination centre but these will not be provided by the University/College. Mobile phones and pagers are not allowed in the examination hall.

Papers, marks and teaching hours allocation :

Paper A :	Statistical Physics and Thermodynamics	(45 + 5*) marks	Total Teaching hrs.** 60
Paper B :	Optics and Lasers	(45 + 5*) marks	Total Teaching hrs.** 60
Paper C :	Quantum Physics	(45 + 5*) marks	Total Teaching hrs.** 60
Physics Practicals :		(45 + 5*) marks	Total Teaching hrs. 90

* marks allotted for internal assessment.

**within the prescribed teaching load, teachers must give two assignments per paper per academic term (July – September, October – December, January – March) with stress on problem solving to enhance the skill component of the students.

Paper A : STATISTICAL PHYSICS AND THERMODYNAMICS (60 Hrs.)

UNIT-I

Basic ideas of Statistical Physics, Scope of Statistical Physics, basic ideas about probability, distribution of four distinguishable particles in two compartments of equal size. Concept of macrostates, microstates, thermodynamic probability, effects of constraints on the system, distribution of n particles in two compartments, deviation from the state of maximum probability, equilibrium state of dynamic system, distribution of distinguishable n particles in k compartments of unequal sizes.

UNIT-II

Phase space and its division into elementary cells, three kinds of statistics. The basic approach in the three statistics. Maxwell-Boltzman statistics applied to an ideal gas in equilibrium, experimental verification of Maxwell-Boltzman's law of distribution of molecular speeds. Need of quantum statistics--B.E. statistics, derivation of Planck's law of radiation, deduction of Wien's displacement law and Stefan's law from Planck's law, F.D. statistics, Comparison of M.B., B.E. and F.D. statistics.

UNIT-III

Statistical definition of entropy, change of entropy of a system, additive nature of entropy, law of increase of entropy, reversible and irreversible processes with examples. Work done in a reversible process. Examples of increase of entropy in natural processes. Entropy and disorder. Brief review of the terms and Laws of Thermodynamics, Carnot's Cycle. Entropy changes in Carnot's Cycle. Applications of thermodynamics to thermoelectric effect, change of entropy along a reversible path in a P.V. diagram, entropy of a perfect gas. Equation of state of ideal gas from simple statistical consideration. Heat death of the universe.

UNIT-IV

Derivation of Maxwell's thermodynamical relations, cooling produced by adiabatic stretching, adiabatic compression, change of internal energy with volume. Expression for (C_p-C_v), change of state and Clayperon Equation. Thermodynamical treatment of Joule-Thomson effect. Use of Joule-Thomson effect for liquification of helium. Production of very low temperature by adiabatic demagnetisation.

UNIT-V

Spread over the entire syllabi of all the four units above.

Books Suggested :

Essential Readings :

1. Bhatia, V.S. : *Statistical Physics and Thermodynamics,* Shoban Lal Nagin Chand, Jalandhar.

2. Saha and Srivastava : *A Treatise on Heat,* Indian Press, Ahmedabad, 1972.

Further Readings :

1. Kittel, C. & Kroemer, H. : *Thermal Physics,* CBS Pub., 1987.

2. Garg, S.C., Bansal, R.M. and Ghosh, C.K. : *Thermal Physics,* TMH, 2000.

Paper-B : OPTICS AND LASERS (60 Hrs.)

UNIT-I

Interference : Concept of coherence, spatial and temporal coherence, coherence time, coherence length, area of coherence. Conditions for observing interference fringes. Interference by wavefront division and amplitude division. Young's double slit experiment. Michelson interferometer—working, principle and nature of fringes. Interference in thin films, Role of interference in anti-reflection and high reflection dielectric coatings. Multiple beam interference, Fabry-Perot interferometer, nature of fringes, finesse.

UNIT-II

Diffraction : Huygen-Fresnal theory half period zones, zone plates. Distinction between Fresnel and Fraunhofer diffraction. Fraunhofer diffraction at rectangular and circular apertures. Effects of diffraction in optical imaging, resolving power of optical instruments. The diffraction grating, its use as a spectroscopic element, resolving power. Polarization : Concept and analytical treatment of unpolarised, plane polarized and elliptically polarized light. Double refraction, Nicol prism, sheet polarisers, retardation plates. Production and analysis of polarized light (quarter and half wave plates).

UNIT-III

Laser Fundamentals : Derivation of Einstein relations, concept of stimulated emission and population inversion. Broadening of spectral lines, natural, collision and doppler broadening. Line width, line profile. Absorption and amplification of a parallel beam of light passing through a medium. Threshold condition. Three level and four level laser schemes. Elementary theory of optical cavity, longitudinal and transverse modes.

UNIT-IV

Laser Systems and Fiber Optics : Types of lasers, Ruby and Nd : YAG lasers. He–Ne and CO_2 lasers – construction, mode of creating population inversion and output characteristics. Semiconductor lasers, Applications of lasers—a general outline, Holography. Fiber fundamentals, wave propagation and dissipation of energy applications.

UNIT-V

Spread over the entire syllabi of all the four units above.

Books Suggested :

Essential Readings :

1. Jenkins and White : *Optics,* McGraw Hill.
2. Ghatak, Ajoy : *Optics,* McMillan India.
3. Silfvast, W.T. : *Laser Fundamentals,* Foundation Books.
4. Laud, B.B. : *Lasers and Non-linear Optics,* Wiley Eastern Limited.
5. Keiser : *Optical Fiber Communication,* MH.

Further Readings :

1. Born and Wolf : *Optics,* Pergmon.
2. Svelto : *Lasers,* Pergmon.

Paper-C : QUANTUM PHYSICS **(60 Hrs.)**

UNIT-I

Formalism of Wave Mechanics :

(i) Brief introduction to need and development of quantum mechanics, wave-particle duality (photon as particle, de-Broglie hypothesis, particle diffraction, particle interference), wave packet, indeterminacy, complementarity.

(Chapter 1 of book 1 in particular section 1.13 to 1.18)

(ii) Schrödinger equation for a free particle, operator correspondence and equation for a particle subject to forces. Normalization and probability interpretation of wave function, superposition principle, expectation value, probability current and conservation of probability, admissibility conditions on the wave function, Ehrenfest theorem.

(Chapters 2A, 2B of book 1 or 2.8 to 2.10, 4.4–4.6 of book 2)

(iii) Fundamental postulates of wave mechanics. Eigenfunctions and eigenvalues, operator formalism, orthogonal systems, expansion in eigenfunctions, Hermitian operators, simultaneous eigenfunctions, equation of motion.

(Section 3.2A of book 1 and section 6.1-6.8, 6.11 of book 2)

(iv) Uncertainty of position and momentum, monochromatic waves, Gaussian wave packet.

(Section 3.1-3.6 of book 2 or 3.11-3.12 of book 1)

UNIT-II

Problems in One and Three Dimensions :

(i) Time dependent Schrödinger equation, Application to stationary states for one dimension, Potential step, potential barrier, rectangular potential well, degeneracy, orthogonality, linear harmonic oscillator.

(ii) Schrödinger equation for spherically symmetric potential, spherical harmonics, hydrogen atom energy levels and eigenfunctions, degeneracy, angular momentum.

(Section 5.1–5.4, 5.8–5.11, 7.1– 7.3, 7.7 of book 2)

UNIT-III

One Electron Atomic Spectra :

(i) Interaction with radiation, transition probability, spontaneous transition, selection rules and life times.

(Section 6.3, 6.4 of book 4 or 11.7 of book 1)

(ii) Spectrum of hydrogen atom, fine structure, normal Zeeman effect, electron spin, Stern-Gerlach experiment, spin orbit coupling, electron magnetic moment, total angular momentum, Hyperfine structure, examples of one electron systems, Anomalous Zeeman effect, Lande-g factor (sodium D-lines).

(Section 4.1– 4.6 of book 4 or chapters 6 & 7 of book 3)

UNIT-IV

Many Electron System Spectra :

(i) Exchange symmetry of wave functions, exclusion principle, shells, subshells in atoms, atomic spectra (Helium), LS coupling, JJ coupling, selection rules, regularities in atomic spectra.

(ii) X-ray spectra, Mosley law, absorption spectra, Auger effect.

(iii) Molecular bonding, molecular spectra, selection rules, symmetric structures, rotational, vibrational electronic level and spectra of molecules, magnetic resonance experiments, Raman spectra.

(Section 5.1, 5.2, 5.4–5.7, 6.7 of book 4 and chapters 7 & 8 of book 3)

UNIT-V

Spread over the entire syllabi of all the four units above.

Books Suggested :

Essential Readings :

1. Mathews, P.M. and Venkatesan, K. : *A Text Book of Quantum Mechanics,* T.M.H., 1997.
2. Powell, J.L. and Craseman : *Quantum Mechanics,* Narosa Pub., 1997.
3. Beiser, A. : *Concepts of Modern Physics,* McGraw Hill, 4th Edition.
4. Patil, S.H. : *Elements of Modern Physics,* McGraw Hill, 1998.

Further Readings :

1.	Merzbacher, E.	:	*Quantum Mechanics,* II Ed., John Wiley.
2.	Eisberg, R. & Resnick, R.	:	*Quantum Physics of Atoms,* Molecular . . . , John Wiley, 1974.
3.	Banwell, C.N.	:	*Fundamentals of Molecular Spectroscopy,* T.M.H., 1972.
4.	Kuhn, H.G.	:	*Atomic Spectra,* Academic Press, N.Y.

PRACTICALS (90 Hours)

Total Marks : 50
Practical : 45
Int. Asses. : 05

General Guidelines for Physics Practical Examination, 2015

1. The distribution of Practical (45) marks is as follows :

I	One full experiment out of Section-A requiring the student to take some data, analyse it and draw conclusions. (candidates are expected to state their results with limits of error).	20 marks
II	One exercise based on experiment or Computer Programming (To be allotted by the external examiner at the time of examination)	10 marks
III	Viva-Voce	10 marks
IV	Record (Practical file)	05 marks

Note for Examiners :
The marks scored under each head must be clearly so written on the answer sheet.

2. There will be one session of 4 hours duration. The paper will have two sections. Section A will consist of 8 experiments out of which an examinee will mark 6 experiments and one of these is to be allotted by the external examiner.

3. Section-B will consist of exercises which will be set by the external examiner on the spot. The length of the exercises should be such that any of these could be completed in one hour.

4. The examiner should take care that the experiment allotted to an examinee from Section-A and exercises allotted from Section-B are not directly related to each other.

5. *Number of candidates in a group for practical examination should not exceed 12.*

6. In a single group, no experiment be allotted to more than three examinees in the group.

List of Experiments :

I Statistical Physics and Thermodynamics :

Activities :

(i) Adiabatic expansion of a gas.

(ii) Thermal expansion of crystal using interference fringes.

(iii) Thermal conduction in poor conductor (variation with geometry) by Lee's method.

(iv) Thermo emf calibration, comparison.

(v) Total radiation law, temperature dependence of radiation.

(vi) Probability distribution using coloured dice, coins.

II Optics and Lasers :

Activities :

(vii) To determine the refractive index of a liquid using spectrometer.

(viii) To determine the Cauchy's constants.

(ix) To study the refractive index of a doubly refracting prism.

(x) Study of rotation of plane of polarization with a polarimeter.

(xi) Set up Newton's rings to determine wave length of sodium light.

(xii) To determine the wave length of a given light using biprism.

(xiii) To determine the wave length and dispersive power using plane diffraction grating (use Hg source).

(xiv) To determine the resolving power of a telescope.

(xv) To determine the resolving power of a grating.

(xvi) To measure an inaccessible height using sextant.

(xvii) To determine the principal points of a lens system.

(xviii) To determine the divergence and wave length of a given laser source.

III Quantum Physics :

Activities :

(xix) Study the Photoelectric effect and determine the value of Planck's constant.

(xx) To study the gas discharge spectrum of hydrogen.

(xxi) To study the absorption spectra of iodine vapours.

(xxii) To determine the ionization potential of mercury.

Exercises : Any exercise out of the following list of sixteen exercises :

General exercises :

1. To measure the angle of rotation of plane of polarization for the given liquid.
2. To measure the diameter of Newton's rings.
3. To determine the least count and setup the spectrometer for minimum deviation position of the prism.
4. Study of variation of light intensity using photovoltaic cell/inverse square law.
5. To measure the thermo e.m.f.
6. To determine the heating efficiency of an electric kettle with varying voltages.
7. To determine the angle of wedge using interference method.
8. To measure the angle of elevation of a tall building.

Computer Based Activities :

9. Motion of particle in a central force field.
10. Calculation of days between two dates of a year.
11. To check if triangle exists and the type of the triangle.
12. To find the sum of the sine and cosine series and print out the curve.
13. Motion of a projective using computer simulation.

Text and Reference Books :

1. Khandelwal, D.P. : *A Laboratory Manual of Physics for Undergraduate Classes.*
2. Arora, C.L. : *B.Sc. Practical Physics,* S. Chand.
3. Dixon, C. : *Numerical Analysis.*
4. Lipsdutz, S. and Poe, A. : *Schaum's Outline of Theory and Problems of Programming with Fortran,* McGraw-Hill Int.

CHEMISTRY

B.Sc. (GENERAL) SECOND YEAR EXAMINATION, 2015

Scheme of Teaching and Examination

Paper	*Course*	*Teaching Hrs.*		*Max. Marks*
V	Inorganic Chemistry	60	3 periods per week	45 +5 *Int. Assessment*
VI	Organic Chemistry	60	3 periods per week	45 +5 *Int. Assessment*
VII	Physical Chemistry	60	3 periods per week	45 +5 *Int. Assessment*
VIII	Laboratory Practicals		6 periods per week	45 +5 *Int. Assessment*
	Total :		**15 Periods/week**	**200**

Paper- V : INORGANIC CHEMISTRY

Max. Marks : 50
Theory : 45
Internal Assessment : 05
Time : 3 Hours
60 hours (2Hrs./Week)
3 periods/Week

OBJECTIVE OF THE COURSE :

To teach the fundamental concepts of Chemistry and their applications. The syllabus pertaining to B.Sc. (General) (3 Year course) in the subject of Chemistry has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Chemistry working in the Panjab University, Chandigarh and affiliated colleges. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

UNIT-I 15 Hrs.

Chemistry of Elements of First Transition Series :

Characteristic properties of *d*-block elements.

Properties of the elements of the first transition series, their simple compounds and complexes, illustrating relative stability of their oxidation states, coordination number and geometry.

Chemistry of Elements of Second and Third Transition Series :

General characteristics, comparative treatment with their 3*d*-analogues in respect of ionic radii, oxidation states, magnetic behaviour, spectral properties and stereochemistry.

UNIT-II 15 Hrs.

Chemistry of Coordination Compounds :

Werner's coordination theory and its experimental verification, effective atomic number concept, chelates, nomenclature of coordination compounds, isomerism in coordination compounds, valence bond theory of transition metal complexes. Properties of Coordination compounds i.e. magnetic properties, colours (Qualitative approach only), use of coordination compounds.

UNIT-III 15 Hrs.

Chemistry of Lanthanide Elements :

Electronic structure, oxidation states and ionic radii and lanthanide contraction, complex formation, occurrence and isolation, lanthanide compounds.

Chemistry of Actinides :

General features and chemistry of actinides, chemistry of separation of Np, Pu and Am from U, similarities between the later actinides and the later lanthanides.

Acids and Bases :

Arrhenius, Bronsted-Lowry, the Lux-Flood, solvent system and Lewis concepts of acids and bases.

UNIT-IV 15 Hrs.

Oxidation and Reduction :

Use of redox potential data – analysis of redox cycle, redox stability in water – Frost, Latimer and Pourbaix diagrams. Principles involved in the extraction of the elements.

Non-aqueous Solvents :

Physical properties of a solvent, types of solvents and their general characteristics, reactions in non-aqueous solvents with reference to liquid NH_3 and liquid SO_2.

INSTRUCTIONS FOR PAPER SETTERS AND CANDIDATES :

(i) Examiner will set total of ***nine*** questions comprising ***two*** questions from each unit and ***one*** compulsory question of short answer type covering whole syllabi.

(ii) The students are required to attempt ***five*** questions in all, ***one*** question from each unit and the Compulsory question.

(iii) All questions carry equal marks.

Books Suggested :

1. Cotton, F.A., Wilkinson, G., Gaus, P.L. : *Basic Inorganic Chemistry;* 3rd Edition, Pubs: John Wiley Sons, 1995.

2. Lee, J.D. : *Concise Inorganic Chemistry;* 4th Edition, Pubs., Chapman Hall Ltd., 1991.

3. Shriver, D.E., Alkins, P.W., Langford, C.H. : *Inorganic Chemistry;* 4th Edition, Oxford Publisher: Oxford University Press, 2006.

4. Douglas, B. McDamiel, D., Alexander, J. : *Concepts and Models of Inorganic Chemistry;* 3rd edition, Pubs: John Wiley and Sons Inc., 1994.

5. Porterfield, W.W., Wesley, A. : *Inorganic Chemistry;* Pubs: Addison-Wesley Publishing Company, 1984.

6. Miessler, G.L., Larr, D.A. : *Inorganic Chemistry;* 3rd Edition, Pubs: Pearson Education Inc., 2004.

7. Jolly, W.L. : *Modern Inorganic Chemistry;* 2nd Edition, Pubs: McGraw-Hill Publishing Company Limited, 1991.

8. Purcell, K.F., Kotz, J.C. : *Inorganic Chemistry;* Pubs: W.B. Saunders Company, 1977.

9. Puri, B.R., Sharma, L.R., Kalia, K.C. : *Principles of Inorganic Chemistry;* 30th Edition, Pubs: Milestones 2006-07.

Paper-VI : ORGANIC CHEMISTRY

Max. Marks	:	50
Theory	:	45
Internal Assessment	:	05
Time	:	3 Hours
60 hours (2Hrs./Week)		
3 periods/Week		

OBJECTIVE OF THE COURSE :

To teach the fundamental concepts of Chemistry and their applications. The syllabus pertaining to B.Sc. (General) (3 Year course) in the subject of Chemistry has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Chemistry working in the Panjab University, Chandigarh and affiliated colleges. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

UNIT-I 15 Hrs.

Alcohols :

Classification and nomenclature

Monohydric alcohols - Nomenclature, methods of formation by reduction of aldehydes, ketones, carboxylic acids and esters. Hydrogen bonding. Acidic nature. Reactions of alcohols.

Dihydric alcohols - Nomenclature, methods of formation, chemical reactions of vicinal glycols. Oxidative cleavage [$Pb(OAc)_4$ and HIO_4] and pinacol-pinacolone rearrangement. Trihydric alcohols - Nomenclature and methods of formation, Chemical reactions of glycerol.

Phenols :

Nomenclature, structure and bonding. Preparation of phenols, physical properties and acidic character. Comparative acidic strengths of alcohols and phenols, resonance stabilization of phenoxide ion. Reactions of phenols - electrophilic aromatic substitution, acylation and carboxylation. Mechanisms of Fries rearrangement, Claisen rearrangement, Gatterman synthesis, Hauben-Hoesch reaction, Lederer-Manasse reaction and Reimer-Tiemann reaction.

UNIT-II 15 Hrs

Aldehydes and Ketones :

Nomenclature and structure of the carbonyl group. Synthesis of aldehydes and ketones with particular reference to the synthesis of aldehydes from acid chlorides, synthesis of aldehydes and ketones using 1, 3-dithianes, synthesis of ketones from nitriles and from carboxylic acids. Physical properties.

Mechanism of nucleophilic additions to carbonyl group with particular emphasis on benzoin, aldol, Perkin and Knoevenagel condensations. Condensation with ammonia and its derivatives. Wittig reaction, Mannich reaction.

Use of acetals as protecting group. Oxidation of aldehydes, Baeyer-Villiger oxidation of ketones, Cannizzaro reaction, MPV, Clemmensen, Wolff-Kishner, $LiAlH_4$ and $NaBH_4$ reductions.

UNIT-III 15 Hrs.

Carboxylic Acids :

Nomenclature, structure and bonding, physical properties, acidity of carboxylic acids, effects of substitutions on acid strength. Preparations of carboxylic acids. Reactions of carboxylic acids. Hell-Volhard-Zelinsky reaction. Synthesis of acid chlorides, esters and amides, Reduction of carboxylic acids. Mechanism of decarboxylation. Methods of formation and chemical reactions of halo acids. Hydroxyl acids : Malic, tartaric and citric acid (structural features only).

Methods of formation and chemical reactions of unsaturated monocarboxylic acids.

Dicarboxylic Acids: Methods of formation and effect of heat and dehydrating agents.

Carboxylic Acid Derivatives :

Structure and nomenclature of acid chlorides, esters, amides and acid anhydrides. Relative stability & reactivity of acyl derivatives. Physical properties, interconversion of acid derivatives by nucleophilic acyl substitution.

Preparation of carboxylic acid derivatives, chemical reactions. Mechanisms of esterification and hydrolysis (acidic and basic).

Ethers and Epoxides :

Nomenclature of ether and methods of their formation, physical properties. Chemical reaction - cleavage and autoxidation, Ziesel's method.

Synthesis of epoxides. Acid and base-catalyzed ring opening of epoxides, orientation of epoxide ring opening, reactions of Grignard and organolithium reagents with epoxides.

UNIT-IV 15 Hrs.

Fats, Oils and Detergents :

Natural fats, edible and industrial oils of vegetable origin, common fatty acids, glycerides, hydrogenation of unsaturated oils. Saponification value, iodine value, acid value. Soaps, synthetic detergents; alkyl and aryl sulphonates.

Organic Compounds of Nitrogen :

Preparation of nitroalkanes and nitroarenes. Chemical reactions of nitroalkanes. Mechanisms of nucleophilic substitution in nitroarenes and their reductions in acidic, neutral and alkaline media. Picric acid.

Structure and nomenclature of amines, physical properties. Stereochemistry of amines. Separation of a mixture of primary, secondary and tertiary amines. Structural features effecting basicity of amines. Amine salts as phase-transfer catalysis. Preparation of alkyl and aryl amines (reduction of nitro compounds, nitriles), reductive amination of aldehydic and ketonic compounds. Gabriel-phthalimide reaction, Hofmann bromamide reaction.

INSTRUCTIONS FOR PAPER SETTERS AND CANDIDATES :

(i) Examiner will set total of ***nine*** questions comprising ***two*** questions from each unit and ***one*** compulsory question of short answer type covering whole syllabi.

(ii) The students are required to attempt ***five*** questions in all, ***one*** question from each unit and the Compulsory question.

(iii) All questions carry equal marks.

Books suggested :

1. Morrison, R.T.; Boyd, R.N. : *Organic Chemistry;* 6th Edition, Pubs. : Prentice-Hall, 1992.

2. Wade Jr., L.G.; Singh, M.S. : *Organic Chemistry;* 6th Edition, Pubs. : Pearson Education, 2008.

3. Mukherji, S.M.; Singh, S.P., Kapoor, R.P. : *Organic Chemistry;* Pubs. : Wiley Eastern Limited, 1985, Vols. I, II, III.

4. Solomons, T.W.; Fryhle, C.B. : *Organic Chemistry;* 9th Edition, Pubs. : Wiley India, 2007.

5. Carey, F.A. : Organic Chemistry; 4th Edition, Pubs. : McGraw-Hill, 2000.

6. Streitwieser, A.; Clayton, Jr. Heathcock, H. : *Introduction to Organic Chemistry;* 3rd Edition, Pubs. : Macmillan Publishing Company, 1989.

Paper-VII : PHYSICAL CHEMISTRY

Max. Marks	:	50
Theory	:	45
Internal Assessment	:	05
Time	:	3 Hours
60 hours (2Hrs./Week)		
3 periods/Week		

OBJECTIVE OF THE COURSE :

To teach the fundamental concepts of Chemistry and their applications. The syllabus pertaining to B.Sc. (General) (3 Year course) in the subject of Chemistry has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Chemistry working in the Panjab University, Chandigarh and affiliated colleges. The syllabus contents are duly arranged unit wise and contents are included in such a manner so that due importance is given to requisite intellectual and laboratory skills.

UNIT-I 15 Hrs.

Liquid State :

Intermolecular forces, structure of liquids (a qualitative description).

Structural differences between solids, liquids and gases.

Liquid Crystals : Difference between liquid crystal, solid and liquid. Classification, structure of nematic and cholestric phases. Thermography and seven segment cell.

Chemical Equilibrium : .

Equilibrium constant and free energy, Thermodynamic derivation of law of mass action, Le - Chatelier's principle.

Reaction isotherm and Reaction isochore, Clapeyron equation and Clausius–Clapeyron equation, applications.

UNIT-II 15 Hrs.

Thermodynamics-II :

Second Law of Thermodynamics: Need for the law, different statements of the law, Carnot cycle and its efficiency, Carnot theorem. Thermodynamic scale of temperature.

Concept of Entropy: Entropy as a state function, entropy as a function of V & T, entropy as a function of P & T, entropy change in physical change, Clausius inequality, entropy as a criteria of spontaneity and equilibrium, Entropy change in ideal gases and mixing of gases.

Thermodynamics-III :

Third Law of Thermodynamics: Nernst heat theorem, statement and concept of residual entropy, evaluation of absolute entropy from heat capacity data, Gibbs and Helmholtz functions; Gibbs function (G) and Helmholtz functions (A) as thermodynamic quantities, A & G as criteria for thermodynamic equilibrium and spontaneity, their advantage over entropy change, Variation of G and A with P, V and T.

UNIT-III 15 Hrs.

Phase Equilibrium :

Statement and meaning of the terms – phase, component and degree of freedom, derivation of Gibbs phase rule, phase equilibria of one component system—water, CO_2 and S systems.

Phase equilibria of two component system – solid–liquid equilibria, simple eutectic – Bi-Cd system, desilverisation of lead.

Solid solutions—Compound formation with congruent melting point (Mg-Zn) and incongruent melting point, (NaCl-H_2O) system. Freezing mixtures, acetone-dry ice.

Partially Miscible Liquids–Phenol-water, trimethylamine – water, nicotine –water systems.

Lower and upper consolute temperature, Effect of impurity on consolute temperature, immiscible liquids, steam distillation, Nernst distribution law - thermodynamic derivation, applications.

Electrochemistry–I :

Electrical transport – Conduction in metals and in electrolyte solutions, specific conductance and equivalent conductance, measurement of equivalent conductance, variation of equivalent and specific conductance with dilution.

Migration of ions and Kohlrausch Law, Arrhennius theory of electrolyte dissociation and its limitations, weak and strong electrolytes, Ostwald's dilution law, its uses and limitations. Debye-Huckel-Onsager's equation for strong electrolytes (elementary treatment only), Transport number, definition and determination by Hittorf method and moving boundary method.

UNIT-IV 15 Hrs.

Electrochemistry-II :

Types of reversible electrodes – gas metal – ion, metal –insoluble salt – anion and redox electrodes. Electrode reactions, Nernst equation, derivation of cell E.M.F. and single electrode potential, standard hydrogen electrode – reference electrodes – standard electrode potential, sign conventions, electrochemical series and its significance.

Electrolytic and Galvanic cells – reversible and irreversible cells, conventional representation of electrochemical cells.

E.M.F. of a cell and its measurements, Computation of cell E.M.F., Calculation of thermodynamic quantities of cell reactions (ΔG, ΔH and K), Polarization, over potential and hydrogen overvoltage.

Concentration cell with and without transport, liquid junction potential, application of concentration cells, valency of ions, solubility product and activity coefficient, potentiometric titrations.

INSTRUCTIONS FOR PAPER SETTERS AND CANDIDATES :

(i) Examiner will set total of ***nine*** questions comprising ***two*** questions from each unit and ***one*** compulsory question of short answer type covering whole syllabi.

(ii) The students are required to attempt ***five*** questions in all, ***one*** question from each unit and the Compulsory question.

(iii) All questions carry equal marks.

Books Suggested :

1. Atkins, P., Paula, J.de : *Atkins Physical Chemistry;* 8th Edition, Pubs: Oxford University Press, 2008.
2. Puri, B.R., Sharma, L.R., Pathania, M.S. : *Principles of Physical Chemistry;* 43rd Edition, Pubs: Vishal Publishing Co., 2008.
3. Barrow, G.M. : *Physical Chemistry;* 6th Edition, Pubs: McGraw Hill Companies Inc., 1996.
4. Rao, C.N.R. : *University General Chemistry;* Pubs: Macmillan of India, 1985.
5. Berry, R.S., Rice, S.A., Ross, J. : *Physical Chemistry;* 2nd Edition, Pubs: Oxford University Press, 2000.
6. Albert, R.A., Silbey, R.J. : *Physical Chemistry;* 1st Edition, Pubs: John Wiley & Sons Inc., 1992.
7. Dogra, S.K., Dogra, S. : *Physical Chemistry Through Problems,* Pubs: Wiley Eastern Limited, 1991.
8. Levine, I.N. : *Physical Chemistry;* 5th Edition, Pubs: Tata McGraw Hill Publishing Co. Ltd., 2002.
9. Moore, W. J. : *Basic Physical Chemistry;* Pubs: Prentice Hall of India Pvt. Ltd., 1983.
10. Metz, C.R. *Theory and Problems of Physical Chemistry;* Schaum's Outline Series, 2nd Edition, Pubs: McGraw-Hall Book Company, 1989.

Paper-VIII : LABORATORY PRACTICALS

Max. Marks : 45+5

6 Periods/week

Inorganic Chemistry

Quantitative Analysis

Volumetric Analysis:

(a) Estimation of calcium content in chalk as calcium oxalate by permanganometry.

(b) Estimation of hardness of water by EDTA.

(c) Estimation of ferrous and ferric by dichromate method.

(d) Estimation of copper using sodium thiosulphate.

Gravimetric Analysis :

Analysis of Cu as CuSCN and Ni as Ni (dimethylgyoxime)$_2$.

Organic Chemistry

Laboratory Techniques :

(a) Determination of R_f values and identification of organic compounds. Separation of isometric mixture of ortho and para–nitroaniline using hexane and ethyl acetate (8.5 : 1.5) by thin layer chromatography.

(b) Extraction of caffeine from tea leaves.

Qualitative Analysis :

Detection of elements (N, S and halogens) and functional groups (phenolic, carboxylic, carbonyl, esters, carbohydrates, amines, amides, nitro and anilide), in simple organic compounds.

Physical Chemistry

Thermochemistry :

(a) To determine the solubility of benzoic acid at different temperatures and to determine ΔH of the dissolution process.

(b) To determine the enthalpy of neutralization of a weak acid/weak base versus strong base/strong acid and determine the enthalpy of ionization of the weak acid/weak base. pH of a Buffer solution; Determination of ionization constant of a weak acid.

General Instructions to the Examiners :

Note : Practical examination will be of four hours duration & shall consist of the following questions:

Q. No.	I	Organic Chemistry: (analysis of organic compound/TLC experiment/ extraction of caffeine)	13 marks
Q. No.	II	Quantitative Analysis	10 marks
Q. No.	III	Physical Chemistry	10 marks
Q. No.	IV	Viva-Voce	6 marks
		Ask three questions (2 marks each) related to chemistry practicals.	
Q. No.	V	Note Book	6 marks

Books Suggested (Laboratory Courses) :

1. Furniss, B.S., Hannaford, A.J., Rogers, V., Smith P.W.G., Tatchell, A.R. : *Vogel's Text Book of Practical Organic Chemistry;* 4th Edition, Pubs: Longman Group, 1978.

2. Bansal, R.K. : *Laboratory Manual of Organic Chemistry;* 3rd Edition, Pubs : Wiley Eastern Limited, New Delhi, 1994.

3. Furniss, B.S., Hannaford, A.J., Smith, P.W.G., Tatchell, A.R. : *Vogel's Textbook of Practical Organic Chemistry;* 5th Edition, Pubs: Dorling Kindersley (India) Pvt. Ltd., Delhi, 2006.

4. Khosla, B.D., Garg, V.C., Gulati, A. : *Senior Practical Physical Chemistry;* 11th Edition, Pubs: R. Chand & Co., New Delhi, 2002.

5. Das, R.C., Behra, B. : *Experimental Physical Chemistry;* Pubs: Tata McGraw Hill Publishing Co. Ltd., 1983.

6. Levitt, B.P. : *Findlays Practical Physical Chemistry;* 8th Edition, Pubs: Longman Group Ltd., London & New York, 1978.

BOTANY

B.Sc. (GENERAL) SECOND YEAR EXAMINATION, 2015

		Time	*Theory*	*Int. Assess.*	*Max. Marks*
Theory Paper A :	Diversity of Seed Plants and their Systematics	3 hrs.	68	7	75
Theory Paper B :	Structure, Development and Reproduction in Flowering Plants.	3 hrs.	68	7	75
One practical paper pertaining to entire syllabus included in both theory papers.		4 hrs.	45	5	50
				Total :	**200**

Note : 1. The number of teaching hours for theory and practical per session shall be 120 hrs. and 200 hrs. respectively.

2. There will be a total of nine questions in each theory paper A & B. Question No. 1 will be compulsory and will consist of 20 parts (one mark each) comprising 10 MCQ and the rest 10 parts will be of fill-in the blanks covering the entire syllabus in both the theory papers A & B. The remaining 8 questions in papers A & B shall include two questions from each unit. Candidates shall be required to attempt one question from each Unit. Question No. 1 will carry 20 marks and the rest of 8 questions will be of 12 marks each.

Paper A : DIVERSITY OF SEED PLANTS AND THEIR SYSTEMATICS

Max. Marks	:	75
Theory	:	68 Marks
Internal Assessment	:	7 Marks
Time	:	3 Hours

Objective :

This paper deals with most advance and evolved group of plants i.e. Gymnosperms and Angiosperms. The study of gradual transition from seed less plants (studied in first year) to seed plants would make students familiar with origin of structural and functional complexity in plant Kingdom. The systematic part of this paper is in fact backbone of the study of Botany. Without having knowledge of taxonomy and species concept, no further research work can be pursued. The identification, nomenclature and classification of the concerned plants make the first step of any research work in Botany.

Teaching Methodology :

Teaching methodology includes series of lectures making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc. In practicals, students would be provided with fresh/preserved materials for their morphological and anatomical studies making use of microscopes and binoculars and hands-on tools/equipment etc. Students will be taken for field excursions to various hill stations/forests to familiarize them with the flora of the area covering various plant forms.

SECTION-I : GYMNOSPERMS

UNIT-I

1. General characteristics, basis of classification and economic importance of gymnosperms.

2. Fossil gymnosperms : Brief account of fossils, their formation and types (excluding details) *Lyginopteris :* Introduction, external structure of stem; internal structure of primary stem, root and leaf; reproduction.

   Williamsonia : Introduction, external morphology; internal structure; reproductive organs, male and female flowers.

UNIT-II

Structure, reproduction (male and female strobilus; structure of ovule; development of male and female gametophytes; pollination, fertilization, development of embryo and structure of seed) and life cycle of :

(a) *Cycas*

(b) *Pinus*

(c) *Ephedra*

SECTION-II : ANGIOSPERMS

UNIT-III

1. General characters of Angiosperms.

2. Plant nomenclature and International Code of Botanical Nomenclature : Principles and rules; taxonomic ranks; type concept; principle of priority, aims and objectives of plant taxonomy; a brief introduction to taxonomy as synthetic discipline.

3. Salient features of Bentham and Hooker's, Hutchinson's and Engler & Prantl's system of classification (Details of Bentham and Hooker's system only).

4. Terminology pertaining to floral description.

5. Criteria for determining primitive and advanced nature of flower.

6. General account and diagnostic features of the following families (excluding economic importance) :

Liliaceae	:	*Asphodelus/Asparagus*
Orchidaceae	:	*Zeuxine*
Gramineae (Poaceae)	:	*Triticum*

UNIT-IV

General account and diagnostic features of the following families (excluding economic importance) :

Ranunculaceae	:	*Ranunculus, Delphinium*
Brassicaceae	:	*Brassica*
Rutaceae	:	*Citrus, Murraya*
Malvaceae	:	*Hibiscus*
Fabaceae	:	*Lathyrus, Cassia, Acacia*
Cucurbitaceae	:	*Luffa*
Umbelliferae (Apiaceae)	:	*Coriandrum*
Compositae (Asteraceae)	:	*Helianthus, Ageratum/Parthenium*
Asclepiadaceae	:	*Calotropis*
Solanaceae	:	*Solanum, Petunia*
Labiatae (Lamiaceae)	:	*Ocimum*
Chenopodiaceae	:	*Chenopodium*
Euphorbiaceae	:	*Euphorbia*

Suggested Readings :

1. Bhatnagar, S.P. and Moitra, A. : *Gymnosperms,* New Age International Limited, New Delhi, 1996.

2. Davis, P.H. and Heywood, V.H. : *Principles of Angiosperm Taxonomy,* Oliver and Boyd, London, 1963.

3. Gifford, E.M. and Foster, A.S. : *Morphology and Evolution of Vascular Plants,* W.H. Freeman & Company, New York, 1988.

4. Goyal, J.P. and Saini, A. : *Angiosperms : Structure Development and Reproduction,* Trueman Book Co. Jalandhar & New Delhi, 2006.

5. Heywood, V.H. and Moore, D.M. (eds.) : *Current Concepts in Plant Taxonomy,* Academic Press, London, 1984.

6. Jeffrey, C. : *An Introduction to Plant Taxonomy,* Cambridge University Press, London, 1982.

7. Jones, S.B., Jr. and Luchsinger A.E. : *Plant Systematics (2nd Edition),* McGraw Hill Book Co., New York, 1986.

8. Maheshwari, J.K. : *Flora of Delhi,* CSIR, New Delhi, 1963.

9. Radford, A.E. : *Fundamentals of Plant Systematics,* Harper and Row, New York, 1986.

10. Singh, G. : *Plant Systematics : Theory and Practice,* Oxford and IBH Pvt. Ltd., New Delhi, 1999.

11. Sporne, K.R. : *The Morphology of Gymnosperms,* Hutchinson & Co. (Publishers) Ltd., London, 1965.

12. Srivastava, H.N. : *Diversity of Seed Plants and their Systematics, Vol. III,* Pradeep Publications, Jalandhar, 2006.

13. Stace, C.A. : *Plant Taxonomy and Biosystematics (2nd Edition),* Edward Arnold, London, 1989.

14. Stewart, W.M. : *Paleobotany and the Evolution of Plants,* Cambridge University Press, Cambridge, 1983.

Paper B : STRUCTURE, DEVELOPMENT AND REPRODUCTION IN FLOWERING PLANTS

Max. Marks : 75
Theory : 68 Marks
Internal Assessment : 7 Marks
Time : 3 Hours

Note : 1. The number of teaching hours for theory and practical per session shall be 120 hrs. and 200 hrs. respectively.

2. There will be a total of nine questions in each theory paper A & B. Question No. 1 will be compulsory and will consist of 20 parts (one mark each) comprising 10 MCQ and the rest 10 parts will be of fill-in the blanks covering the entire syllabus in both the theory papers A & B. The remaining 8 questions in papers A & B shall include two questions from each unit. Candidates shall be required to attempt one question from each Unit. Question No. 1 will carry 20 marks and the rest of 8 questions will be of 12 marks each.

Objective :

This paper deals with structure development and reproduction in flowering plants – the most fascinating group of plants on earth. The course material of this paper deals with internal structure of various plant parts, their growth patterns and abnormalities in structural development. The vast range of variation found in this group of plants provides a platform to students for acquiring basic knowledge of diversity in flowering plants which makes a foundation of applied branches like horticulture, floriculture, olericulture and arboriculture.

Teaching Methodology :

Teaching methodology includes series of lectures making use of charts, transparencies, LCD, Models, slides, practical demonstrations, extension lectures from experts, field visits, discussions, quiz competitions etc. In practicals, students would be provided with fresh/preserved materials for their morphological and anatomical studies making use of microscopes and binoculars and hands-on tools/equipment etc.

UNIT-I

1. The basic body plan of a flowering plant.

2. Diversity in plant form in annuals, biennials and perennials.

3. *Leaf :* Origin, development, arrangement and diversity in size and shape, and modifications; internal structure (dicot and monocot leaves).

4. *Root System :* Structural modifications for storage, respiration and reproduction.

UNIT-II

1. *Tissue Systems :* Meristematic, shoot apex, root apex; simple and complex permanent tissues, special tissues; internal structure of primary dicot stem & root (*Helianthus*); secondary growth in dicot stem & root (*Helianthus*); internal structure of monocot stem & root (Maize).

2. Anomalous stem structure in *Boerhaavia, Nyctanthes, Mirabilis* and *Dracaena.*

3. Various methods of vegetative propagation.

UNIT-III

Flower : As a modified shoot, functions; Structure of anther and pistil; Male and female gametophytes; Double fertilization and its significance; Different types of ovules and embryo-sacs.

UNIT-IV

Types of pollination, Advantages and disadvantages of self and cross-pollination, Contrivances for self and cross-pollination, various agencies to bring about cross-pollination, characters of flowers pollinated by different agencies; Seed formation : development of endosperm and embryo; Fruit development and maturation. Significance of Seed : Suspended animation; Ecological adaptation; Dispersal strategies.

Suggested Readings :

1. Bhojwani, S.S. and Bhatnagar, S.P. : *The Embryology of Angiosperms,* 4th Revised and Enlarged Edition, Vikas Publishing House, Delhi, 2000.
2. Cutter, E.G. : *Part I, Cells and Tissues,* Edward Arnold, London, 1969.
3. Cutter, E.G. : *Plant Anatomy : Experiments and Interpretation, Part II, Organs,* Edward Arnold, London, 1971.
4. Esau, K. : *Anatomy of Seed Plants, 2nd Edition,* John Wiley & Sons, New York, 1977.
5. Fageri, K., and Van Der Pijl : *The Principles of Pollination Ecology*, Pergamon Press, Oxford, 1979.
6. Fahn, A. : *Plant Anatomy,* 2nd Edition, Pergamon Press, Oxford, 1974.
7. Hartmann, H.T., and Kestler, D.E. : *Plant Propagation : Principles and Practices,* 3rd Edition, Prentice Hall of India Pvt. Ltd., New Delhi, 1976.
8. King, J. : *Reaching for the Sun : How Plants Work,* Cambridge University Press, Cambridge, U.K., 1997.
9. Mauseth, J.D. : *Plant Anatomy,* The Benjamin/Cummings Publishing Company Inc., Menlo Park, California, U.S.A., 1988.
10. Proctor, M. and Yeo, P. : *The Pollination of Flowers,* William Collins Sons, London, 1973.
11. Raven, P.H., Evert, R.F. and Eichhorn, S.E. : *Biology of Plants,* 5th Edition, W.H. Freeman and Co., Worth Publishers, New York, 1999.
12. Thomas, P. : *Trees : Their Natural History,* Cambridge University Press, Cambridge, 2000.

Suggested Laboratory Exercises :

1. *Cycas :*
   (i) Habit, armour of leaf bases on the stem, young and old foliage leaves, scale leaves, male cone, megasporophyll, seed.
   (ii) Preparation of permanent stained slides of *Cycas* (T.S. leaflet, rachis and coralloid root).

2. *Pinus :*
   (i) Dwarf shoot; Male cone; Female cones (first year, second year and third year); Seed.
   (ii) Preparation of permanent slide of T.S. needle and Pollen grains.
   (iii) Study through permanent slides :
      (a) L.S. male cone.
      (b) L.S. female cone.
      (c) L.S. ovule.

3. *Ephedra :*
   (i) Habit and structure of male and female cones.
   (ii) Preparation of permanent slide of T.S. stem (young and old).
   (iii) Study through permanent slides :
      (a) L.S. female cone.
      (b) L.S. male cone.

4. Preparation of permanent stained slides to show :
   (i) Anomalous internal structure of stem in *Boerhaavia, Nyctanthes, Mirabilis* and *Dracaena.*
   (ii) T.S. Dicot stem & root (*Helianthus*).
   (iii) T.S. Monocot stem & root (Maize).

5. Description of flowers in technical terms, drawing F.D., V.S., T.S. ovary, writing F.F. and reference to their respective families of the genera mentioned in different families in theory syllabus.

Note : To make the students familiar with plants, (preferably pertaining to syllabus) the teachers are required to organize excursions to forests and hills.

Guidelines for Botany Practical Examination:

Max. Marks	:	50
Practical	:	45 Marks
Internal Assessment	:	5 Marks
Time	:	4 Hours

1.	Cut T.S. of specimen A. Make its permanent stained slide. Draw its well labelled diagram and show it to the examiner. Identify the slide giving at least two reasons.	9
2.	Describe specimen B or C in technical terms and proper sequence. Draw F.D. and V.S. of flower B or C. Cut T.S. of ovary and V.S. of flower and show them to the examiner. Write F.F. also. Refer specimen B or C to its respective family giving at least two reasons (The specimen B or C is to be given to the alternately seated students).	10
3.	Write illustrated morphological note on specimen D & E (from Gymnosperms).	8
4.	Identify slides F, G & H giving at least two reasons.	6
5.	Practical note book.	4
6.	Plant album.	3
7.	*Viva-Voce.*	5

ZOOLOGY

B.Sc. (GENERAL) SECOND YEAR EXAMINATION, 2015

Paper-A	:	Biodiversity -III (Chordates) & Evolution	75	(67 + 8 marks)
Paper-B	:	Biochemistry & Animal Physiology	75	(67 + 8 marks)
Practical		(Covering the entire syllabus) of 4 hours duration	50	(45 + 5 marks)
		Total :	**200 marks**	

Note : The number of hours for Theory and Practical per week shall be 5 hours and 4 hours respectively.

OBJECTIVES OF THE COURSE :

The syllabus pertaining to B.Sc. (General) Part-II, in the subject of Zoology has been upgraded as per provision of the UGC module and demand of the academic environment. The course contents have been revised from time to time as per suggestions of the teachers of the Zoology working in the Panjab University, Chandigarh and affiliated colleges.

The syllabus contents are duly arranged section wise as well as unit wise. The contents are included in such manner so that due importance may be given to skill oriented components.

The course contents are also given due stress for excursion/field trips to Zoological Parks, Sea-shores, Hill Stations, Museum, Fossil Park and Apiary/godowns for better academic outlook. The Department of Zoology, P.U., Chandigarh usually organizes workshop/seminars from time to time for updating the teachers.

PAPER–A : BIODIVERSITY III (CHORDATES) & EVOLUTION (ZOO 201)

Max. Marks	:	75
Theory	:	67 Marks
Internal Assessment	:	8 Marks
Time	:	3 Hours

Note : Nine questions are to be set. Question No.1 is **compulsory** consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1½ marks each. Two questions are to be set from each unit. One question is to be attempted from each unit. In all, *Five* questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts.

UNIT-I

Chordates – Origin, Parental care and migration.

Protochordates – Urochordata – Type Study – *Herdmania* except development.

Cephalochordata - Type Study – *Amphioxus* (except development).

Cyclostomata – External Characters of *Petromyzon* & affinities of Cyclostomata.

Pisces-Type study - *Labeo*, Scales & fins.

Classification of the animals up to orders relating to the following groups along with brief ecological notes of the following :

Protochordates	:	*Herdmania, Molgula, Pyrosoma, Doliolum, Salpa, Oikopleura, & Amphioxus (*excluding development*).*
Cyclostomata	:	*Myxine, Petromyzon* & *Ammocoetes* larva.
Chondrichthyes	:	*Zygaena* (Hammer headed shark), *Pristis* (saw fish), *Narcine* (electric ray), *Trygon*, *Rhinobatus* and *Chimaera* (rabbit fish).
Actinopterygii		*Polypterus, Acipenser, Lepisosteus, Muraena, Mystus, Catla, Hippocampus, Syngnathus, Exocoetus, Anabas, Diodon, Tetradon, Echeneis* and *Solea.*
Dipneusti (Dipnoi)	:	*Protopterus* (lung fish)

UNIT-II

Amphibia – Type study-Frog.

Reptilia-Type study-*Uromastix*, Poisonous and non-poisonous snakes, Poison apparatus.

Classification of the animals up to orders relating to the following groups along with brief ecological notes of the following :

Amphibia	:	*Uraeotyphlus, Necturus, Ambhiuma, Amblystoma* and its Axolotl Larva, *Triton, Salamandra, Hyla, Rhacophorus.*
Reptilia	:	*Chelone* (turtle) and *Testudo* (Tortoise), *Hemidactylus* (wall lizard), *Calotes, Draco, Varanus, Phrynosoma, Chamaeleon, Typhlops, Python, Eryx, Bungarus, Naja, Hydrus, Vipera, Crocodilus, Gavialis* and Alligator.

UNIT-III

Aves-Type study–Pigeon, Flight adaptations.

Mammals - Type study-Rat, Dentition.

Classification of the animals up to orders relating to the following groups along with brief ecological notes of the following :

Aves	:	*Ardea, Milvus, Pavo, Tyto, Alcedo, Eudynamis* and *Casuarius*.
Mammalia	:	*Ornithorhynchus, Echidna, Didelphis, Macropus* (Kangaroo), *Loris, Macaca, Manis* (Scaly ant eater), *Hystrix* (porcupine), *Funambulus* (Squirrel) *Panthera, Canis, Herpestes* (Mongoose), *Capra, Pteropus*.

UNIT-IV

Concept and evidences of organic evolution.

Theories of organic evolution.

Origin of life.

Concept of micro, macro and mega-evolution.

Biological concept of species.

Fossils and evolutionary rate.

Evolution of man.

PAPER B : BIOCHEMISTRY & ANIMAL PHYSIOLOGY (ZOO 202)

Max. Marks	:	75
Theory	:	67 Marks
Internal Assessment	:	8 Marks
Time	:	3 Hours

Note : Nine questions are to be set. Question No.1 is **compulsory** consisting of short answer type questions covering the whole syllabus. It will have 10 parts of 1½ marks each. Two questions are to be set from each unit. One question is to be attempted from each unit. In all, *Five* questions are to be attempted including compulsory one. 50% of the questions are to be split up into 2-4 sub-parts.

UNIT-I

Biochemistry and its scope.

Carbohydrates, proteins, lipids and nucleic acids: their classifications and functions.

Carbohydrate metabolism : The Embden Meyerhoff, Parnas pathway, (glycolysis), the tricarboxylic acid cycle, the hexose monophosphate shunt, glycogenesis and glycogenolysis.

UNIT-II

Enzymes : Nature, their classification and coenzymes.

Lipid metabolism : β-oxidation of fatty acids, rate of glycerol and gluconeogenesis, interaction of carbohydrates and lipids, lipogenesis in tissues, ketosis.

Protein metabolism : Protein metabolism of amino acids (Oxidative deamination, transamination and decarboxylation; hydrolysis of proteins and Ornithine cycle).

UNIT-III

Digestion : Digestion of dietary constituents, regulation of digestive processes and absorption. extra and intra cellular digestion, enzymatic digestion and symbiotic digestion.

Blood : Composition and functions of blood and lymph. Function of haemoglobin, blood clotting. Blood groups including Rh. Factor.

Heart : Origin and regulation of heart beat, cardiac cycle, electrocardiogram, cardiac output, blood pressure and micro-circulation.

Respiration : Transport of O_2 and CO_2, Oxygen dissociation curve of haemoglobin, Bohr effect, chloride (-) shift, Haldane effect and control of breathing.

UNIT-IV

Excretion	:	Urine formation, osmoregulation and counter current mechanism.
Muscles	:	Ultrastructure, chemical and physiological basis of skeletal muscle contraction.
Neural Integration	:	Structure of neuron, resting membrane potential, origin and propagation of impulse along the axon, synapse and myoneural function.
Endocrine System	:	Structure, hormones and functions of thyroid, parathyroid, adrenal, pineal, hypothalamus, pituitary, pancreas, gonads, thymus, hormones of alimentary canal and kidney.

PRACTICALS : Practical based on Theory Papers ZOO 201 & 202 (ZOO 251)

1. Classification up to orders habits, habitats, distinctive characters and economic importance (if any) of the following animals :

Urochordata	:	*Herdmania, Molgula, Pyorosoma, Doliolum, Salpa & Oikopleura.*
Cephalochordata	:	*Amphioxus.*
Cyclostomata	:	*Myxine, Petromyzon & Ammocoetes* Larva.
Chondrinchthyes	:	*Zygaena* (Hammer headed shark), *Pristis* (Saw fish), *Narcine* (Electric ray), *Trygon*, *Rhinobatus*, and *Chimaera* (Rabbit fish).
Actinoptergii	:	*Polypterus, Acipenser, Lepisosteus, Muraena, Mystus, Catla, Hippocampus, Syngnathus, Exocoetus, Anabas, Diodon, Tetradon, Echeneis* and *Solea.*
Dipneusti (Dipnoi)	:	*Protopterus.*
Amphibia	:	*Uraeotyphlus, Necturus, Amphiuma, Amblystoma* and its Axolotl larva, *Triton, Salamandra, Hyla, Rhacophorus.*
Reptilia	:	*Chelone* (Turtle) and *Testudo* (Tortoise), *Hemidactylus, Calotes, Draco, Varanus, Phrynosoma, Chamaeleon, Typhlops, Python, Eryx, Ptyas, Bungarus, Naja, Hydrus, Vipera, Crocodilus, Gavialis.*
Aves	:	*Casuarius, Ardea, Anas, Milvus, Pavo, Eudynamis, Tyto* and *Alcedo*.

Mammalia : *Ornithorhynchus, Echidna, Didelphis, Macropus, Loris, Macaca, Manis, Hystrix, Funambulus, Panthera, Canis, Herpestes, Capra, Pteropus.*

2. Examine and dissect the following animals :

Herdmania	:	General anatomy
Labeo	:	Digestive, reproductive systems, heart, afferent and efferent branchial arteries and cranial nerves.
Chick	:	Digestive, arterial, venous and urinogenital systems.
White Rat	:	Digestive, arterial, venous and urinogenital systems.

3. Study of the skeleton of *Labeo, Rana* (Frog), *Varanus, Gallus* and Rat/*Oryctolagus* (Rabbit).

4. Study of the following prepared slides :

   T.S. *Amphioxus* through various regions.

   Spicules & pharynx of *Herdmania* & pharynx of *Amphioxus*.

   Histology of rat/rabbit (compound tissues).

5. Make temporary preparations of the following :

   Temporary preparation of spicules of *Herdmania.*

   Permanent preparation of whole mount of pharynx of *Herdmania* and *Amphioxus.*

   Blood smear of a mammal.

6. Identification of food stuffs : Starch, glucose, proteins and fats in solution.

7. Demonstration of osmosis and diffusion.

8. Demonstrate the presence of amylase in saliva, denaturation by pH and temperature.

9. Determination of coagulation and bleeding time of blood in man/rat/rabbit.

10. Determination of blood groups of human blood sample.

11. Recording of blood pressure of man.

12. Analysis of urine for urea, chloride, glucose and uric acid.

13. Estimation of haemoglobin content.

Field study : Visit to a fossil Park/Museum.

Familarity with the local vertebrate fauna and report.

Guidelines for conduct of Practical Examination :

Max. Marks : 50
Practical Exam. : 45 Marks
Internal Assessment : 5 Marks
Time : 4 Hours

1. Dissect the animal provided so as to expose its _____ system. Make labelled sketch of the dissection and demonstrate the same to the Examiner. (7)

2. Identify the given bones A & B. Make labelled sketches of their respective _____views. (6)

3. Make temporary mount of the material C. Identify and make labelled sketch. Show the preparation to the Examiner. (3)

OR

Minor experiment of physiology such as blood groups determination/identification of some food stuffs/presence of chloride or glucose in urine etc.

4. Perform the given physiology experiment, write the procedure and show it to the examiner. (5)

5. Identify the slides D & E giving two reasons for each identification. (2+2=4)

6. Identify and classify the specimens F to I upto orders. Write a short note on habitat of F, special feature of G, feeding habit of H and economic importance of specimen I.

(1+1+1) each × 4 = (12)

7. Viva-Voce (3)

8. Note book and Project Report (3+2 =5)

Suggested Readings :

1. Parker, T.J. and Haswell, W.A. : *Text Book of Zoology, Vol. II (Vertebrates),* ELBS and Macmillan Press Ltd., 1981.

2. Dobzhansky, T., Ayala, F.J., Stebbins, G.L., Valentine, J.W. : *Evolution,* WH Freeman, 1952.

3. Colbert, E.H. : *Evolution of Vertebrates,* IInd Edition, Wiley Eastern Ltd., 1989.

4.	Dhami, P.S. & Dhami, J.K.	:	*Vertebrates,* R. Chand & Co., New Delhi, 1998.
5.	Taneja, S.K.	:	*Biochemistry & Animal Physiology,* Trueman Book Co.,1997.
6.	Guyton, A.S.	:	*Text Book of Medical Physiology,* 7th Edition, W.B. Saunders Company, 1994.
7.	Robert, K., Murray, R.K., Daryl, M., Granner, K., Victor, W. and Woodwell	:	*Harper's Biochemistry,* 22nd Edition, Prentice–Hall International Inc., 1990.
8.	Nelson, D.L. and Cox, M.M.	:	*Lehninger Principles of Biochemistry,* 5th Edition, W.H., Freeman and Company, New York, 2008.
9.	Bhamrah, H.S., Juneka, K.	:	*Cytogenetics & Evolution,* Anmol Publication Pvt. Ltd., 1993.

BIOCHEMISTRY

B.Sc. (GENERAL) SECOND YEAR EXAMINATION, 2015

Scheme of Examination

			Duration	*Max. Marks*
Theory Paper-A	:	Metabolism	3 hrs.	67+8
Theory Paper-B	:	Biochemical Techniques	3 hrs.	67+8
One practical pertaining to the entire syllabus included in Theory Papers A & B			3 hrs.	45+5
		Total Marks :		**200**

Paper-A : METABOLISM

Max. Marks	:	75
Theory	:	67 Marks
Internal Assessment	:	8 Marks
Time	:	3 Hours
(Three Periods per week)		

INSTRUCTIONS FOR PAPER-SETTER AND CANDIDATES :

1. Total No. of questions will be nine.
2. Q. Paper will be divided into 4 sections.
3. Attempt 5 questions in all selecting one question from each section. Q. No. 1 is **compulsory.**
4. Q. No. 1 will consist of 10 short questions covering entire syllabus and will carry 15 marks.
5. All other questions may contain 2-3 parts and will be of 13 marks each.

Objectives :

To learn digestion and absorption of major classes of biomolecules and major metabolic pathways pertaining to them.

SECTION-I

Digestion & Absorption of Carbohydrates : (Lectures : 14)

Metabolic Pathways of Carbohydrates, Glycolysis and alcoholic fermentation, The Pentose Phosphate Pathway, Glucuronate and glyoxylate pathway, TCA cycle, Glycogenolysis & Glycogenesis, Gluconeogenesis, Biosynthesis of starch, Biosynthesis of Ascorbic acid.

SECTION-II

Digestion & Transport of Lipids : (Lectures : 14)

β-Oxidation of fatty acids including odd chain fatty acids. $\alpha$and ω oxidation of fatty acids Degradation of triglycerides and phospholipids. Formation and utilization of ketone bodies. Biosynthesis of saturated and unsaturated fatty acids. Biosynthesis of triglycerides and phospholipids, biosynthesis of cerebrosides; sulfatides and gangliosides. Biosynthesis of Cholesterol. Biosynthesis of Prostaglandins, Thromboxanes, Leukotrienes, Lipoxins and Prostacyclins.

SECTION-III

Digestion of Proteins : (Lectures : 11)

General Reactions of Amino Acids : Deamination, transamination and decarboxylation. Urea cycle.

Catabolism of Carbon Skeletons of Amino Acids :

Glycine and Alanine, Serine and threonine, Phenylalanine and Tyrosine, Tryptophan, Histidine, Leucine, Valine and Isoleucine, Cysteine and Methionine, Lysine, Glutamic acid and Glutamine, Aspartic acid and Asparagine.

Biosynthesis of Nutritionally Non-Essential Amino Acids :

Glutamate and Glutamine, Aspartate and Asparagine, Proline, Alanine, Cysteine & Selenocysteine, Tyrosine, Serine, Glycine.

SECTION-IV

Nucleic Acids : (Lectures : 11)

Degradation of purines and pyrimidines. Biosynthesis of purines, pyrimidines and nucleotides. Catabolism of Heme & Formation of Bile pigments. Biosynthesis of porphyrins and heme. Conjugation of bilirubin and its clinical significance.

Books Suggested :

1. *Biochemistry* by Lubert Stryer, 5th Ed., 2002, W.F. Freeman and Co., New York.

2. *Lippincott's Illustrated Reviews : Biochemistry* by P.C. Champe, R.A. Harvey & D.R. Ferrier, Indian Edition, 2005.

3. *Harper's Illustrated Biochemistry* by R.K. Murray, P.A. Mayes, D.K. Granner and V.W. Rodwell, 27th Ed., 2006, Prentice Hall International Ltd. (U.K.).

Paper- B : BIOCHEMICAL TECHNIQUES

Max. Marks : 75
Theory : 67 Marks
Internal Assessment : 8 Marks
Time : 3 Hours
(Three periods per week)

INSTRUCTIONS FOR PAPER-SETTER AND CANDIDATES :

1. Total No. of questions will be nine.

2. Q. Paper will be divided into 4 sections.

3. Attempt 5 questions in all selecting one question from each section. Q. No. 1 is **compulsory.**

4. Q. No. 1 will consist of 10 short questions covering entire syllabus and will carry 15 marks.

5. All other questions may contain 2-3 parts and will be of 13 marks each.

Objectives :

Principles and applications of spectroscopic, electrophoretic, sedimentation, chromatographic and radioisotopic techniques.

SECTION-I

Spectroscopic Techniques : (Lectures : 13)

Beer-Lambert's Law. Light absorption and its transmittance. Determination and application of extinction coefficient. Applications of following spectroscopic techniques in elucidating structure of Biomolecules-Visible, U.V., infra-red and fluorescence spectroscopy. ORD, C.D. and N.M.R.

SECTION-II

Electrophoretic Techniques : (Lectures : 13)

Principles and applications of the following electrophoresis techniques. Paper and gel electrophoresis, high voltage electrophoresis, SDS-PAGE : Discontinuous electrophoresis, isotachophoresis, isoelectric focussing and immunoelectrophoresis.

Centrifugation Techniques :

Various centrifugation techniques and their applications in Biochemistry. Preparative and analytical ultra-centrifugation procedures. Application of partial specific volume, diffusion coefficient and viscosity measurements in the study of macromolecules of biochemical importance.

SECTION-III

Chromatographic Techniques : (Lectures : 12)

General principles of chromatography and the application of following chromatographic procedures in isolation and purification of biomolecules : Absorption, partition, paper and thin layer chromatography. Gas liquid chromatography. High performance liquid chromatography (HPLC), Ion exhange and Exclusion chromatography. Affinity chromatography.

SECTION-IV

Radio Isotopic Techniques : (Lectures : 12)

Nature of isotopes and radioisotopes. Radioactive decay. Properties of radioactive emissions. Units of radioactivity. Techniques used to measure radioactivity; GM counter and liquid scintillation counting and gamma counter. Labelling of Biochemical compounds and autoradiography. Use of radioactive tracers in the study of enzyme reaction mechanisms and metabolic pathways. Radioimmuno assay. Biological hazards of radiation and safety measures in handling radioisotopes.

Books Suggested :

1. *Biochemical Techniques – Theory and Practice* by John. F. Robyt and Bernard J., White Cole Publishing Co. (U.S.A.).

2. *Physical Biochemistry : Application to Biochemistry and Molecular Biology* by David Freifelder, W.H. Freeman and Company (U.S.A.).

3. *A Biologist's Guide to Principles and Techniques of Practical Biochemistry* by K.H. Goulding and Keith Wilson, Edward Arnold (U.K.) Ltd.

PRACTICALS :

Max. Marks : 50
Practical : 45 Marks
Internal Assessment : 5 Marks
(One Practical of 3 hrs.per week)

1. Estimation of blood glucose by the methods of (i) Folin Wu (ii) Nelson Somogyi.
2. Isolation and assay of glycogen from rat liver.
3. Separation and identification of amino acids by (i) Paper chromatography (ii) Thin layer chromatography.
4. Estimation of Ca^{+} in serum.
5. Estimation of total and free cholesterol in serum.
6. Estimation of total lipids in serum by Vanillin method.
7. Estimation of proteins by Lowry's method.
8. Estimation of Lipoproteins in plasma.
9. Colorimetric estimation of inorganic phosphate.
10. Separation of phospholipids by thin layer chromatography.
11. Estimation of lactic acid in blood before and after exercise.
12. Preparation of starch from potato and its hydrolysis by salivary amylase.

ELECTRONICS

B. Sc. (GENERAL) SECOND YEAR EXAMINATION, 2015

Note :

1. There will be two papers of theory and one laboratory (practical) courses.
2. The number of lectures per week will be four for each theory paper.
3. The number of lectures per week will be six for practicals.
4. The examination time for each theory paper will be three hours.
5. The examination time for practical will be 4 hours.
6. The use of Non-programmable calculators will be allowed in the examination centre, but these will not be provided by the University/College.

Distribution of Marks & Total Teaching Hours will be as under :

Paper A :	Principles of Communication Systems	: (67 + 8*) = 75 marks	Total Teaching Hours	75
Paper B :	Microprocessor Based Systems	: (67 + 8*) = 75 marks	Total Teaching Hours	75

* Internal assessment

Electronics Practicals	: 50 marks	Total Teaching Hours	120

Instructions for the Paper-Setter and the Students :

Each theory paper will consist of five units :

UNIT-I : There will be two questions from this unit. Each question will have two parts. Only one question is to be attempted. Each question will carry thirteen marks.

UNIT-II : There will be two questions from this unit. Each question will have two parts. Only one question is to be attempted. Each question will carry thirteen marks.

UNIT-III : There will be two questions from this unit. Each question will have two parts. Only one question is to be attempted. Each question will carry thirteen marks.

UNIT-IV : There will be two questions from this unit. Each question will have two parts. Only one question is to be attempted. Each question will carry thirteen marks.

UNIT-V : There will be eight questions of small answer type covering the syllabi of all the four units (I-IV). Five questions are to be attempted. Each question will carry three marks.

Paper-A : PRINCIPLES OF COMMUNICATION SYSTEMS

UNIT-I

Communication Principles :

Communication bands, basic principles of propagation of e.m. waves through atmosphere and ionosphere: ground wave, sky wave, space waves, Dead Zones, Concept of modulation and its advantages, qualitative treatment of FM, AM (DBS, SSB), PM.

Superhetrodyne Receivers :

Principles, advantages, RF input and AE coupling arrangement, RF amplifier, mixers, local oscillator, IF amplifier, detector, audio amplifier, loud speaker, tuning/alignment of receivers, trouble shooting of radio receiver.

UNIT-II

Tape Recorders :

Principles of magnetic recording, playback principles, need for biasing, tape speed, magnetic tape, head care, head alignment, high frequency response, record replay equalization, noise, noise reduction, tape transport, tape recorder electronics, playback amplifier, record amplifier.

Digital Audio :

Sampling and Quantizing the input signal, recording system, playback, compact disc, CD player (block diagram).

UNIT-III

Elements of T.V. Systems :

Overview of picture transmission, sound transmission, picture reception, sound reception, synchronization.

Composite Video Signal :

Scanning sequence details, sync details of 625 line system, channel band width, vestigial side band transmission, reception of vestigial side band signals, frequency modulation, FM channel band width, allocation of frequency bands for television, signal transmission, television standards.

Picture Tube- Monochromatic :

Beam deflection, screen phosphor, face plate, picture tube characteristics, picture tube circuit controls.

UNIT-IV

Television Receivers :

Block diagram of black and white T.V. receivers, functions of various sections i.e. Balun, RF amplifier, mixer local oscillator, IF amplifier, video detector, video amplifier, sound IF, sound detector and amplifier, sync separators, horizontal and vertical oscillator and their synchronization, EHT section, AGC, DC power supply.

Fundamentals of Colour T.V.:

Concept of primary and complementary colours, colour mixing, beam splitting in camera, colour difference signals (U and V signals), luminance (Y) and chrominance (C) signals, compatibility with black and white signals, colour signal subcarrier, concept of colour picture tube (PIL).

Books Recommended :

1. G.K. Mitthal : Radio Engineering, Khanna Publishers.
2. Sanjay Attri : Audio and Video Systems, BPB Publications.
3. R.R. Gulati : Monochrome and Colour Television, New Age International New Delhi.

Paper- B : MICROPROCESSOR BASED SYSTEMS

UNIT-I

Microcomputer Fundamentals :

Introduction, microcomputer architecture, memory organization, Memory classification, CPU organization, Microcomputer operation.

8085 Microprocessor Fundamentals :

Introduction, The 8085 pin diagram and functions, The 8085 microprocessor architecture, 8085 Bus system, Demultiplexing of 8085 bus system, The 8085 machine cycles.

UNIT-II

Interfacing the Microprocessor :

Introduction to basic interfacing concepts, Interfacing of 8085 microprocessor with ROM and RAM, Interfacing of 8085 microprocessor with input/output devices, Synchronizing I/O data transfer using interrupts, Address decoding, Memory mapping, The 8155 multipurpose programmable device.

UNIT-III

Instruction Set :

Machine and assembly language, Instruction set : data transfer instructions, arithmetic instructions, logical instructions, branch instructions, stack, I/O machine control instructions, addressing modes, Subroutine, CALL and RETURN instructions, Idea of counters, Time delays, Miscellaneous instructions.

UNIT-IV

Programming 8085 Microprocessor :

Straight line programs, Program branching, Program looping, Program indexing, Mathematical programs addition with and without carry, subtraction, multiplication of two 8 bit numbers.

Applications of 8085 Microprocessor :

(i) Traffic Control

(ii) Temperature Control

(iii) Digital Clock

Books Recommended :

1. Ramesh G. Gaonkar : Microprocessor Architecture, Programming and Applications, Prentice Hall Publications.
2. B. Ram : Fundamentals of Microprocessors and Microcomputers, Dhanpat Rai & Sons.
3. Bhupindra Singh Chhabra and Amrit Chhabra : 8085 Microprocessor and its Applications, Dhanpat Rai Publishing Co.

ELECTRONICS PRACTICALS

Guidelines for Electronics Practical Examinations :

I *The distribution of marks is as follows :*

(i)	One full experiment requiring the student to take data, Analyse it and draw conclusions	:	20 marks
(ii)	Brief Theory	:	05 marks
(iii)	One exercise based on experiment (to be allotted by the examiner)	:	10 marks
(iv)	Viva-Voce	:	10 marks
(v)	Record (Practical file)	:	05 marks

II There will be one session of 4 hours duration. The paper will consist of 8 experiments out of which an examinee will mark 6 and one of them will be allotted by the examiner.

III Number of candidates in a group for practical examination should not exceed 12.

IV In a single group, same experiment will not be allotted to more than three examinees in any group.

List of Experiments :

1. (a) Writing a program to add two 8 bit numbers stored in memory locations NNNN and NNNN+1 and place the result in the subsequent memory location NNNN+2.

   (b) Writing a program to add two 8 bit numbers stored in memory locations NNNN and NNNN+1 and place the result (16 bits) in the subsequent memory location NNNN+2 and NNNN+3.

2. Writing a program to subtract one 8 bit number from the other 8 bit number stored in memory locations NNNN and NNNN+1 and place the result in the subsequent memory location NNNN+2.

3. Writing a program to add more than two 8 bit numbers stored in memory locations starting from NNNN and place the result (16 bits) in the subsequent memory locations.

4. Writing a program to multiply two 8 bit numbers stored in memory locations NNNN and NNNN+1 and place the result in the subsequent memory location NNNN+2 and carry at NNNN+3.

5. Writing a program to find smaller/larger of two 8 bit numbers stored in memory locations NNNN and NNNN+1 and place the result in the subsequent memory location NNNN+2.

6. Writing a program to transfer bulk of data from memory locations starting from NNNN to memory locations starting from MMMM.

7. Writing a program to find number of positive and negative numbers in the given data. Assume zeros as positive numbers.

8. To locate various blocks of T.V. as P.S., H and V deflection, Video Amplifier, Audio section, Chroma section, IF section, Tuner, Tube, deflection yokes etc.

9. VIF and SIF section—Testing, alignment and troubleshooting.

10. Vertical Amplifier section—Testing and troubleshooting.

11. Tuner—Testing and troubleshooting.

12. P. S. and Picture tube—Testing and troubleshooting.

13. Study of Modulo 10 Counters.

14. Study of operation of shift registers.

15. Study of DA and AD converters.

MICROBIOLOGY

B.Sc. (GENERAL) SECOND YEAR EXAMINATION, 2015

Scheme of Examination	*Duration*	*Max. Marks*
Theory : MIC 201 : Introduction to Medical Microbiology	3 hrs.	75 (67 + 8*)
Theory : MIC 202 : Introductory Immunology	3 hrs.	75 (67 + 8*)
One Practical pertaining to MIC-201 and MIC-202.	3 hrs.	50 (45 + 5*)
Total Marks :		**200**

Note : *Denotes marks for the Internal Assessment.

OUTLINES OF TESTS AND SYLLABI

MIC 201 : INTRODUCTION TO MEDICAL MICROBIOLOGY

Max. Marks : 75
Theory : 67 Marks
Internal Assessment : 8 Marks
Time : 3 Hours
(Three periods per week)

Note : The question paper will consist of four sections (A-D). There will be nine questions and five questions have to be attempted. Question 1 will span the complete syllabus and will be **compulsory.** Rest eight questions will be from different sections of the syllabus. There will be two questions from each of the four sections and one is to be attempted. Each question should be sub-divided into 2-4 sub-parts.

Objectives :

The paper deals with diversification of microorganisms, various infectious diseases including their modes of transmission and events in infection process. It also provides knowledge about latest developments in chemotherapy.

SECTION-A

Brief introduction to terminology of Infectious diseases, Frequency of disease, Recognition of Infectious disease, Infections, Disease cycle, Virulence and mode of transmission, Emerging and reemerging Infectious diseases, Global travel & Health considerations, Nosocomial Infectious.

SECTION-B

Nomenclature and classification of microbes of medical importance. Origin of the Normal Flora, Germfree and Gnotobiotic Life, Distribution and occurrence of Normal Flora of Skin, Eye, Respiratory Tract, Mouth, Intestinal Tract & Genitourinary Tract.

SECTION-C

Microbial adherence, Passive Penetration into body, Active Penetration into body, Events in Infection following penetration, Microbial virulence factors.

SECTION-D

Development of chemotherapy, General characteristics of antimicrobial drugs, Determining level of antimicrobial activity, Mechanism of action of antimicrobial agents, factors influencing the effectiveness of antimicrobial drugs, Antibacterial drugs viz sulfonamides, Quinolones, Penicillins, Cephalosporins, Tetracyclines, Erythromycin, Chloramphenicol, Drug Resistance, Antifungal and Antiviral drugs.

Reference Books :

1. *Bergey's Manual of Systematic Bactriology. Vol-I (2001),* The Archae and the Deeply Branching Phototrophic Bacteria, David, R. Boone and Richard, W. Castenholz (author) George Garrity (editor).

2. Prescott, L.M., Harley, J.P., Klein, D.A. (2002), *Microbiology,* Mc Graw Hill.

3. Tortora, G.J., Funke, B.R., Case, C.L. (2004), *Microbology, An Introduction,* Pearson Education.

4. Mims, C., Playfair J., Roitt I., Wakelin D., Williams R., (2000), *Medical Microbiology,* Mosby, London, St. Louis.

MIC 202 : INTRODUCTORY IMMUNOLOGY

Max. Marks	:	75
Theory	:	67 Marks
Internal Assessment	:	8 Marks
Time	:	3 Hours
(Three periods per week)		

Note : The question paper will consist of four sections (A-D). There will be nine questions and five questions have to be attempted. Question 1 will span the complete syllabus and will be **compulsory.** Rest eight questions will be from different sections of the syllabus. There will be two questions from each of the four sections and one is to be attempted. Each question should be sub-divided into 2-4 sub-parts.

Objectives :

The paper focuses on history of immunology, various kinds of defense systems operative against microbial diseases and latest techniques used in immunology.

SECTION-A

Introduction and history of Immunology, Non-specific Defense; Physical Barriers, Chemical Barriers, Phagocytosis, Inflammation, Fever, Types of Immunity, Active & Passive Immunity, Immunological memory, Primary & Secondary Lymphoid organs, Mucosa Associated Lymphoid tissue (MALT), Cutaneous Associated Lymphoid Tissue (CALT), Lymphocyte Traffic, Cells of immune system, Antigens; factors affecting Immunogenicity, epitopes, haptens.

SECTION-B

Humoral Immune Response, Antibodies/Immunoglobulins, Structure, function and type of antibodies, Antigentic-combining regions of antibodies, factors influencing antibody production, Genetic model, Multigene Organisation, generation of antibody diversity.

SECTION-C

Cell Mediated Immune System, Mechanism of CMI, Types of effector T Cells, Helper T-cells, Suppressor, T-cells, cytotoxic T cells, Killer T cells, Cytokines, Lymphokines, Colony Stimulating factors, Tumour Necrosis factor, Interferons, Accessory cells (Macrophages), the Complement System, Classical and Alternate pathway, HLA, Monoclonal antibody technology and its applications, Interactions between B and T lymphocytes.

SECTION-D

Antigen-Antibody Interactions : Precipitation reaction, Immuno-diffusion test, counter current Immuno electrophoresis, complement fixation tests, Widal test, Wasserman's test, Weil Felix reaction, Western Blotting, Types of vaccines.

Reference Books :

1. T.J. Kindt, R.A.G.B.A. Osborne, J. Kuby (2006), *Immunology,* W.H. Freeman and Company, USA.

2. N. Shetty (2005), *Immunology, Introductory Text Book,* New Age International (P.) Limited, (Formerly Wiky Eastern Limited).

3. Paul, W.E. (2008), *Fundamental Immunology,* Lippincott Williams & Wilkins.

4. Roitt, I.M., Brostoff, J., Male, D.K. (2001), *Immunology,* Illustrated Publishers, Mosby.

5. Lydyard, P.M., Whelan, A., Fanger, M.W. (2003), *Instant Notes in Immunology,* Viva Books Private Limited.

6. Kannan, I. (2007), *Immunology,* M.J.P. Publishers.

7. Hannigan, B.M. (2008), *Immunology,* Viva Books Private Limited.

8. Cappuccino, J.G., Sherman N. (2008), *Microbiology, A Laboratory Manual,* Tara Art Printers (P.) Limited.

PRACTICALS

Max. Marks : 50
Practical : 45 Marks
Internal Assessment : 5 Marks
(One practical of 3 hrs. per week)

1. Stainings – Gram's, Alberts, ZNCF.
2. Isolation and Maintenance of Pure Cultures.
3. Physiological characteristics of bacteria and its use for their identification.
4. Assay of antimicrobials.
5. Preparation of serum/plasma.
6. Sterilization – Introduction to autoclave, hot air oven, filter sterilization.
7. Demonstration of Immune organs in dissected animal.
8. Demonstration of Immune cells in the smears prepared from Immune organs.
9. Complement fixation.
10. Antigen-antibody interactions :
    - Agglutination
    - Precipitation
    - Blood grouping
    - Immunodiffusion

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